

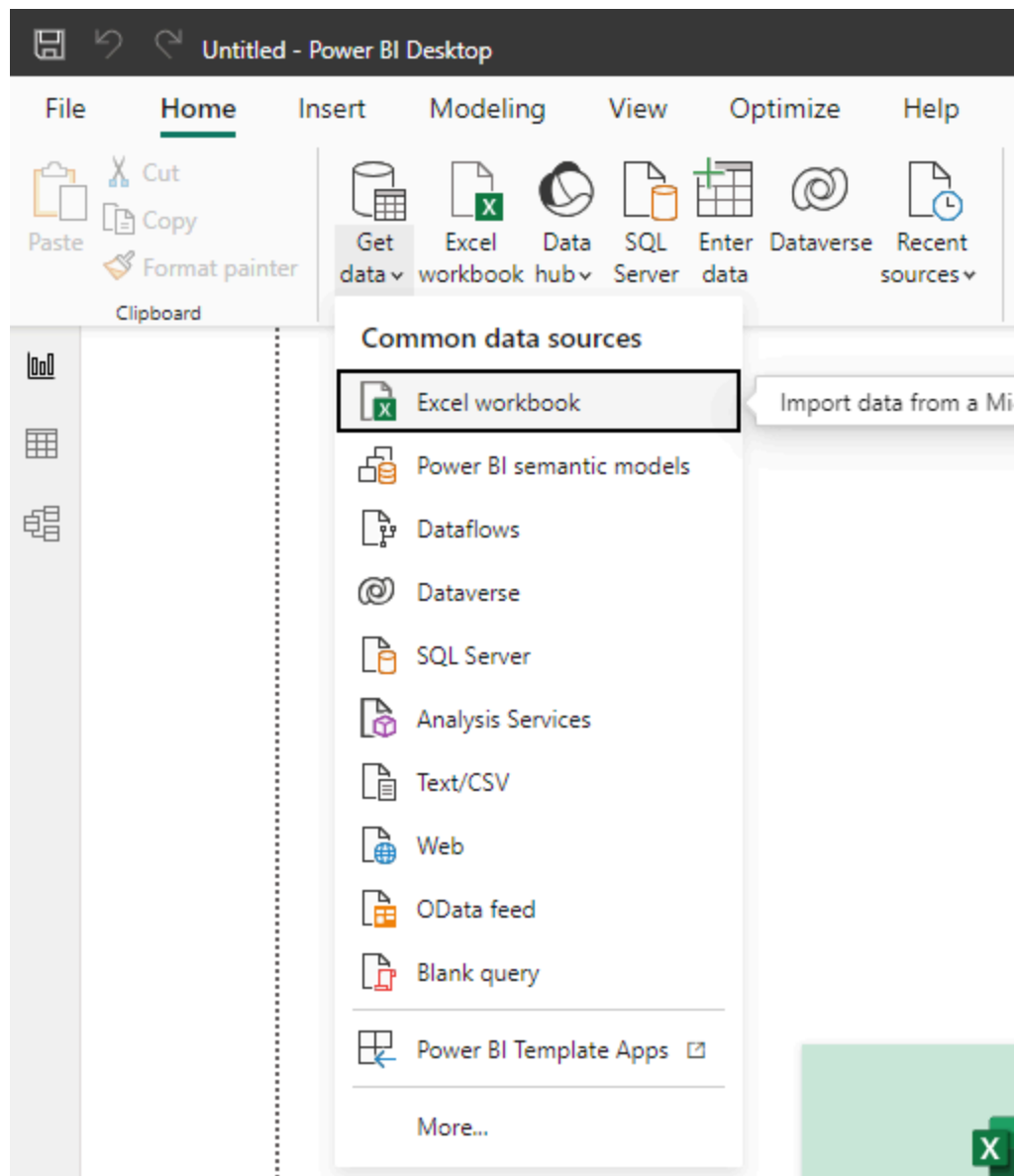
PRACTICAL 1

Aim: Import the legacy data from different sources such as (Excel, SQL Server, Oracle etc.) and load in the target system.

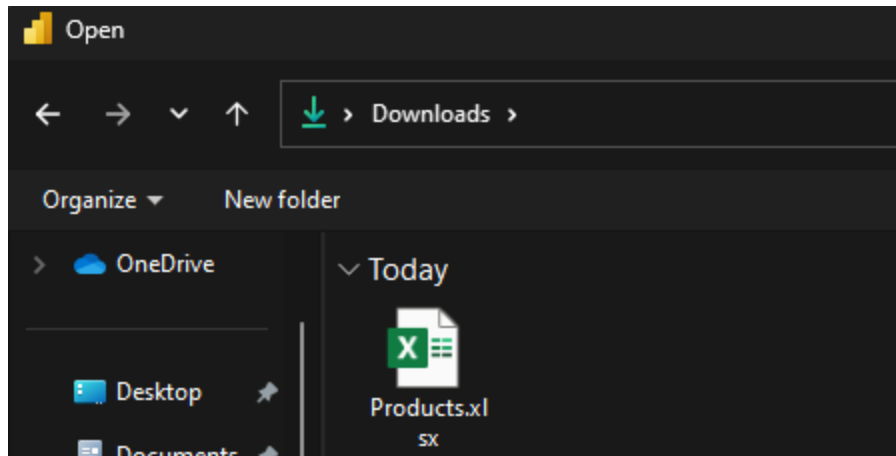
Steps: -

A) IMPORTING EXCEL DATA TO POWER BI

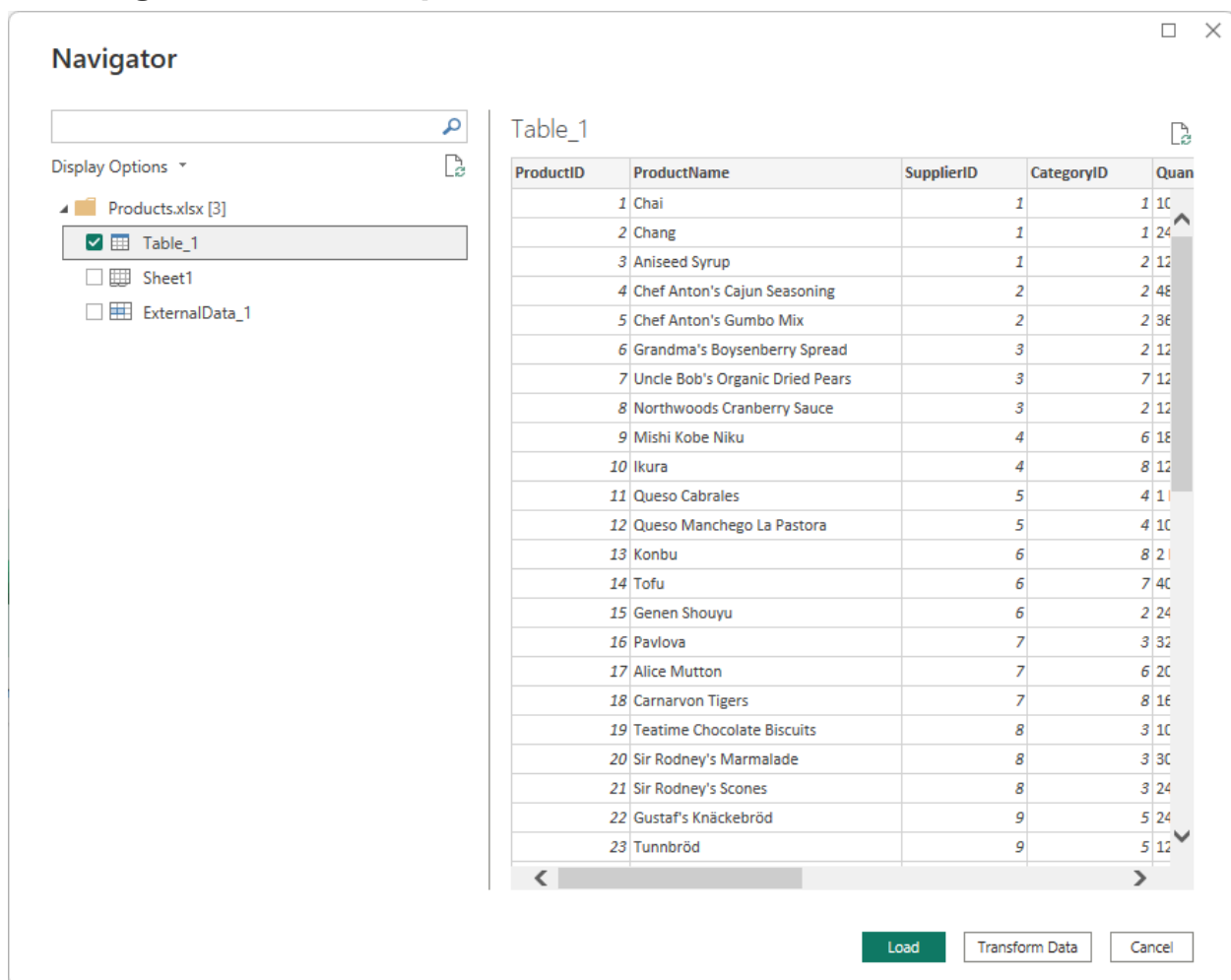
- 1. Open Power BI.**
- 2. Click on Get data following list will be displayed → select Excel**



3. Select the Products.xlsx file



4. Navigator window opens, select table1 and then load



5. Excel file:-

Untitled - Power BI Desktop

File Home Help Table tools

Name Table_1

Make an empty table v

Manage relationships

New Quick measure measure column table

Structure

Table tools

Structure

Table_1

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder	ReorderLevel	Discontinued
1	Chai	1	1	10 boxes x 20 bags	18	39	0	10	False
2	Chang	1	1	24 - 12 oz bottles	19	17	40	25	False
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10	13	70	25	False
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0	False
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35	0	0	0	True
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	25	False
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30	15	0	10	False
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0	False
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97	29	0	0	True
10	Ikura	4	8	12 - 200 ml jars	31	31	0	0	False
11	Quesco Calasque	5	4	1 kg pkg.	21	22	30	30	False
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38	86	0	0	False
13	Konbu	6	8	2 kg box	6	24	0	5	False
14	Tofu	6	7	40 - 100 g pkgs.	23.25	35	0	0	False
15	Genen Shouyu	6	2	24 - 250 ml bottles	15.5	39	0	5	False
16	Pavlova	7	3	32 - 500 g boxes	17.45	29	0	10	False
17	Alice Mutton	7	6	20 - 1 kg tins	39	0	0	0	True
18	Carnarvon Tigers	7	8	16 kg pkg.	62.5	42	0	0	False
19	Taiyume Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2	25	0	5	False
20	Sir Rodney's Marmalade	8	3	30 gift boxes	81	40	0	0	False
21	Sir Rodney's Scones	8	3	24 pkgs. x 4 pieces	10	3	40	5	False
22	Gustaf's Knäckebröd	9	5	24 - 500 g pkgs.	21	104	0	25	False
23	Tunnbröd	9	5	12 - 250 g pkgs.	9	61	0	25	False
24	Guaraná Fantástica	10	1	12 - 355 ml cans	4.5	20	0	0	True
25	NuNuCa Nuß-Nougat-Creme	11	3	20 - 450 g glasses	14	76	0	30	False
26	Gummibär Gummibärchen	11	3	100 - 250 g bags	31.23	15	0	0	False
27	Schoggi Schokolade	11	3	100 - 100 g pieces	43.9	49	0	30	False
28	Rössle Sauerkraut	12	7	25 - 825 g cans	45.6	26	0	0	True
29	Thüringer Rostbratwurst	12	6	50 bags x 30 sausgs.	123.79	0	0	0	True
30	Nord-Ost Matjeshering	13	8	10 - 200 g glasses	25.89	10	0	15	False
31	Gorgonzola Telino	14	4	12 - 100 g pkgs	12.5	0	70	20	False
32	Mascarpone Fabioli	14	4	24 - 200 g pkgs.	32	9	40	25	False
33	Gelbst	15	4	500 g	2.5	112	0	20	False
34	Sasquatch Ale	16	1	24 - 12 oz bottles	14	111	0	15	False
35	Steeleye Stout	16	1	24 - 12 oz bottles	18	20	0	15	False
36	Inlagd Sill	17	8	24 - 250 g jars	19	112	0	20	False
37	Gravad lax	17	8	12 - 500 g pkgs.	26	11	50	25	False
38	Öst	18	1	12 - 75 cl bottles	362.5	17	0	14	False

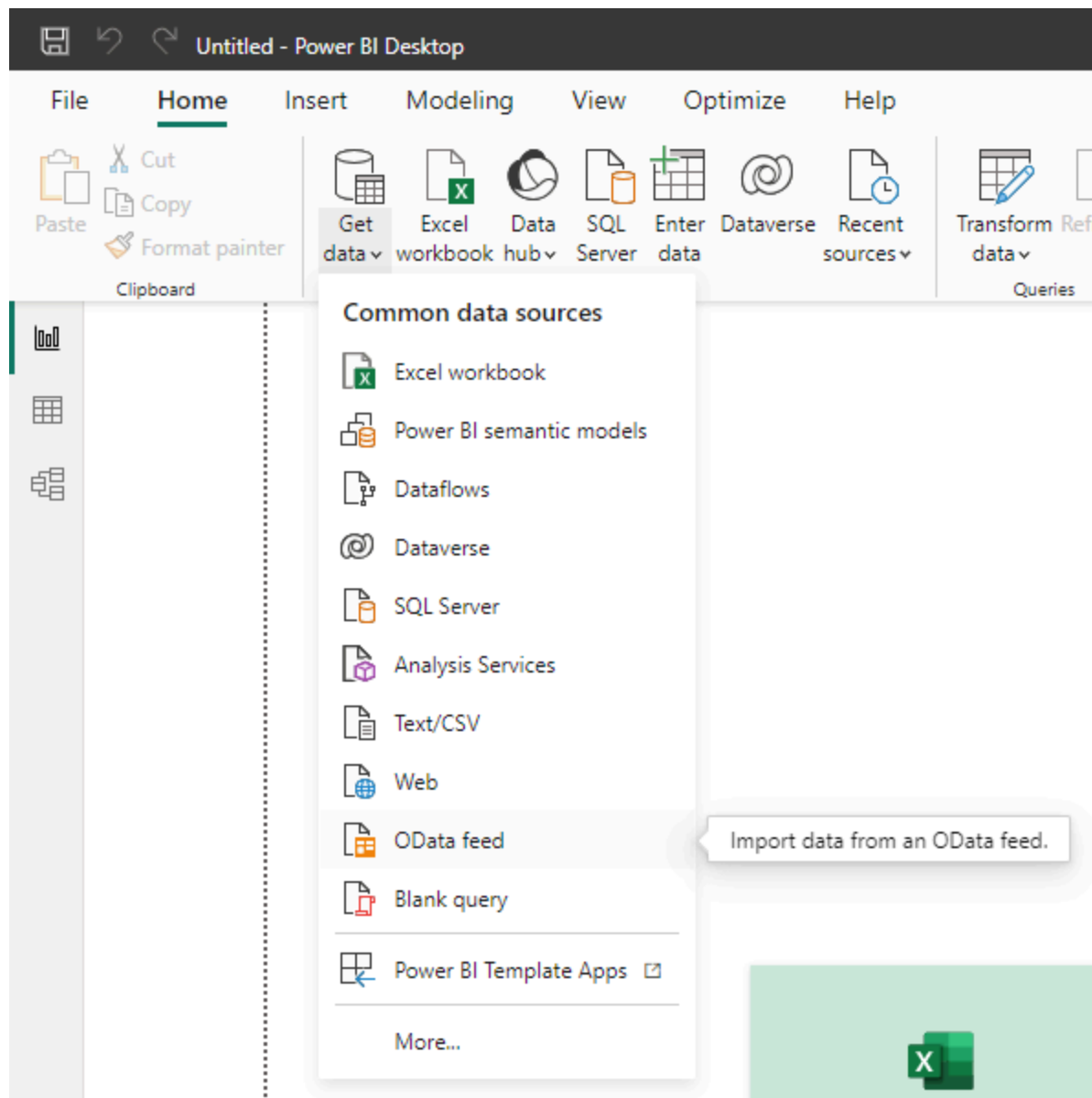
Table: Table_1 (77 rows)

18 / 19 rows

Table_1

B) IMPORTING ODATA TO POWER BI

1. Again, go to Get Data and select OData feed



2. Paste URL as powse & Click OK



3. Select orders table And Click on edit to view table

Navigator

http://services.odata.org/V3/Northwind/No...

☐ Alphabetical_list_of_products

☐ Categories

☐ Category_Sales_for_1997

☐ Current_Product_Lists

☐ Customer_and_Suppliers_by_Cities

☐ CustomerDemographics

☐ Customers

☐ Employees

☐ Invoices

☐ Order_Details

☐ Order_Details_Extendeds

☐ Order_Subtotals

☒ Orders

☐ Orders_Qries

☐ Product_Sales_for_1997

☐ Products

☐ Products_Above_Average_Prices

☐ Products_by_Categories

☐ Regions

Orders

Preview downloaded on 25 January 2024

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate
10248	VINET	5	04-07-1996 00:00:00	01-08-1996
10249	TOMSP	6	05-07-1996 00:00:00	16-08-1996
10250	HANAR	4	08-07-1996 00:00:00	05-08-1996
10251	VICTE	3	08-07-1996 00:00:00	05-08-1996
10252	SUPRD	4	09-07-1996 00:00:00	06-08-1996
10253	HANAR	3	10-07-1996 00:00:00	24-07-1996
10254	CHOPS	5	11-07-1996 00:00:00	08-08-1996
10255	RICSU	9	12-07-1996 00:00:00	09-08-1996
10256	WELLI	3	15-07-1996 00:00:00	12-08-1996
10257	HILAA	4	16-07-1996 00:00:00	13-08-1996
10258	ERNSH	1	17-07-1996 00:00:00	14-08-1996
10259	CENTC	4	18-07-1996 00:00:00	15-08-1996
10260	OTTIK	4	19-07-1996 00:00:00	16-08-1996
10261	QUEDE	4	19-07-1996 00:00:00	16-08-1996
10262	RATTC	8	22-07-1996 00:00:00	19-08-1996
10263	ERNSH	9	23-07-1996 00:00:00	20-08-1996
10264	FOLKO	6	24-07-1996 00:00:00	21-08-1996
10265	BLONP	2	25-07-1996 00:00:00	22-08-1996
10266	WARTH	3	26-07-1996 00:00:00	06-09-1996
10267	FRANK	4	29-07-1996 00:00:00	26-08-1996
10268	GROSR	8	30-07-1996 00:00:00	27-08-1996
10269	WHITC	5	31-07-1996 00:00:00	14-08-1996

Select Related Tables

Load

Transform Data

Cancel

FileHomeHelpTable tools

NameOrders

Mark as data table +

Manage relationships

New measure measure column

Quick

New

New

New table

StructureRelationshipsCalculations

OrderID

CustomerID

EmployeeID

OrderDate

RequiredDate

ShippedDate

ShipVia

Freight

ShipName

ShipAddress

ShipCity

ShipRegion

ShipPostalCode

ShipCountry

10249

TOMSP

6

05-07-1996 00:00:00

16-08-1996 00:00:00

10-07-1996 00:00:00

1

116.1

Toms Spezialitäten

Luitenstr. 48

Münster

44087

Germany

10260

OTTIK

4

19-07-1996 00:00:00

16-08-1996 00:00:00

29-07-1996 00:00:00

1

55.09

Ottlies Käsereien

Mehrheimerstr. 369

König

50739

Germany

10267

FRANK

4

29-07-1996 00:00:00

26-08-1996 00:00:00

06-08-1996 00:00:00

1

208.58

Frankenservand

Berliner Platz 43

München

80805

Germany

10275

QUICK

3

05-08-1996 00:00:00

02-09-1996 00:00:00

12-08-1996 00:00:00

3

76.07

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10277

HORCK

2

09-08-1996 00:00:00

06-09-1996 00:00:00

13-08-1996 00:00:00

3

125.77

Margareten Grundstüdt

Lipfert 22

Lipsitz

05358

Germany

10279

LEHMS

8

13-08-1996 00:00:00

10-09-1996 00:00:00

16-09-1996 00:00:00

2

25.83

Lehmans Marktstand

Magazinweg 7

Frankfurt a.M.

65538

Germany

10284

LEHMS

4

19-08-1996 00:00:00

16-09-1996 00:00:00

27-08-1996 00:00:00

1

76.56

Lehmans Marktstand

Magazinweg 7

Frankfurt a.M.

65538

Germany

10285

QUICK

1

20-08-1996 00:00:00

17-09-1996 00:00:00

26-08-1996 00:00:00

2

76.83

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10286

QUICK

8

21-08-1996 00:00:00

18-09-1996 00:00:00

30-08-1996 00:00:00

3

229.24

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10301

WANDK

8

09-09-1996 00:00:00

07-10-1996 00:00:00

17-09-1996 00:00:00

2

45.08

Die Wandende Kuh

Adenauerallee 900

Stuttgart

70563

Germany

10312

WANDK

2

23-09-1996 00:00:00

21-10-1996 00:00:00

03-10-1996 00:00:00

2

40.26

Die Wandende Kuh

Adenauerallee 900

Stuttgart

70563

Germany

10313

QUICK

2

24-09-1996 00:00:00

23-10-1996 00:00:00

04-10-1996 00:00:00

2

1.98

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10323

KÖNE

4

07-10-1996 00:00:00

04-11-1996 00:00:00

14-10-1996 00:00:00

1

4.98

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10325

KÖNE

4

09-10-1996 00:00:00

23-10-1996 00:00:00

14-10-1996 00:00:00

3

64.86

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10337

FRANK

4

24-10-1996 00:00:00

21-11-1996 00:00:00

29-10-1996 00:00:00

3

108.26

Frankenservand

Berliner Platz 43

München

80805

Germany

10342

FRANK

4

30-10-1996 00:00:00

28-11-1996 00:00:00

04-11-1996 00:00:00

2

54.83

Frankenservand

Berliner Platz 43

München

80805

Germany

10343

LEHMS

4

31-10-1996 00:00:00

28-11-1996 00:00:00

06-11-1996 00:00:00

2

110.37

Lehmans Marktstand

Magazinweg 7

Frankfurt a.M.

65538

Germany

10345

QUICK

2

04-11-1996 00:00:00

02-12-1996 00:00:00

11-11-1996 00:00:00

2

249.05

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10348

WANDK

4

07-11-1996 00:00:00

05-12-1996 00:00:00

15-11-1996 00:00:00

2

0.78

Die Wandende Kuh

Adenauerallee 900

Stuttgart

70563

Germany

10356

WANDK

6

18-11-1996 00:00:00

16-12-1996 00:00:00

27-11-1996 00:00:00

2

36.71

Die Wandende Kuh

Adenauerallee 900

Stuttgart

70563

Germany

10361

QUICK

1

12-11-1996 00:00:00

20-12-1996 00:00:00

03-12-1996 00:00:00

2

183.17

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10363

DRACD

4

26-11-1996 00:00:00

24-12-1996 00:00:00

04-12-1996 00:00:00

3

30.54

Drachenblut Delikatessen

Walserweg 21

Aachen

52066

Germany

10391

DRACD

3

23-12-1996 00:00:00

20-01-1997 00:00:00

31-12-1996 00:00:00

3

5.45

Drachenblut Delikatessen

Walserweg 21

Aachen

52066

Germany

10396

FRANK

1

27-12-1996 00:00:00

10-01-1997 00:00:00

06-01-1997 00:00:00

3

13

Frankenservand

Berliner Platz 43

München

80805

Germany

10407

OTTIK

4

10-01-1997 00:00:00

04-02-1997 00:00:00

20-01-1997 00:00:00

2

97.48

Ottlies Käsereien

Mehrheimerstr. 369

König

50739

Germany

10418

QUICK

1

17-01-1997 00:00:00

14-02-1997 00:00:00

24-01-1997 00:00:00

1

17.55

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10438

TOMSP

3

06-02-1997 00:00:00

06-03-1997 00:00:00

14-02-1997 00:00:00

2

8.24

Toms Spezialitäten

Luitenstr. 48

Münster

44087

Germany

10446

TOMSP

6

14-02-1997 00:00:00

14-03-1997 00:00:00

19-02-1997 00:00:00

1

14.68

Toms Spezialitäten

Luitenstr. 48

Münster

44087

Germany

10451

QUICK

4

19-02-1997 00:00:00

05-03-1997 00:00:00

12-03-1997 00:00:00

3

189.09

QUICK-Stop

Tauchestraße 10

Cunewalde

01307

Germany

10456

KÖNE

8

25-02-1997 00:00:00

08-04-1997 00:00:00

09-02-1997 00:00:00

2

8.12

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10457

KÖNE

6

25-02-1997 00:00:00

25-03-1997 00:00:00

03-03-1997 00:00:00

1

11.57

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10468

FRANK

3

03-03-1997 00:00:00

04-04-1997 00:00:00

12-03-1997 00:00:00

3

44.12

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10488

FRANK

2

03-03-1997 00:00:00

24-04-1997 00:00:00

02-04-1997 00:00:00

2

4.93

Frankenservand

Berliner Platz 43

München

80805

Germany

10497

LEHMS

7

04-04-1997 00:00:00

02-05-1997 00:00:00

07-04-1997 00:00:00

1

36.21

Lehmans Marktstand

Magazinweg 7

Frankfurt a.M.

65538

Germany

10501

BLAUS

9

09-04-1997 00:00:00

07-05-1997 00:00:00

16-04-1997 00:00:00

3

8.85

Blauer See Delikatessen

Forststr. 57

Mannheim

68306

Germany

10506

KÖNE

5

15-04-1997 00:00:00

13-05-1997 00:00:00

02-05-1997 00:00:00

2

21.19

Königlich Essen

Maubeltz 90

Brandenburg

14776

Germany

10508

OTTIK

1

16-04-1997 00:00:00

14-05-1997 00:00:00

13-05-1997 00:00:00

2

4.99

Ottlies Käsereien

Mehrheimerstr. 369

König

50739

Germany

10509

BLAUS

1

17-04-1997 00:00:00

15-05-1997 00:00:00

30-04-1997 00:00:00

1

0.15

Blauer See Delikatessen

Forststr. 57

Mannheim

68306

Germany

Table Orders

10509

BLAUS

PRACTICAL 2

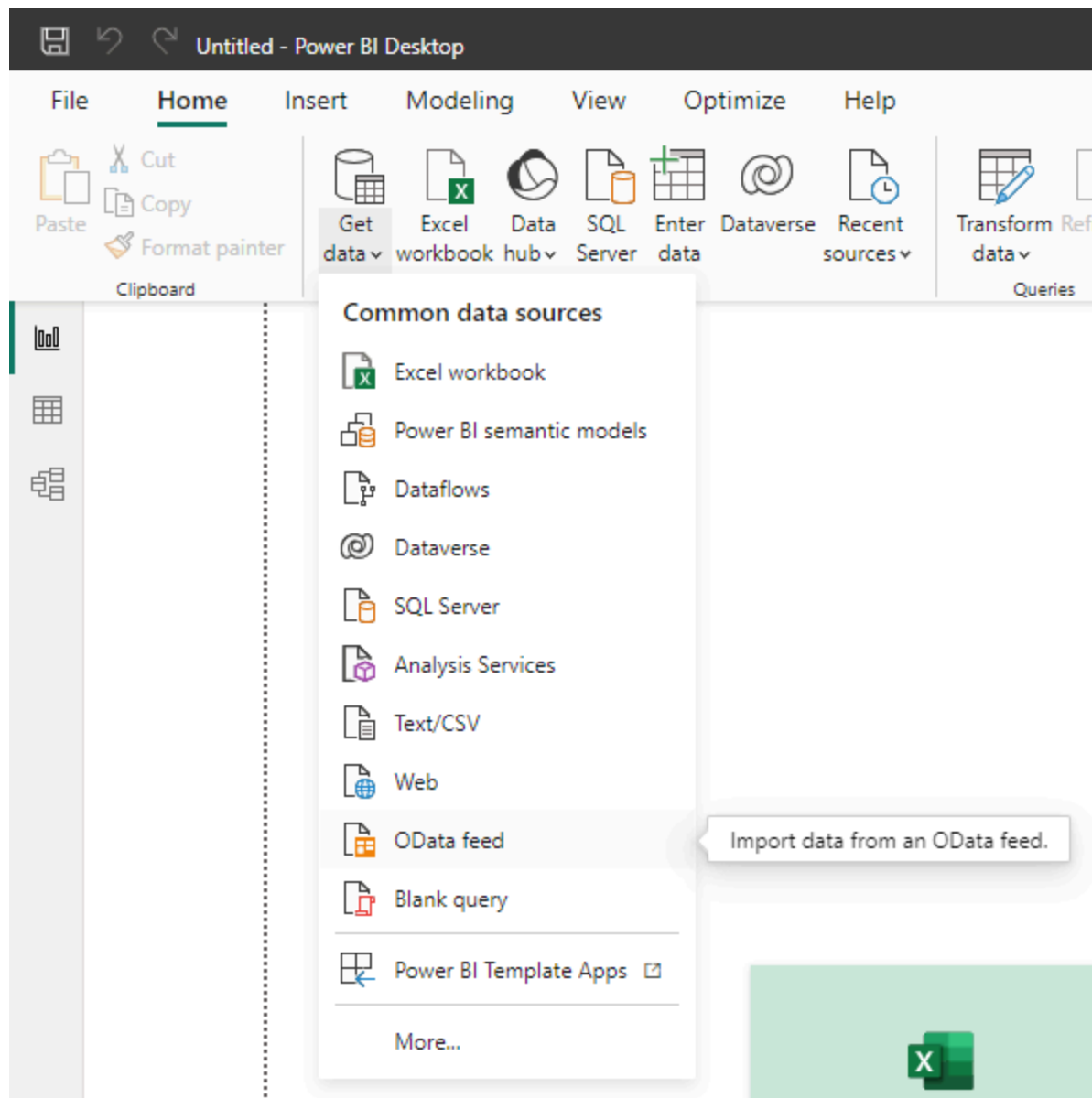
Aim: -

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server / Power BI.

Steps:-

A) ETL PROCESS IN POWER BI

1. Open Power BI, Click on Get Data → OData Feed



2. Paste URL:

<http://services.odata.org/V3/Northwind/Northwind.svc/> & Click OK

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Data Data source settings Data Sources Manage Parameters Refresh Preview Properties Advanced Editor Choose Remove Columns Manage Columns Keep Remove Rows Rows Sort Split Columns Group By Data Type: Whole Number Use First Row as Headers Replace Values Merge Queries Append Queries Use Vision Combine Files Text Analytics Azure Machine Learning All Insights

Queries [1] This preview may be up to 42 days old. Refresh

Products

Source: (Name="Products", Signature="table") [Data]

ProductID	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice	UnitsInStock	UnitsOnOrder	ReorderLevel
1	Chai	1	1	10 boxes x 20 bags	18	39	0	0
2	Orange	1	1	1 24 - 12 oz bottles	19	17	40	0
3	Aniseed Syrup	1	1	12 - 550 ml bottles	10	13	70	0
4	Chief Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22	53	0	0
5	Chief Anton's Gumbo Mix	2	2	36 boxes	21.35	0	0	0
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25	120	0	0
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb ppls.	30	15	0	0
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40	6	0	0
9	Maisie Neko Neko	4	6	18 - 500 g pkgs.	97	29	0	0
10	Buns	4	8	12 - 200 ml jars	31	31	0	0
11	Queso Cabrales	5	4	1 kg pkg.	21	22	30	0
12	Queso Manchego La Pastora	5	4	10 - 500 g pkgs.	38	86	0	0
13	Konbu	6	8	2 kg box	6	24	0	0
14	Tofu	6	7	40 - 100 g pkgs.	23.25	35	0	0
15	Garden of Eatin'	6	2	24 - 250 ml bottles	15.5	39	0	0
16	Pavlova	7	3	12 - 500 g boxes	17.45	29	0	0
17	Alice Mutton	7	6	20 - 1 kg tins	39	0	0	0
18	Carnarvon Tigers	7	8	16 kg pkg.	62.5	42	0	0
19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2	25	0	0
20	Sir Rodney's Marmalade	8	3	30 gift boxes	81	40	0	0
21	Sir Rodney's Scones	8	3	24 pkgs. x 4 pieces	10	3	40	0
22	Gustaf's Knäckebröd	9	5	24 - 500 g pkgs.	21	104	0	0
23	Turkey	9	5	12 - 250 g pkgs.	9	61	0	0
24	Guaraná Fantástica	10	12	355 ml cans	4.5	20	0	0
25	NuNu's Nuts-N-Cream	11	3	20 - 450 g glasses	14	76	0	0
26	Gumbär Gummibärchen	11	3	100 - 250 g bags	31.29	15	0	0
27	Schoggi Schokolade	11	3	100 - 100 g pieces	43.9	49	0	0
28	Rössle Sauerkraut	12	7	25 - 825 g cans	45.6	26	0	0
29	Thüringer Rostbratenwurst	12	6	50 bags x 50 sausgs.	123.79	0	0	0
30	Nori-On Merguezine	13	8	10 - 200 g glasses	25.89	10	0	0
31	Gorgonzola Telino	14	4	12 - 100 g pkgs	12.5	0	70	0
32	Mascarpone Fabboli	14	4	24 - 200 g pkgs.	32	9	40	0
33	Gerdost	15	4	500 g	2.5	112	0	0
34	Sasquatch Ale	16	1	24 - 12 oz bottles	14	111	0	0
35	Steeleye Stout	16	1	24 - 12 oz bottles	18	20	0	0
36	Innoga Sil	17	8	24 - 250 g jars	19	112	0	0

13 COLUMNS, 77 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON 25 JANUARY 2024

Query Settings

PROPERTIES

Name

Products

APPLIED STEPS

Source

X Navigation

4. Remove other columns to only display columns of interest In Query Editor, select the ProductID, ProductName, QuantityPerUnit, and UnitsInStock columns.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Source Sources Data Data source settings Parameters Advanced Editor Choose Remove Columns Columns Manage Columns Keep Remove Rows Rows Sort Split Column Group By Data Type: Decimal Number Use First Row as Headers Replace Values Merge Queries Append Queries Text Analytics Vision Azure Machine Learning AI Insights Combine Files Combine

Queries [1] Table.RemoveColumns(Products_table,{"SupplierID","CategoryID","UnitPrice","UnitsOnOrder","ReorderLevel","Discontinued","Category","Order_Details","Supplier"})

Products

ProductID	ProductName	QuantityPerUnit	UnitsInStock
1	Chai	10 boxes x 20 bags	39
2	Chang	24 - 12 oz bottles	17
3	Aniseed Syrup	12 - 550 ml bottles	13
4	Chef Anton's Cajun Seasoning	48 - 6 oz jars	53
5	Chef Anton's Gumbo Mix	36 boxes	0
6	Grandma's Boysenberry Spread	12 - 8 oz jars	120
7	Uncle Bob's Organic Dried Pears	12 - 1 lb pkgs.	15
8	Northwoods Cranberry Sauce	12 - 12 oz jars	6
9	Mishi Kobe Niku	18 - 500 g pkgs.	29
10	Ikura	12 - 200 ml jars	31
11	Queso Cabrales	1 kg pkg.	22
12	Queso Manchego La Pastora	10 - 500 g pkgs.	86
13	Konbu	2 kg box	24
14	Tofu	40 - 100 g pkgs.	35
15	Genen Shouyu	24 - 250 ml bottles	39
16	Pavlova	32 - 500 g boxes	29
17	Alice Mutton	20 - 1 kg tins	0
18	Carnarvon Tigers	16 kg pkg.	42
19	Teatime Chocolate Biscuits	10 boxes x 12 pieces	25
20	Sir Rodney's Marmalade	30 gift boxes	40
21	Sir Rodney's Scones	24 pkgs. x 4 pieces	9
22	Gustaf's Knäckebröd	24 - 500 g pkgs.	104
23	Tunnbröd	12 - 250 g pkgs.	61
24	Guaraná Fantástica	12 - 355 ml cans	20
25	NuNuCa Nuß-Nougat-Creme	20 - 450 g glasses	76
26	Gumbär Gummibärchen	100 - 250 g bags	15
27	Schoggi Schokolade	100 - 100 g pieces	49
28	Kissle Seuehrst	25 - 825 g cans	26
29	Thüringer Rostbratenwurst	50 bags x 10 sausgs.	0
30	Nord-Ost Marzipanering	10 - 200 g glasses	10
31	Gorgonzola Telino	12 - 100 g pkgs	0
32	Mascarpone Fabboli	24 - 200 g pkgs.	9
33	Gelbst	500 g	112
34	Sauquatch Ale	24 - 12 oz bottles	111
35	Steevie Stout	24 - 12 oz bottles	20
36	Imagü Dai	24 - 250 g jars	112
37	Graved lax	12 - 500 g pkgs.	11
38	Côte de Blaye	12 - 75 cl bottles	17

4 COLUMNS, 77 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED ON 25 JANUARY 2024

Query Settings

PROPERTIES

Name Products

APPLIED STEPS

Source Navigation Removed Columns

5. Change the data type of the UnitsInStock column

a. Select the UnitsInStock column. Change Type:- Whole number

Table.RemoveColumns(Products_table,{"SupplierID","CategoryID","UnitPrice","UnitsOnOrder","ReorderLevel","Discontinued","Category","Order_Details","Supplier"})

Products

ProductID	ProductName	QuantityPerUnit	UnitsInStock
1	Chai	10 boxes x 20 bags	39
2	Chang	24 - 12 oz bottles	17
3	Aniseed Syrup	12 - 550 ml bottles	13
4	Chef Anton's Cajun Seasoning	48 - 6 oz jars	53
5	Chef Anton's Gumbo Mix	36 boxes	0
6	Grandma's Boysenberry Spread	12 - 8 oz jars	120
7	Uncle Bob's Organic Dried Pears	12 - 1 lb pkgs.	15
8	Northwoods Cranberry Sauce	12 - 12 oz jars	6
9	Mishi Kobe Niku	18 - 500 g pkgs.	29
10	Ikura	12 - 200 ml jars	31
11	Queso Cabrales	1 kg pkg.	22
12	Queso Manchego La Pastora	10 - 500 g pkgs.	86
13	Konbu	2 kg box	24
14	Tofu	40 - 100 g pkgs.	35
15	Genen Shouyu	24 - 250 ml bottles	39
16	Pavlova	32 - 500 g boxes	29
17	Alice Mutton	20 - 1 kg tins	0
18	Carnarvon Tigers	16 kg pkg.	42
19	Teatime Chocolate Biscuits	10 boxes x 12 pieces	25
20	Sir Rodney's Marmalade	30 gift boxes	40
21	Sir Rodney's Scones	24 pkgs. x 4 pieces	9
22	Gustaf's Knäckebröd	24 - 500 g pkgs.	104
23	Tunnbröd	12 - 250 g pkgs.	61
24	Guaraná Fantástica	12 - 355 ml cans	20
25	NuNuCa Nuß-Nougat-Creme	20 - 450 g glasses	76

Copy

Remove

Remove Other Columns

Duplicate Column

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Group By...

Fill

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

Drill Down

Add as New Query

Decimal Number

Fixed decimal number

Whole Number

Percentage

Date/Time

Date

Time

Date/Time/Timezone

Duration

Text

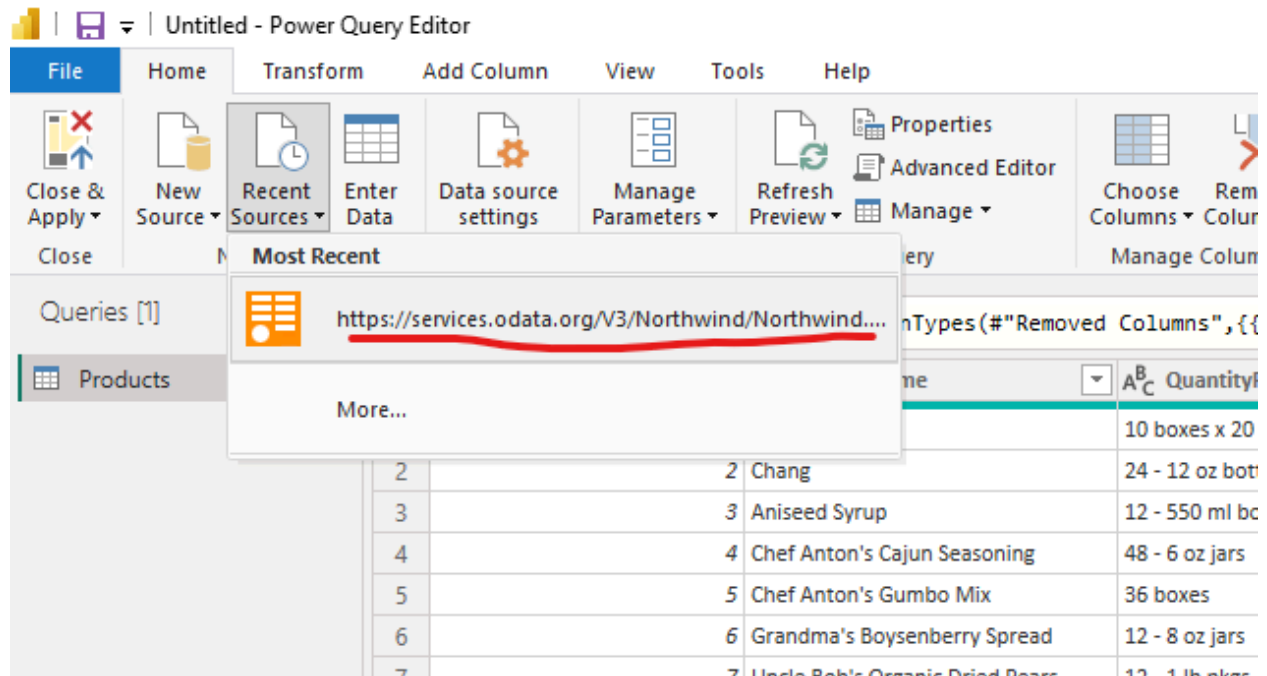
True/False

Binary

Using Locale...

6. Expand Orders table

- a. Once you have loaded a data source ,you can click on **Recent Sources** to select desirable table (Orders)



- b. After selecting the URL, Navigator window will appear from which you can select Orders table ->Click on Load

Manage Parameters Refresh Manage Choose Remove Keep Remove Split Group Replace Values Combine Files Azure Machine Learning Parameters Query Manage Columns Manage Columns Reduce Rows Sort Transform Combine AI Insights

Table.TransformColumnTypes(#"Removed Columns",{{"UnitInStock",Type64.Type}})

Product Name

- Chai
- Chang
- Aniseed Syrup
- Chef Anton's Cajun Seasoning
- Chef Anton's Gumbo Mix
- Grandma's Boysenberry Spread
- Uncle Bob's Organic Dried Pears
- Northwoods Cranberry Sauce
- Mishi Kobe Niku
- Ikura
- Queso Cabrales
- Queso Manchego La Pastora
- Konbu
- Tofu
- Genen Shouyu
- Pavlova
- Alice Mutton
- Carnarvon Tigers
- Teatime Chocolate Biscuits
- Sir Rodney's Marmalade
- Sir Rodney's Scones
- Gustaf's Knäckebröd
- Tunnbröd
- Guaraná Fantástica
- NuNuCa Nuß-Nougat-Creme
- Gumbär Gummibärchen
- Schoggi Schokolade
- Rössle Sauerkraut
- Thüringer Rostbratwurst
- Nord-Ost Matjeshering
- Gorgonzola Telino
- Mascarpone Fabioli
- Gelbstost
- Sasquatch Ale

Navigator

Display Options

https://services.odata.org/V3/Northwind/N...

- Alphabetical_list_of_products
- Categories
- Category_Sales_for_1997
- Current_Product_Lists
- Customer_and_Suppliers_by_Cities
- CustomerDemographics
- Customers
- Employees
- Invoices
- Order_Details
- Order_Details_Extended
- Order_Subtotals
- Orders
- Orders_Qries
- Product_Sales_for_1997
- Products
- Products_Above_Average_Prices
- Products_by_Categories
- Regions

Select Related Tables

Orders

Preview downloaded on 29 January 2024

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate
10248	VINET	5	04-07-1996 00:00:00	01-08-1996
10249	TOMSP	6	05-07-1996 00:00:00	16-08-1996
10250	HANAR	4	08-07-1996 00:00:00	05-08-1996
10251	VICTE	3	08-07-1996 00:00:00	05-08-1996
10252	SUPRD	4	09-07-1996 00:00:00	06-08-1996
10253	HANAR	3	10-07-1996 00:00:00	24-07-1996
10254	CHOPS	5	11-07-1996 00:00:00	08-08-1996
10255	RICSU	9	12-07-1996 00:00:00	09-08-1996
10256	WELLI	3	15-07-1996 00:00:00	12-08-1996
10257	HILAA	4	16-07-1996 00:00:00	13-08-1996
10258	ERNSH	1	17-07-1996 00:00:00	14-08-1996
10259	CENTC	4	18-07-1996 00:00:00	15-08-1996
10260	OTTIK	4	19-07-1996 00:00:00	16-08-1996
10261	QUEDE	4	19-07-1996 00:00:00	16-08-1996
10262	RAITC	8	22-07-1996 00:00:00	19-08-1996
10263	ERNSH	9	23-07-1996 00:00:00	20-08-1996
10264	FOLKO	6	24-07-1996 00:00:00	21-08-1996
10265	BLONP	2	25-07-1996 00:00:00	22-08-1996
10266	WARTH	3	26-07-1996 00:00:00	06-09-1996
10267	FRANK	4	29-07-1996 00:00:00	26-08-1996
10268	GROSR	8	30-07-1996 00:00:00	27-08-1996
10269	WHITC	5	31-07-1996 00:00:00	14-08-1996

c. Query Editor Window will appear

- In the Query View ,scroll to the Order_Details column
- In the Order_Details column,select the expand icon .
- In the Expand drop-down:
 - Select(Select All Columns) to clear all columns
 - Select ProductID,UnitPrice and Quantity.
 - Click OK.

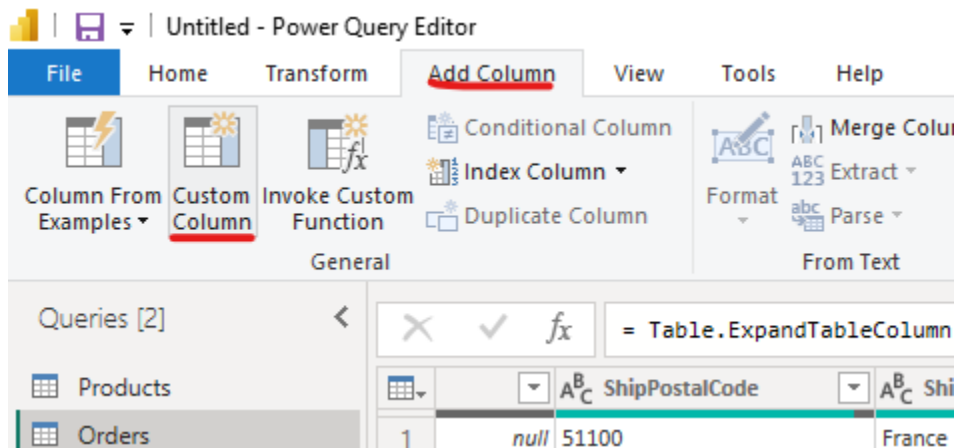
The screenshot displays a data application interface with a modal dialog box open. The background is a table with four columns: 'Customer', 'Employee', 'Order_Details', and 'Shipper'. The 'Order_Details' column is highlighted in green. The dialog box, titled 'Expand', contains a search bar with the text 'Search Columns to Expand'. Below the search bar are two radio buttons: 'Expand' (which is selected) and 'Aggregate'. A list of columns is shown with checkboxes: '(Select All Columns)' (checked), 'OrderID' (unchecked), 'ProductID' (checked), 'UnitPrice' (checked), 'Quantity' (checked), 'Discount' (unchecked), 'Order' (unchecked), and 'Product' (unchecked). At the bottom of the dialog is a checkbox labeled 'Use original column name as prefix' which is also checked. There are 'OK' and 'Cancel' buttons at the bottom right of the dialog. The table in the background has 15 rows, each starting with 'Record' in the 'Customer' column, followed by 'Record' in the 'Employee' column, and 'Table' in the 'Order_Details' and 'Shipper' columns.

"Quantity"}, {"Order_Details.ProductID", "Order_Details.UnitPrice", "Order_Details.Quantity"})

	1.2.3 Order_Details.ProductID	1.2 Order_Details.UnitPrice	1.2 Order_Details.Quantity	Shipper
	11	14	12	Record
	42	9.8	10	Record
	72	34.8	5	Record
	14	18.6	9	Record
	51	42.4	40	Record
	41	7.7	10	Record
	51	42.4	35	Record
	65	16.8	15	Record
	22	16.8	6	Record
	57	15.6	15	Record
	65	16.8	20	Record
	20	64.8	40	Record
	33	2	25	Record
	60	27.2	40	Record
	31	10	20	Record
	39	14.4	42	Record
	49	16	40	Record
	24	3.6	15	Record
	55	19.2	21	Record
	74	8	21	Record
	2	15.2	20	Record
	16	13.9	35	Record
	36	15.2	25	Record
	59	44	30	Record
	53	26.2	15	Record
	77	10.4	12	Record
	27	35.1	25	Record
	39	14.4	6	Record
	77	10.4	15	Record
	2	15.2	50	Record
	5	17	65	Record
	32	25.6	6	Record
	21	8	10	Record
	37	20.8	1	Record
	41	7.7	16	Record
	57	15.6	50	Record
	62	39.4	15	Record

d. Calculate the lines total for each Order_Details row

- In the Add Column ribbon tab, Click Add Custom Column
- In the Custom Column dialog box ,in the Custom Column Formula textbox,enter:-
[Order_Details.UnitPrice]*[Order_Details.Quantity] by selecting from available columns and click on insert for each column]
- In the New column name textbox,enter LineTotal
- Click OK.



Custom Column

Add a column that is computed from the other columns.

New column name

LineTotal

Custom column formula ⓘ

= [Order_Details.UnitPrice]*[Order_Details.Quantity]

Available columns

ShipPostalCode
ShipCountry
Customer
Employee
Order_Details.ProductID
Order_Details.UnitPrice
Order_Details.Quantity
Shipper

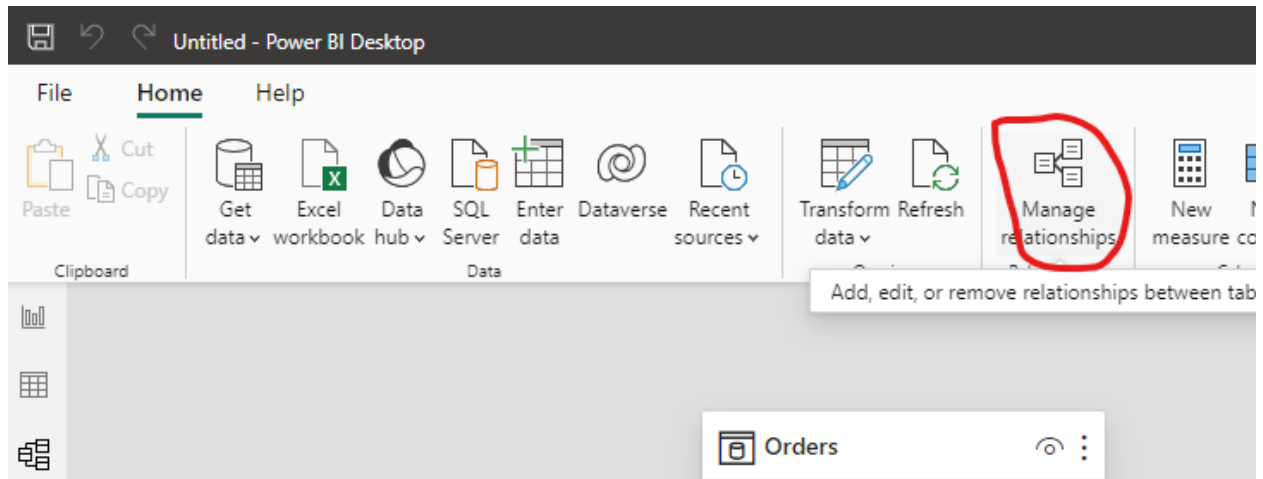
<< Insert

[Learn about Power Query formulas](#)

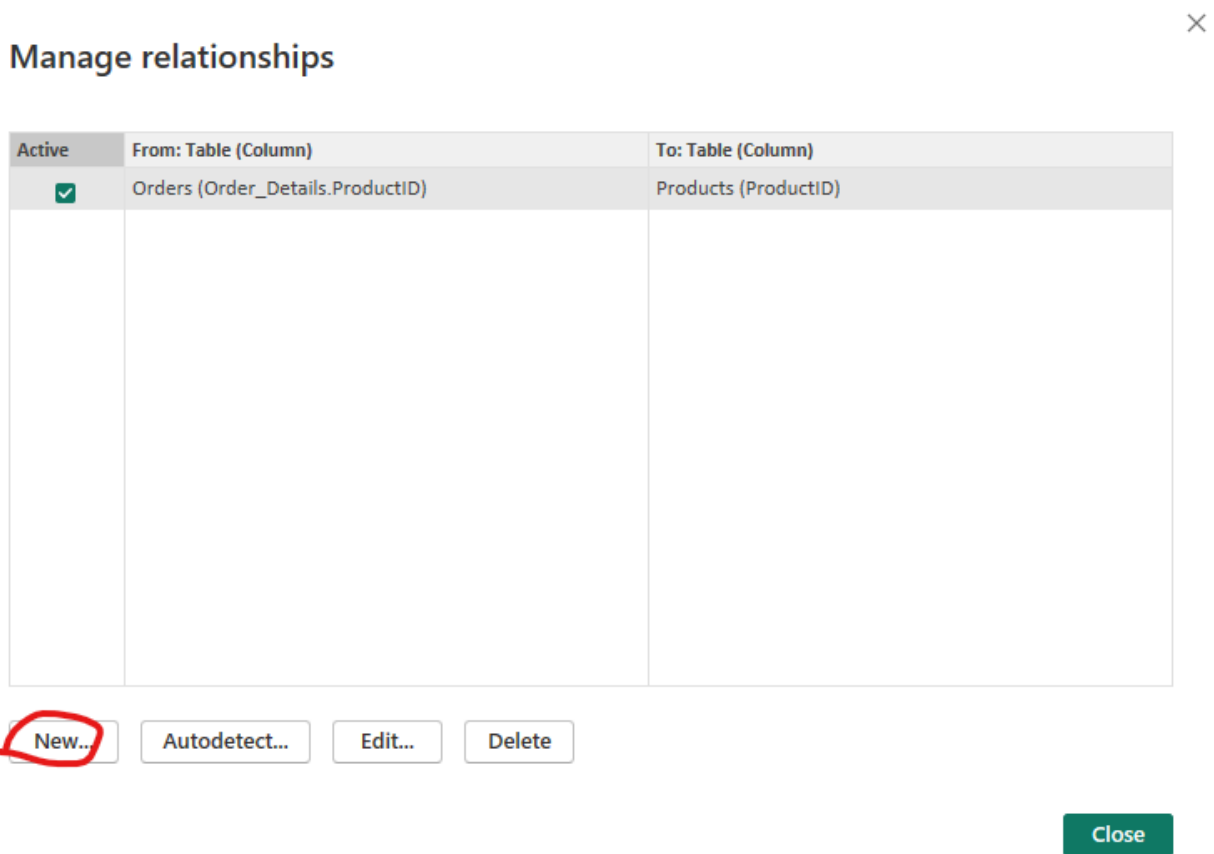
✓ No syntax errors have been detected.

OK

Cancel



Click on new->Create relationship ->click on ok





Create relationship

Select tables and columns that are related.

Products

ProductID	ProductName	QuantityPerUnit	UnitsInStock
1	Chai	10 boxes x 20 bags	39
2	Chang	24 - 12 oz bottles	17
3	Aniseed Syrup	12 - 550 ml bottles	13

Orders

Name	ShipAddress	ShipCity	ShipRegion	ShipPostalCode	ShipCountry	Order_Details.ProductID	OrderID
<-Stop	Taucherstraße 10	Cunewalde	null	01307	Germany	10	
<-Stop	Taucherstraße 10	Cunewalde	null	01307	Germany	31	
<-Stop	Taucherstraße 10	Cunewalde	null	01307	Germany	33	

Cardinality

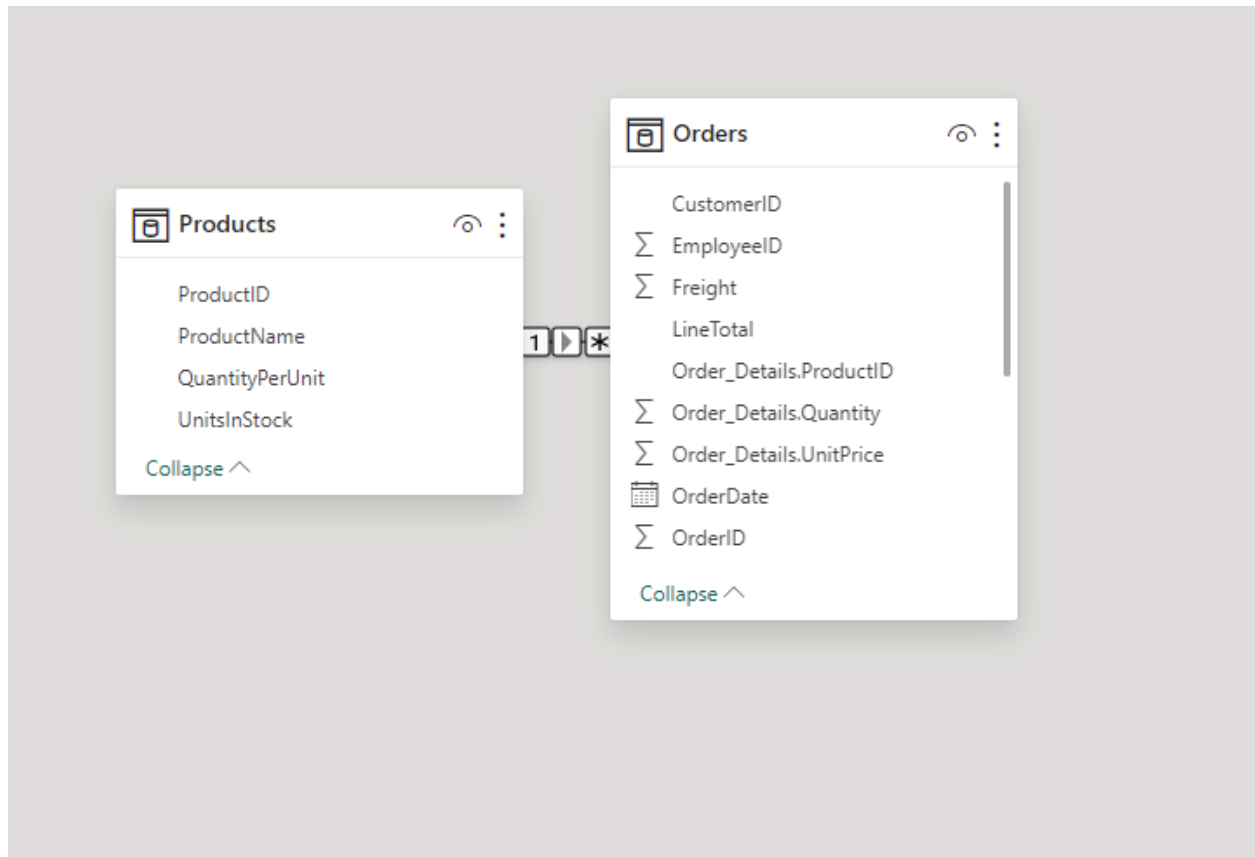
Cross filter direction

☐ Make this relationship active

☐ Apply security filter in both directions

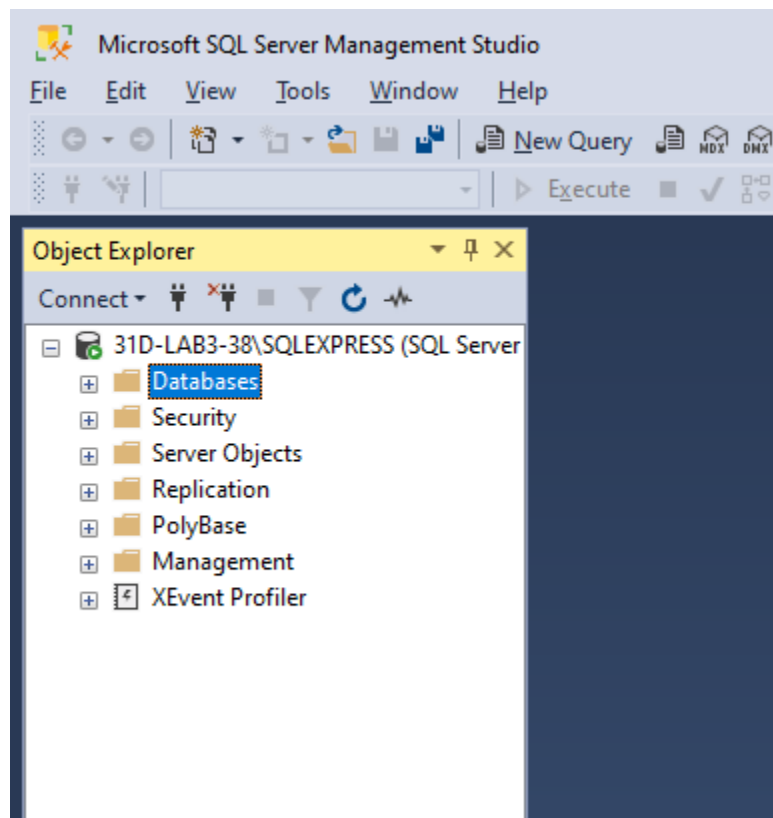
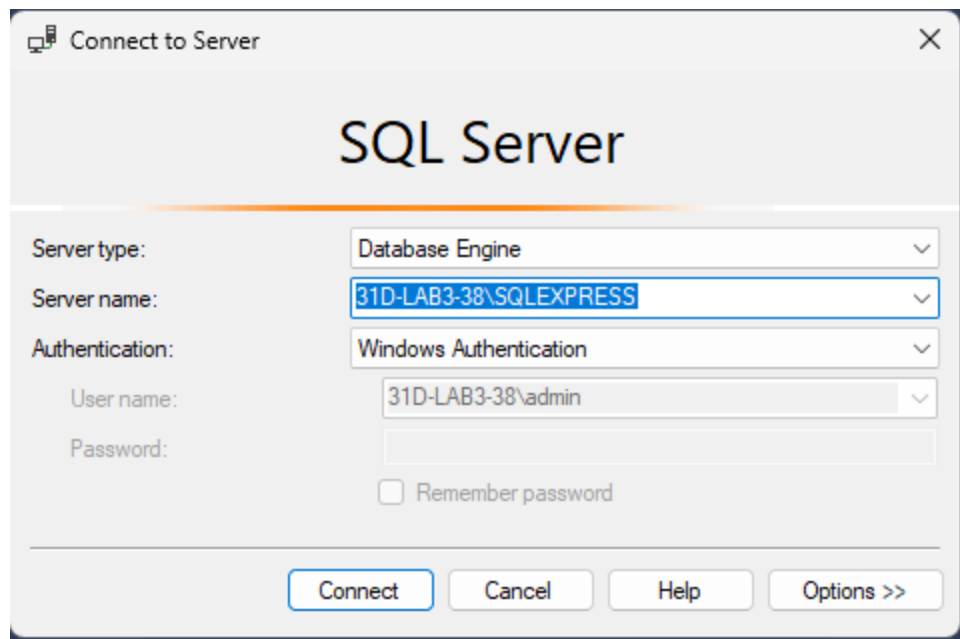
☐ Assume referential integrity

OUTPUT:-

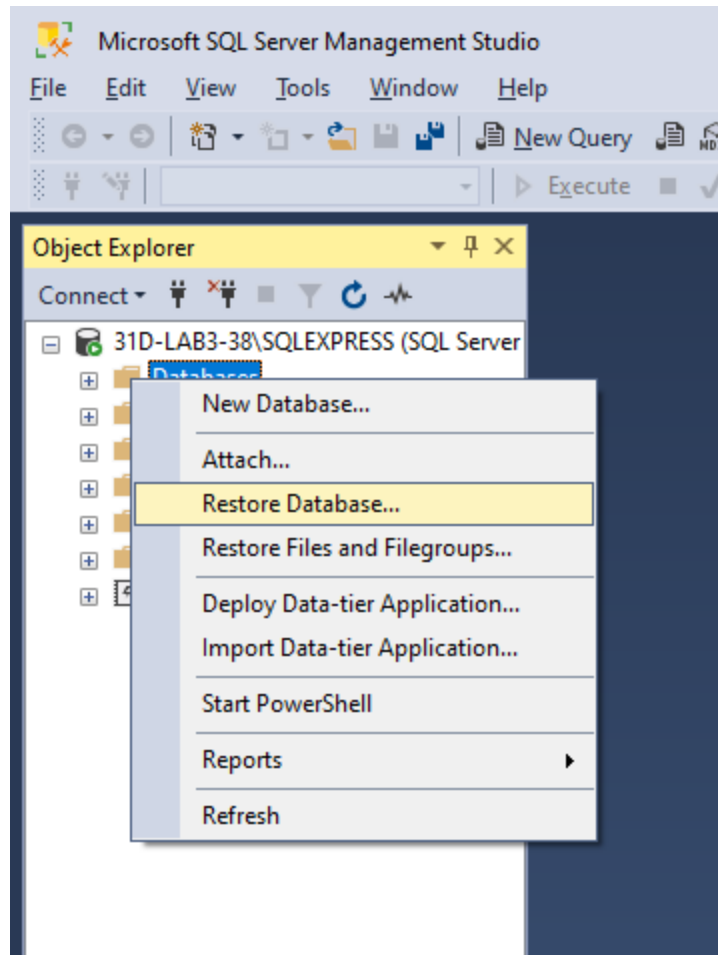


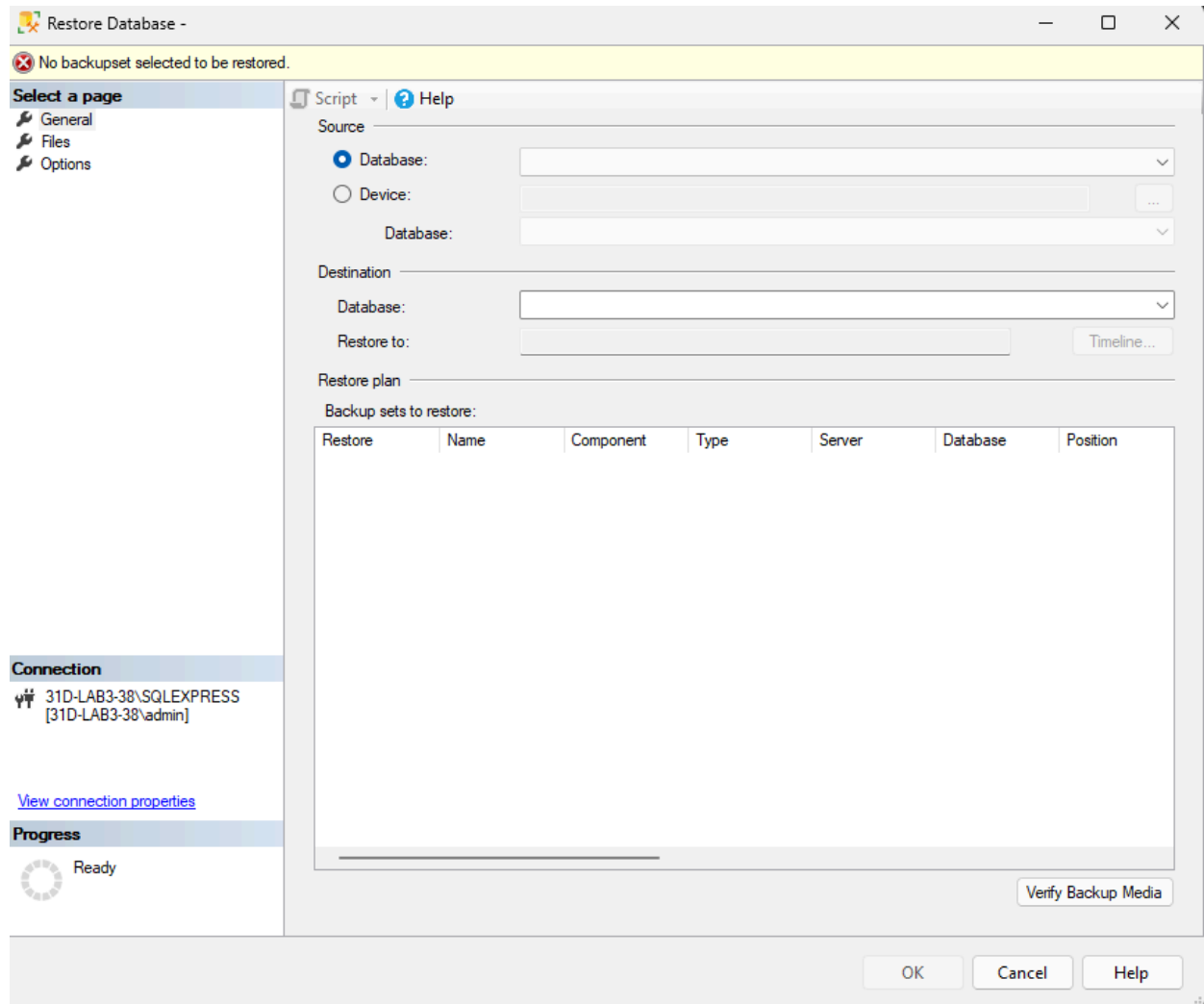
2B) ETL PROCESS IN SQL SERVER

1. Open SQL Server Management Studio to restore backup file(connect to the server first)

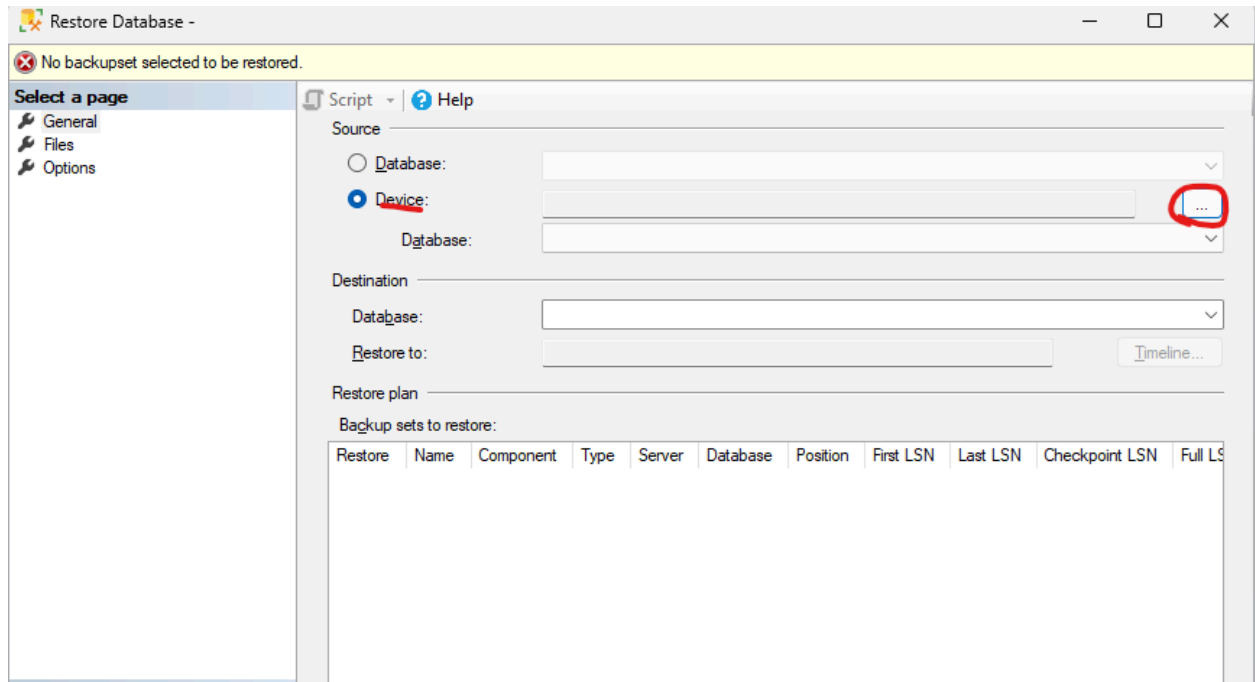


2. Right click on Databases → Restore Database

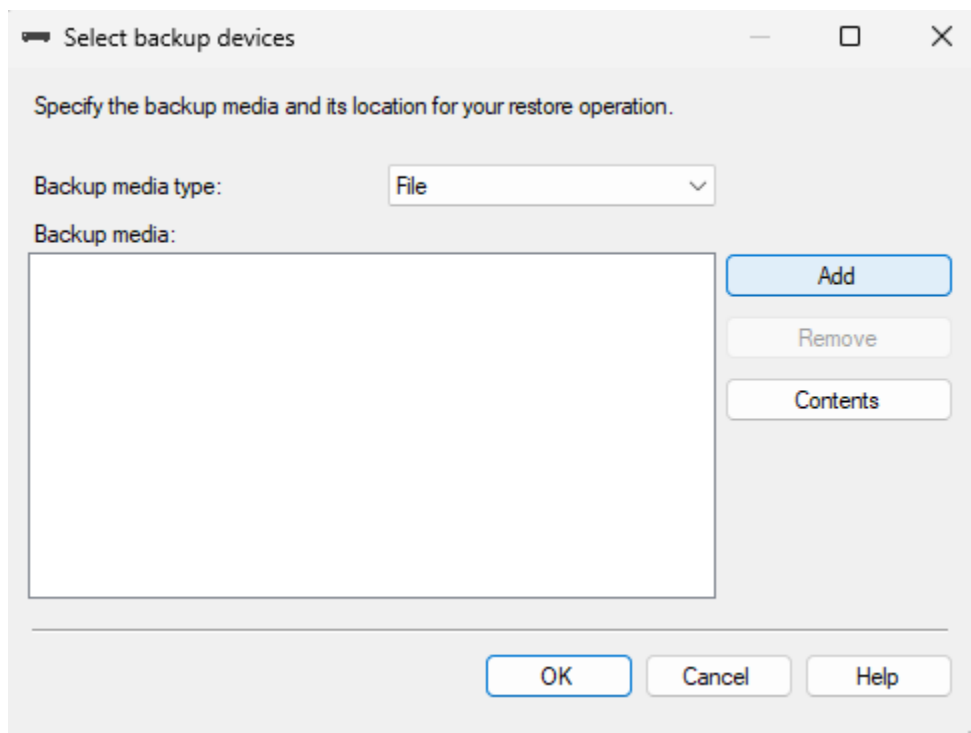




3. Select Device → click on ... icon towards end of device box



4. Click on Add -> Select on folder where you have downloaded your backup files -> Click Ok



Select backup devices

Specify the backup media and its location for your restore operation.

Backup media type:

File

Backup media:

D:\AdventureWorks2012.bak
D:\AdventureWorksDW2012.bak

Add

Remove

Contents

OK

Cancel

Help

Restore Database - AdventureWorks2012

Ready

Select a page

General

Files

Options

Connection

31D-LAB3-38\SQLEXPRESS

[31D-LAB3-38\admin]

View connection properties

Progress

Done

Script

Help

Source

Database:

Device:

D:\AdventureWorks2012.bak, D:\AdventureWorksDW2012.bak

...

Dgtabase:

AdventureWorks2012

Destination

Database:

AdventureWorks2012

Restore to:

The last backup taken (06 November 2017 19:51:36)

Timeline...

Restore plan

Backup sets to restore:

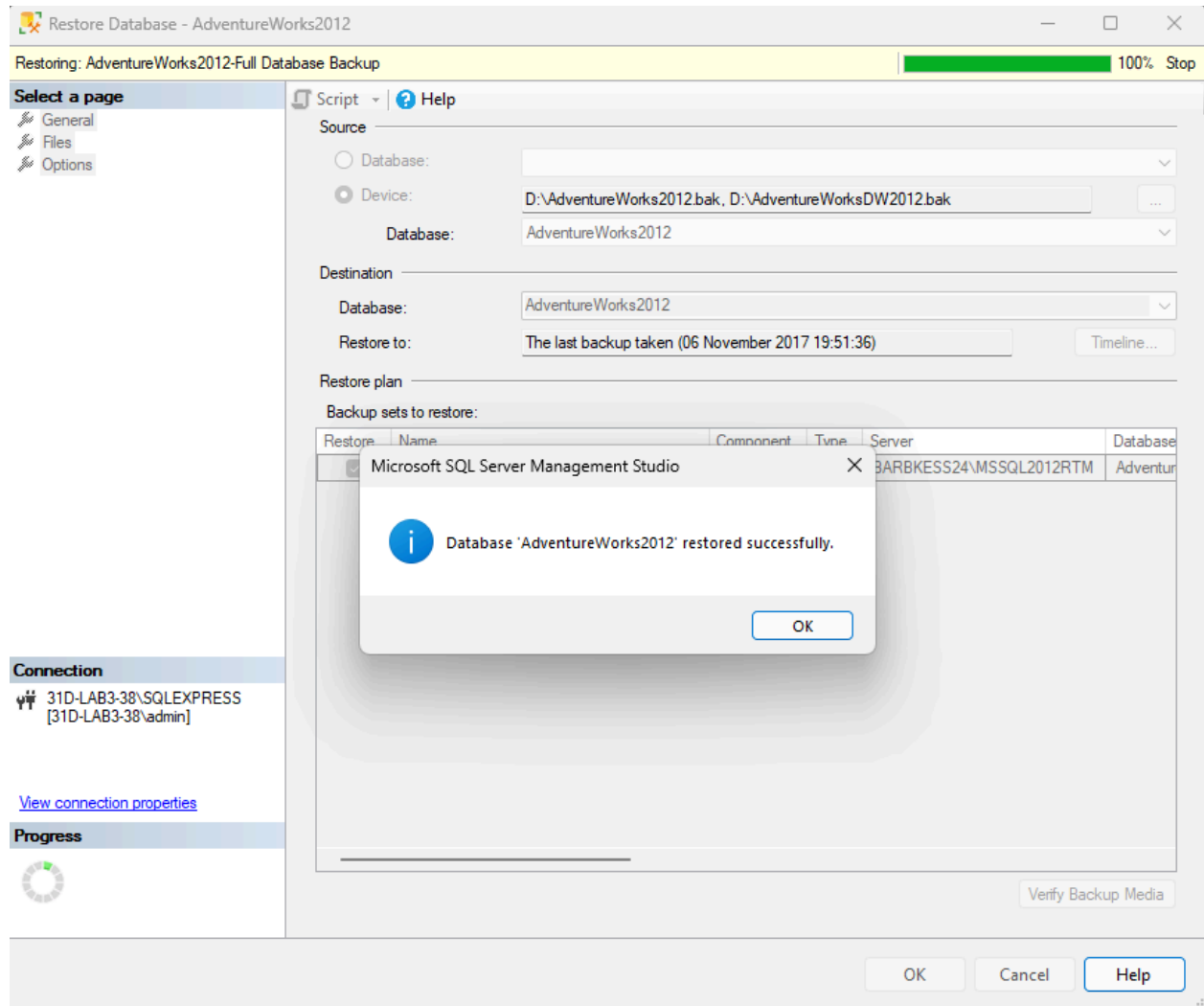
Restore	Name	Component	Type	Server	Database
☒	AdventureWorks2012-Full Database Backup	Database	Full	BARBKESS24\MSSQL2012RTM	Adventur

Verify Backup Media

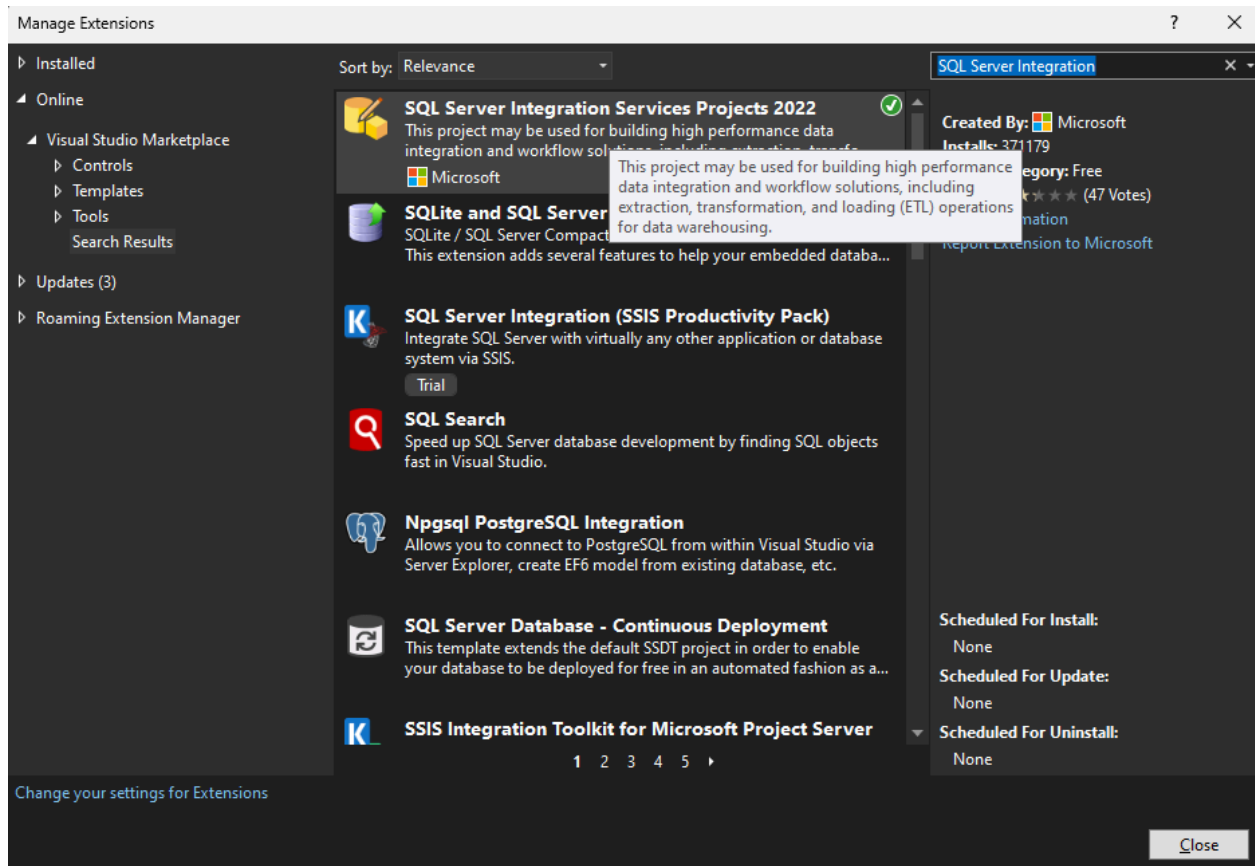
OK

Cancel

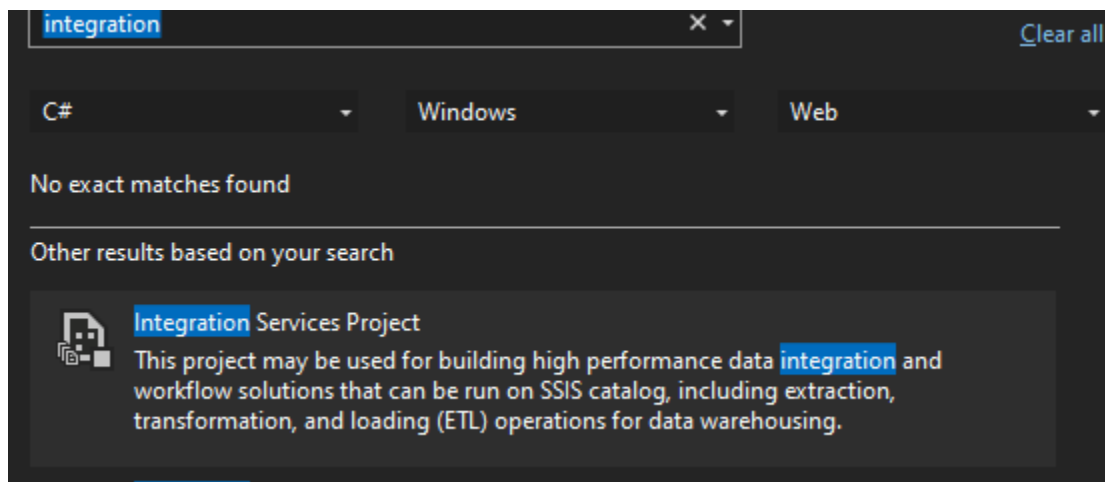
Help



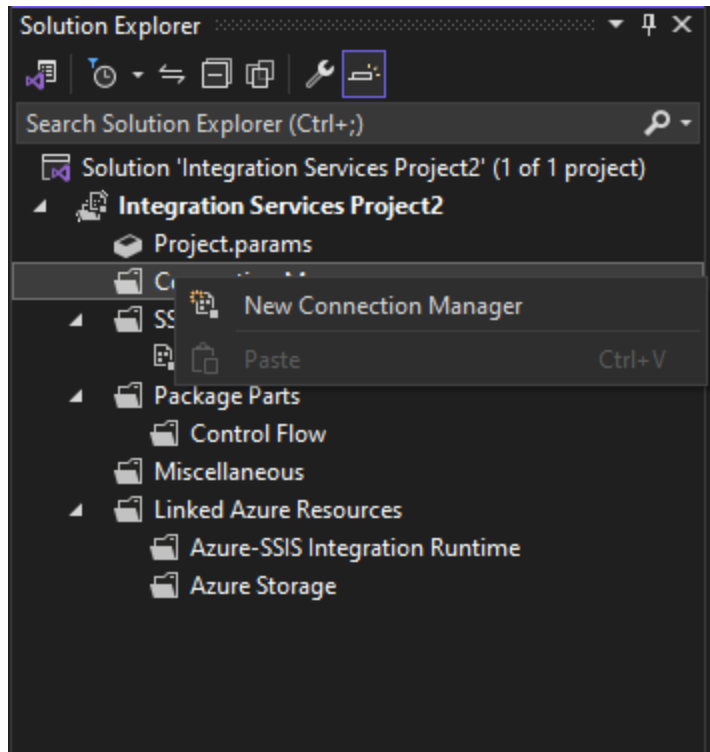
4. Open Visual Studio 2022-> Download & install extension (SQL Server Integration Service Project)



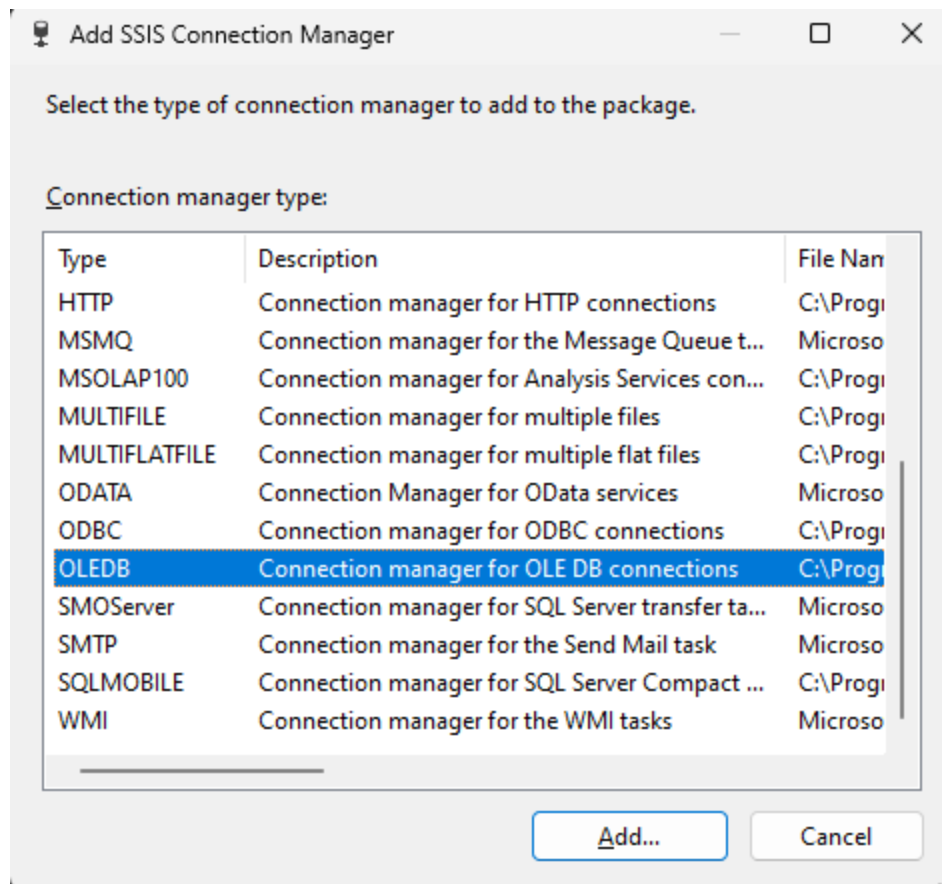
5. Open Visual Studio -> Create a new project -> Integration Services Project & give appropriate project name.



6. Right click on Connection Managers in solution explorer and click on New Connection Manager.



7. Configure OLEDB Connection Manager window appears → Click on Add



Click on New

Connection Manager

Provider: Native OLE DB\SQL Server Native Client 11.0

Connection

All

Server name: Refresh

Log on to the server

Authentication: Windows Authentication

User name:

Password:

☐ Save my password

Connect to a database

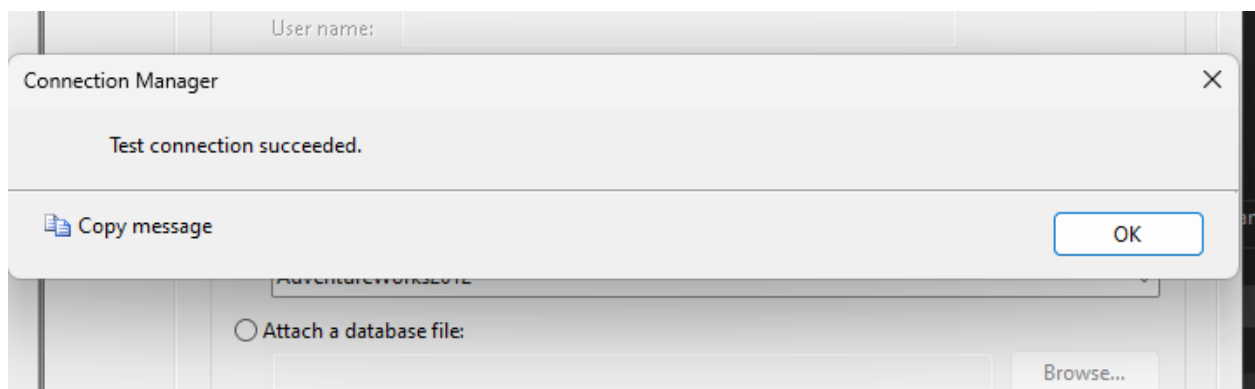
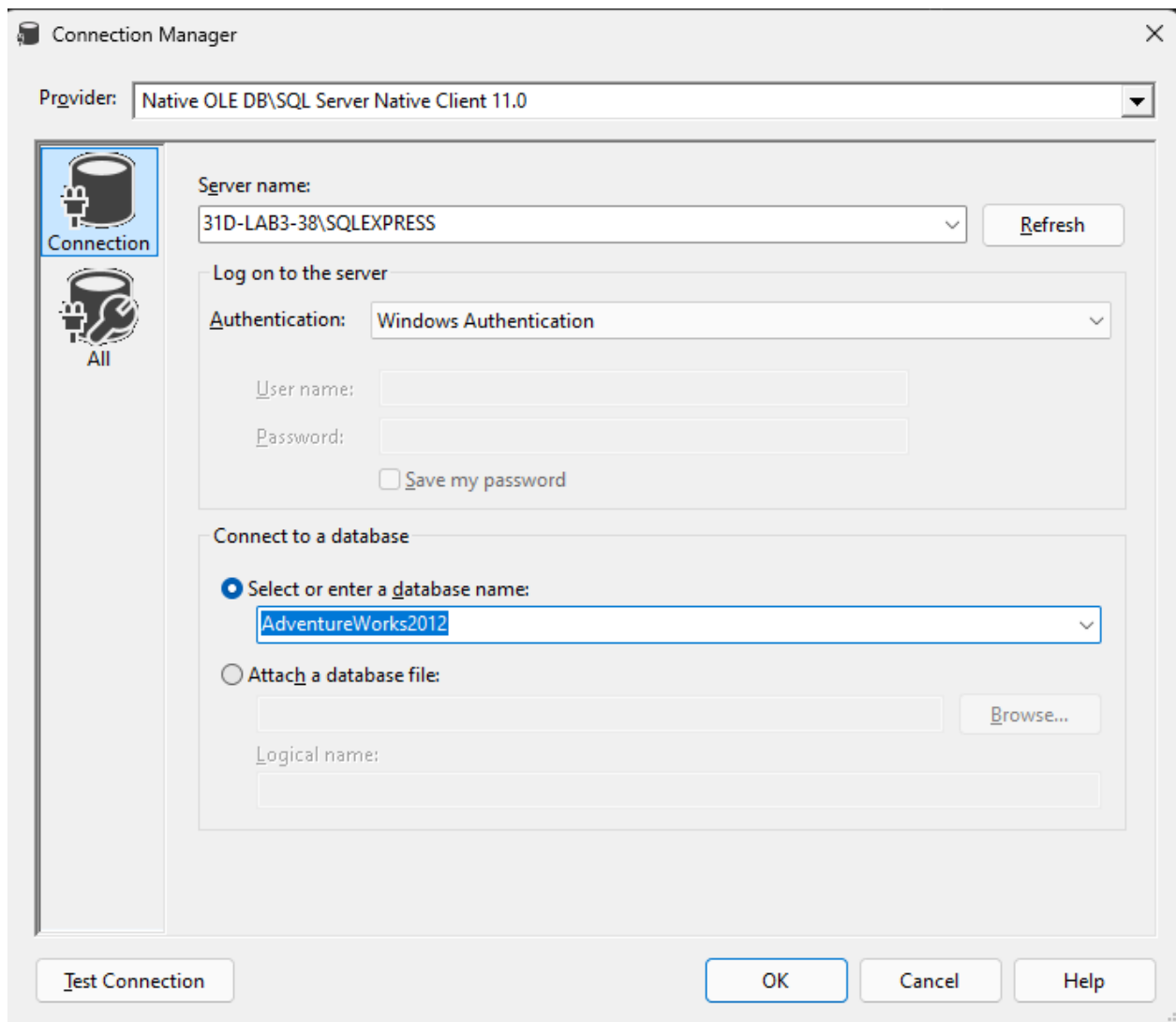
☒ Select or enter a database name:

☐ Attach a database file: Browse...

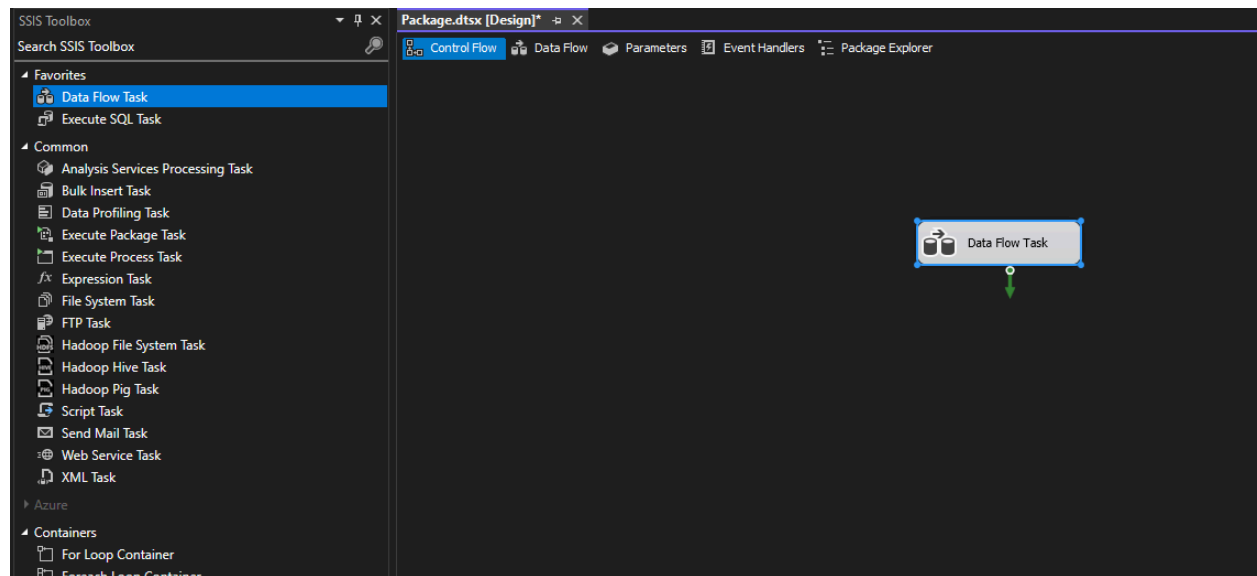
Logical name:

Test Connection OK Cancel Help

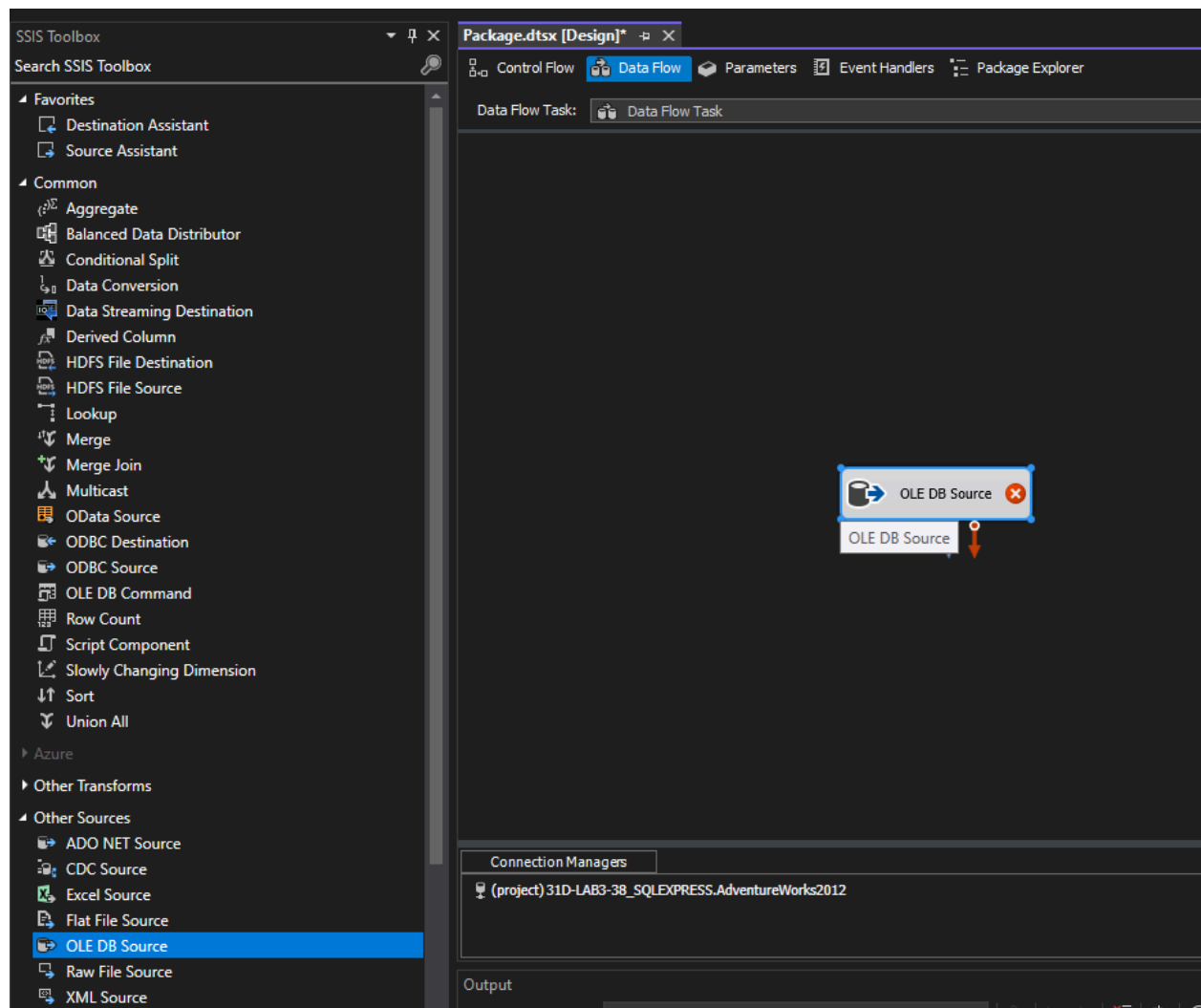
8. Select Server name(Your PC Name) from drop down and database name and click on Test connection.



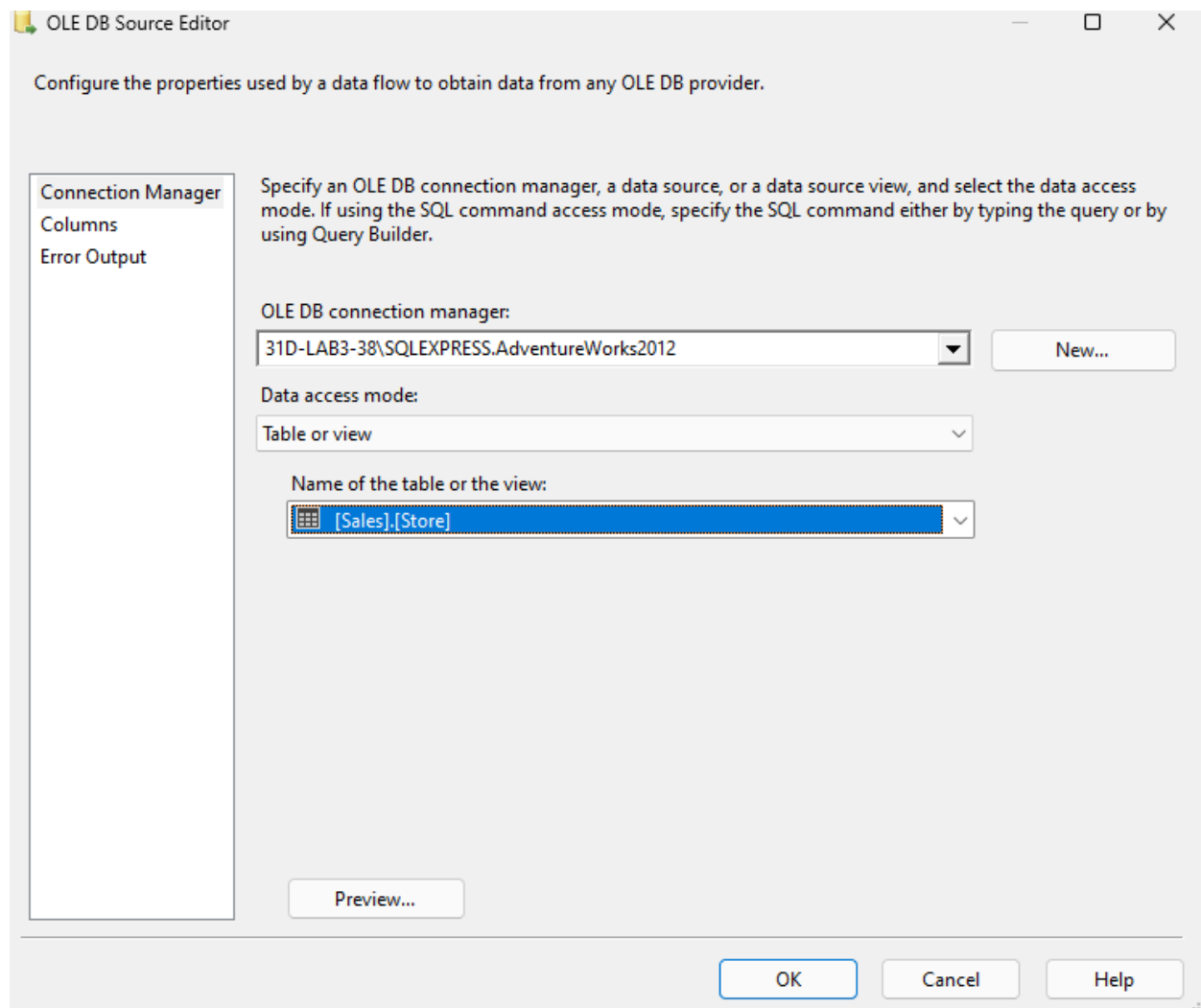
9. Drag and drop Data Flow Task in Control Flow tab



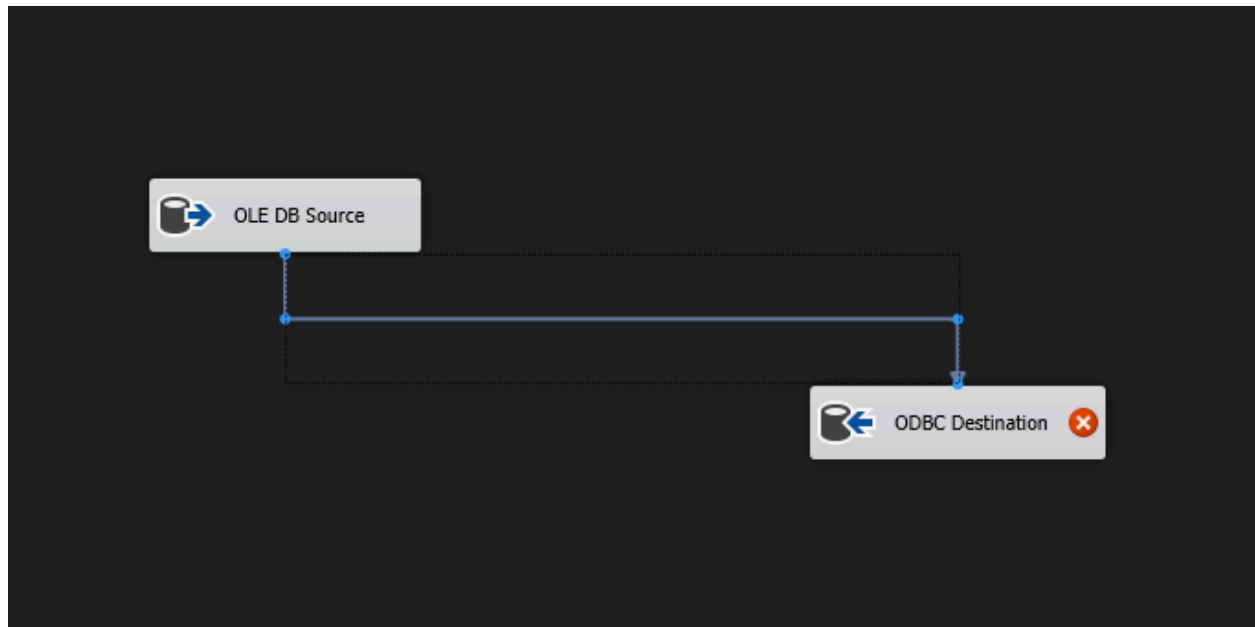
10. Drag OLE DB Source from Other Sources and drop into Data Flow tab



11. Double click on OLE DB source → OLE DB Source Editor appears → click on New to add connection manager. Select [Sales].[Store] table from drop down → ok



12. Drag OLE DB destination from Other Destination in data flow tab and connect both



13. Double click on OLE DB destination Click on New to run the query to get [OLEDB Destination] in Name of the table or the view.

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager

Mappings

Error Output

Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.

OLE DB connection manager:

31D-LAB3-38\SQLEXPRESS.AdventureWorks2012

New...

Data access mode:

Table or view - fast load

Name of the table or the view:

New...

☐ Keep identity

☒ Table lock

☐ Keep nulls

☒ Check constraints

Rows per batch:

Maximum insert commit size:

2147483647

View Existing

Select a table or view from the list.

OK

Cancel

Help

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager
Mappings
Error Output

Specify an OLE DB connection manager, a data source, or a data source view, and select the data access mode. If using the SQL command access mode, specify the SQL command either by typing the query or by using Query Builder. For fast-load data access, set the table update options.

OLE DB connection manager:
31D-LAB3-38\SQLEXPRESS.AdventureWorks2012 New...

Data access mode:
Table or view - fast load

Name of the table or the view:
[OLE DB Destination] New...

☐ Keep identity ☒ Table lock
☐ Keep nulls ☒ Check constraints

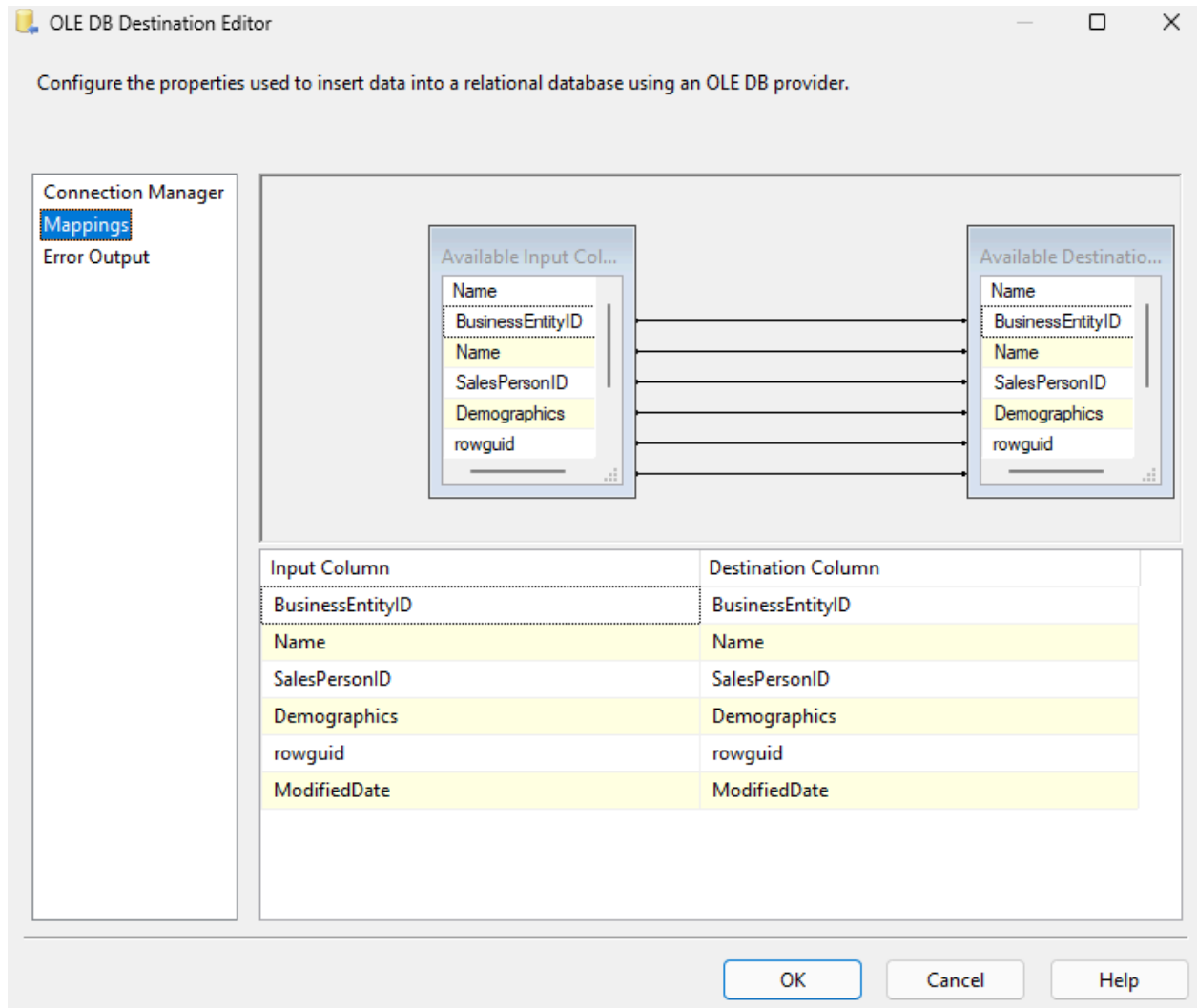
Rows per batch:
Maximum insert commit size: 2147483647

View Existing

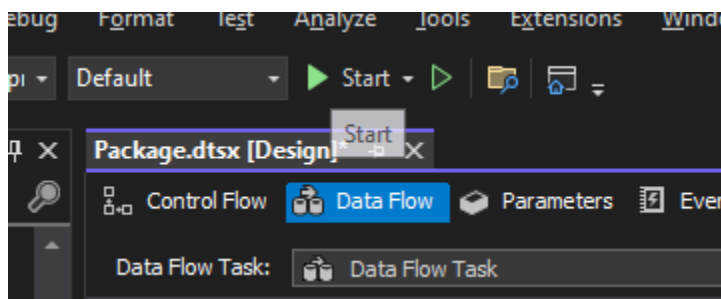
⚠ Map the columns on the Mappings page.

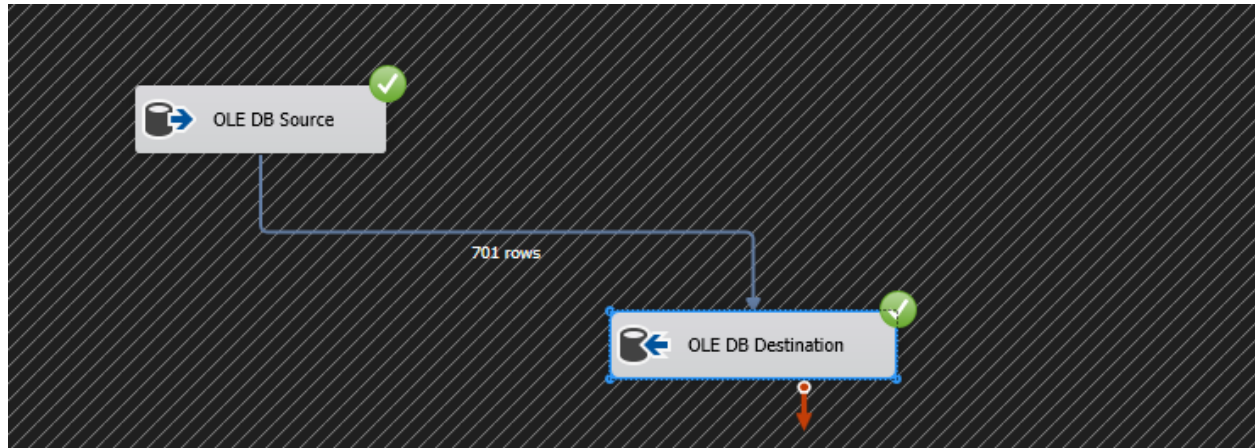
OK Cancel Help

Click on Mappings

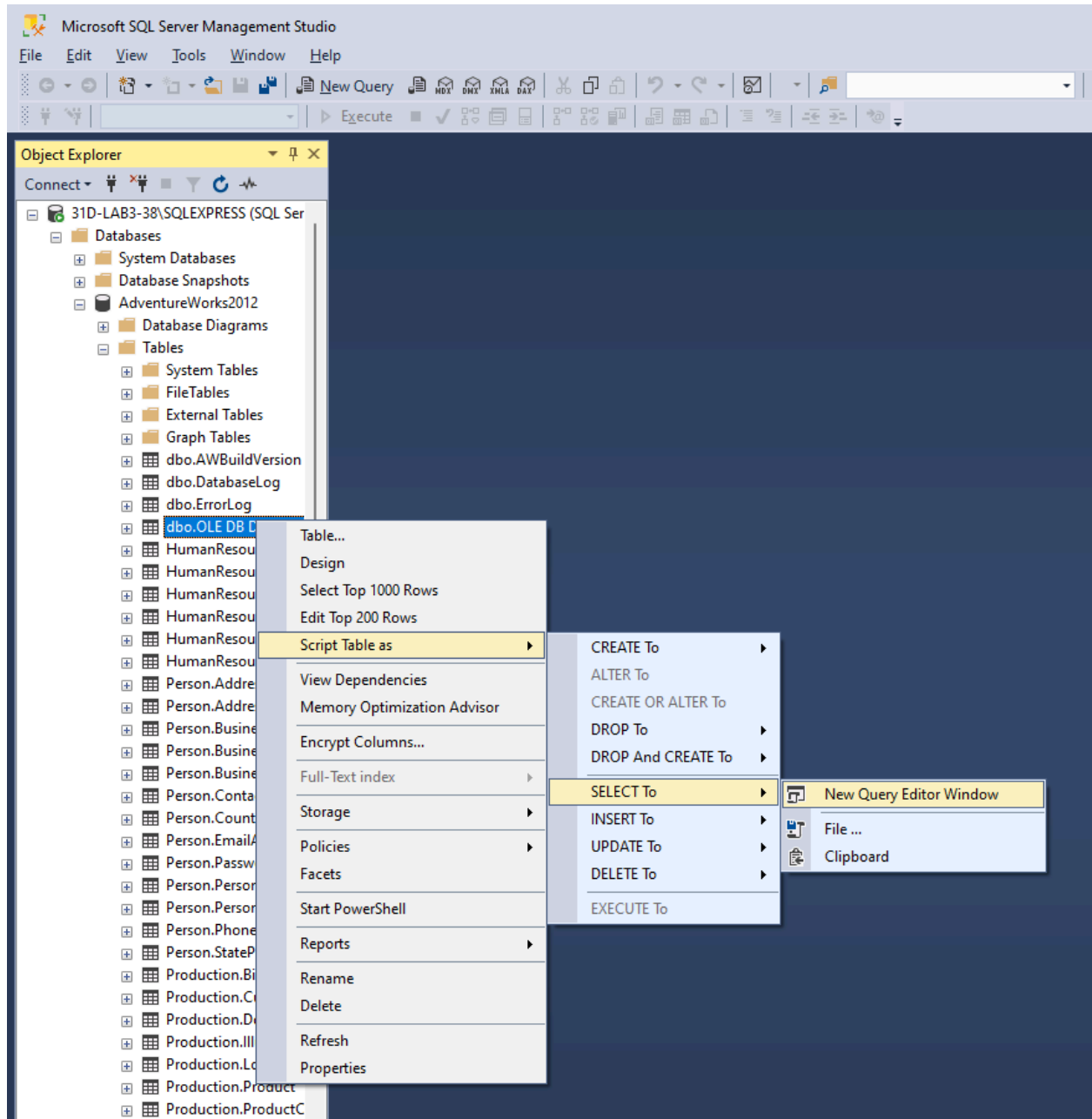


14. Click on start





15. Go to SQL Server Management Studio In database tab → AdventureWorks2019 → Tables → Right click on [dbo].[OLE DB Destination] → Scrip Table as → SELECT To → New Query Editor Window



16. Execute the following query to get output.

```
SQLQuery1.sql - 31...AB3-38\admin (56) - [X]
USE [AdventureWorks2012]
GO

SELECT [BusinessEntityID]
      , [Name]
      , [SalesPersonID]
      , [Demographics]
      , [rowguid]
      , [ModifiedDate]
FROM [dbo].[OLE DB Destination]
GO
```

Execute [Icons]

```
SQLQuery1.sql - 31...AB3-38\admin (56) - [X]
USE [AdventureWorks2012]
GO

SELECT [BusinessEntityID]
      , [Name]
      , [SalesPersonID]
      , [Demographics]
      , [rowguid]
      , [ModifiedDate]
FROM [dbo].[OLE DB Destination]
GO
```

OUTPUT:-

	BusinessEntityID	Name	SalesPersonID	Demographics	rowguid	ModifiedDate
1	292	Next-Door Bike Store	279	<StoreSurvey xmlns="http://schemas.microsoft.co...	A22517E3-848D-4E8E-B909-7437F3432304	2014-09-12 11:15:07.497
2	294	Professional Sales and Service	276	<StoreSurvey xmlns="http://schemas.microsoft.co...	B50CA50B-C601-4A13-B07E-3C63862071B4	2014-09-12 11:15:07.497
3	296	Riders Company	277	<StoreSurvey xmlns="http://schemas.microsoft.co...	337C3688-1339-4E1A-A08A-854823566E49	2014-09-12 11:15:07.497
4	298	The Bike Mechanics	275	<StoreSurvey xmlns="http://schemas.microsoft.co...	7894F278-F0C8-4D16-BD75-213FD8F13023	2014-09-12 11:15:07.497
5	300	Nationwide Supply	286	<StoreSurvey xmlns="http://schemas.microsoft.co...	C3FC9705-A8C4-4F3A-9550-EB2FA87B64D0	2014-09-12 11:15:07.497
6	302	Area Bike Accessories	281	<StoreSurvey xmlns="http://schemas.microsoft.co...	368BE6DD-30E5-498B-9A86-71FD49C58F4E	2014-09-12 11:15:07.497
7	304	Bicycle Accessories and Kits	283	<StoreSurvey xmlns="http://schemas.microsoft.co...	35F40636-5105-49D5-869E-27E231189150	2014-09-12 11:15:07.497
8	306	Clamps & Brackets Co.	275	<StoreSurvey xmlns="http://schemas.microsoft.co...	64D068FC-D060-405C-9C60-C067FE7C67DF	2014-09-12 11:15:07.497
9	308	Valley Bicycle Specialists	277	<StoreSurvey xmlns="http://schemas.microsoft.co...	59386B0C-652E-4668-B44B-4E1717179330	2014-09-12 11:15:07.497
10	310	New Bikes Company	279	<StoreSurvey xmlns="http://schemas.microsoft.co...	47E4B6BD-5CD1-45A3-A231-79D930381C56	2014-09-12 11:15:07.497
11	312	Vinyl and Plastic Goods Corporation	282	<StoreSurvey xmlns="http://schemas.microsoft.co...	DC610525-E373-49B1-B786-EA040EC25C06	2014-09-12 11:15:07.497
12	314	Top of the Line Bikes	288	<StoreSurvey xmlns="http://schemas.microsoft.co...	E290E93F-A980-4BA3-86C3-9858F15C8A6D	2014-09-12 11:15:07.497
13	316	Fun Toys and Bikes	281	<StoreSurvey xmlns="http://schemas.microsoft.co...	6DCDF941-4192-49C7-994A-SADBA53AE095	2014-09-12 11:15:07.497
14	318	Great Bikes	283	<StoreSurvey xmlns="http://schemas.microsoft.co...	956FBC35-5E0D-4175-8045-E08E380BA340	2014-09-12 11:15:07.497
15	320	Metropolitan Sales and Rental	275	<StoreSurvey xmlns="http://schemas.microsoft.co...	0CB4FEF2-5047-40F7-8848-859F7A3F3EEC	2014-09-12 11:15:07.497
16	322	Irregulars Outlet	288	<StoreSurvey xmlns="http://schemas.microsoft.co...	CDE66279-83D8-4340-A83C-E86E15514AC4	2014-09-12 11:15:07.497
17	324	Valley Toy Store	282	<StoreSurvey xmlns="http://schemas.microsoft.co...	6A1BEA56-DCB7-45CF-9C92-3705E12EB2AA	2014-09-12 11:15:07.497
18	326	Worthwhile Activity Store	279	<StoreSurvey xmlns="http://schemas.microsoft.co...	BAD63717-998D-4581-8160-0F17238E42CB	2014-09-12 11:15:07.497
19	328	Purchase Mart	275	<StoreSurvey xmlns="http://schemas.microsoft.co...	A3140349-57A3-46AA-BF32-D7F4753A06D6	2014-09-12 11:15:07.497

Query executed successfully. 31D-LAB3-38\SQLEXPRESS (15.... | 31D-LAB3-38\admin (56) | AdventureWorks2012 | 00:00:00 | 701 rows

PRACTICAL 3

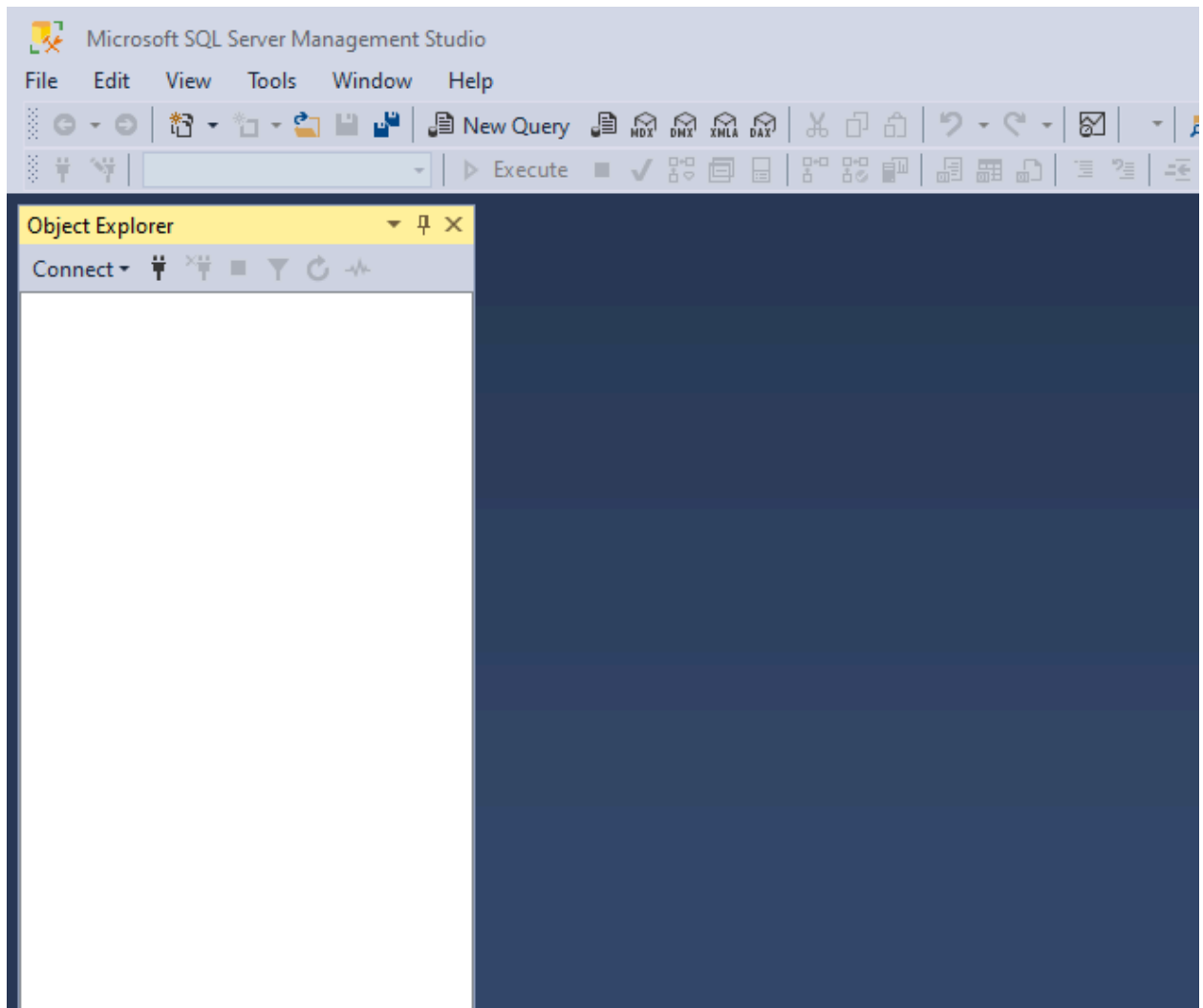
Aim: -

Create the cube with suitable dimension and fact tables based on OLAP

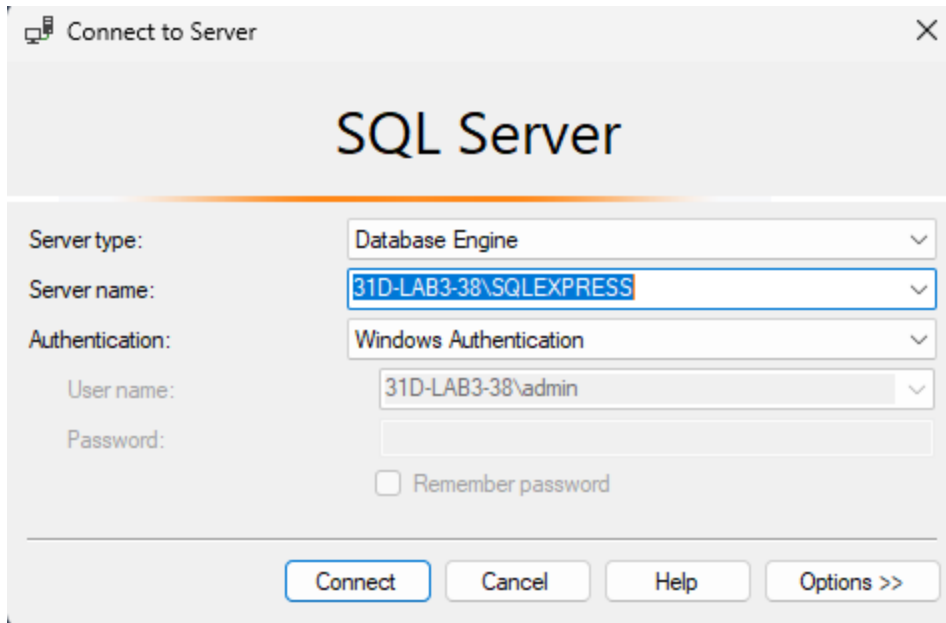
Steps: -

1. Creating Data Warehouse

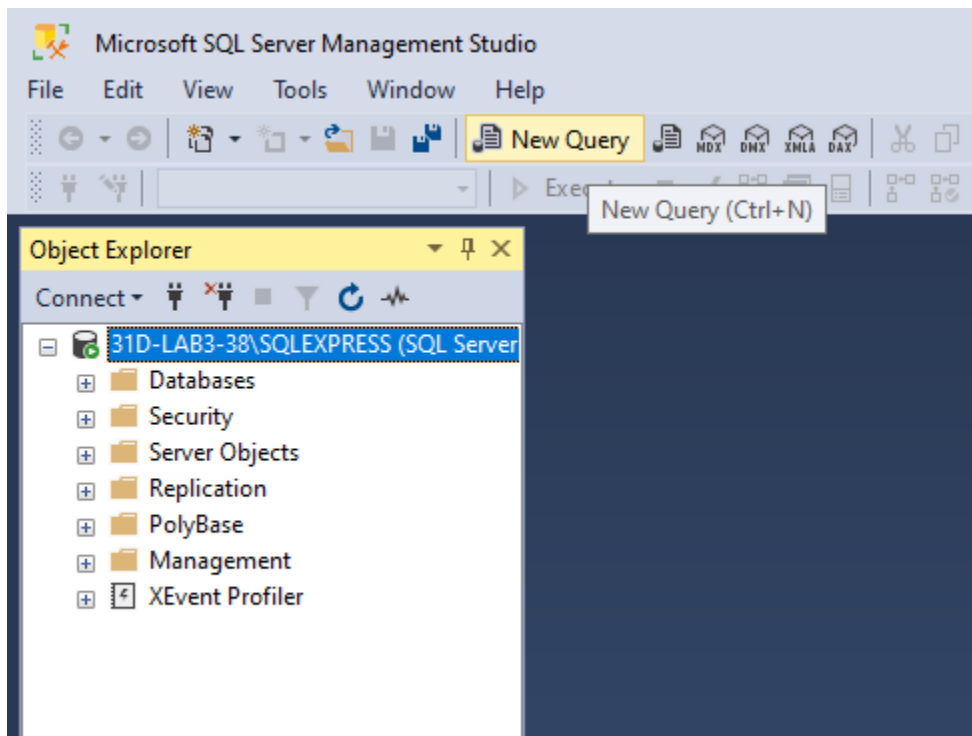
a. Open SQL Server Management Studio 2012

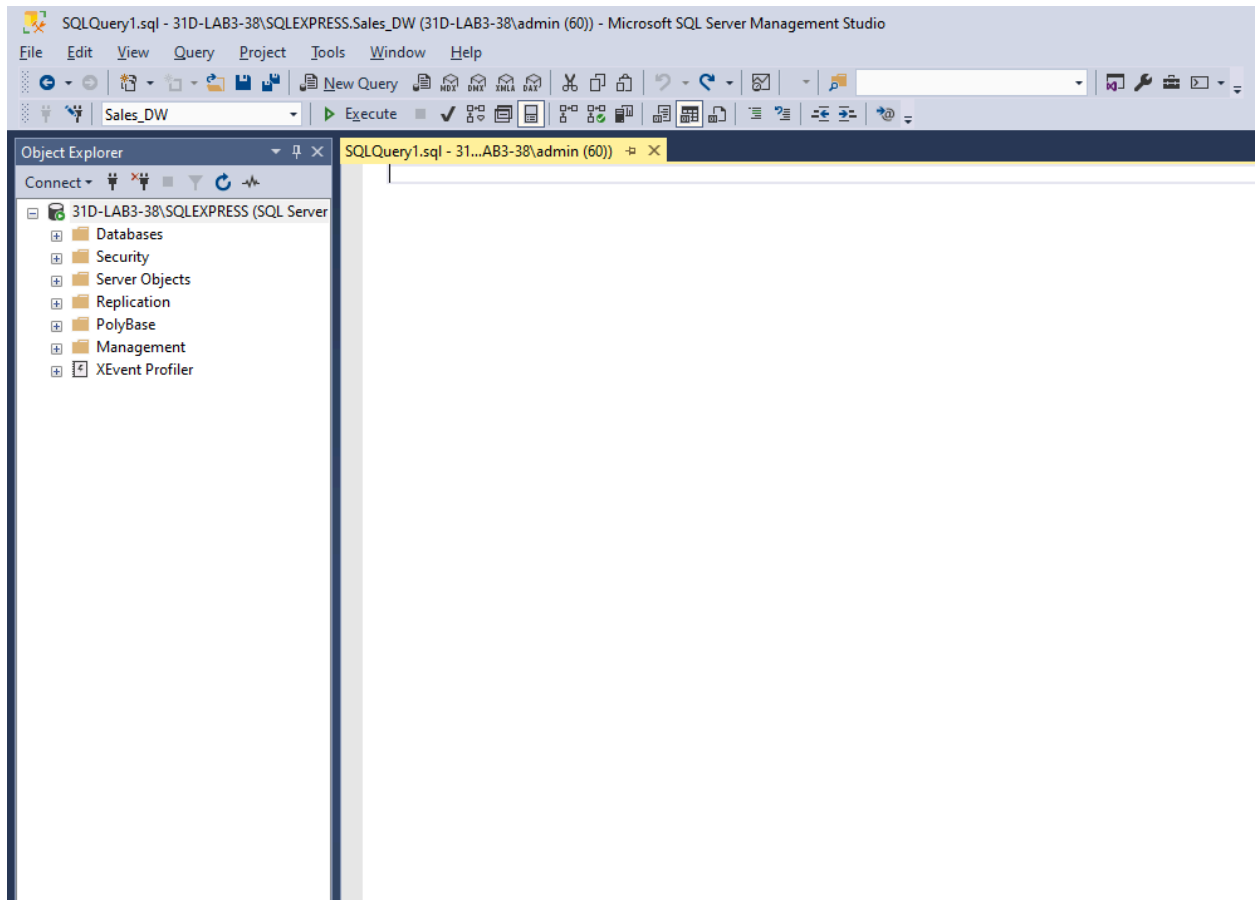


b. Connect Database Engine



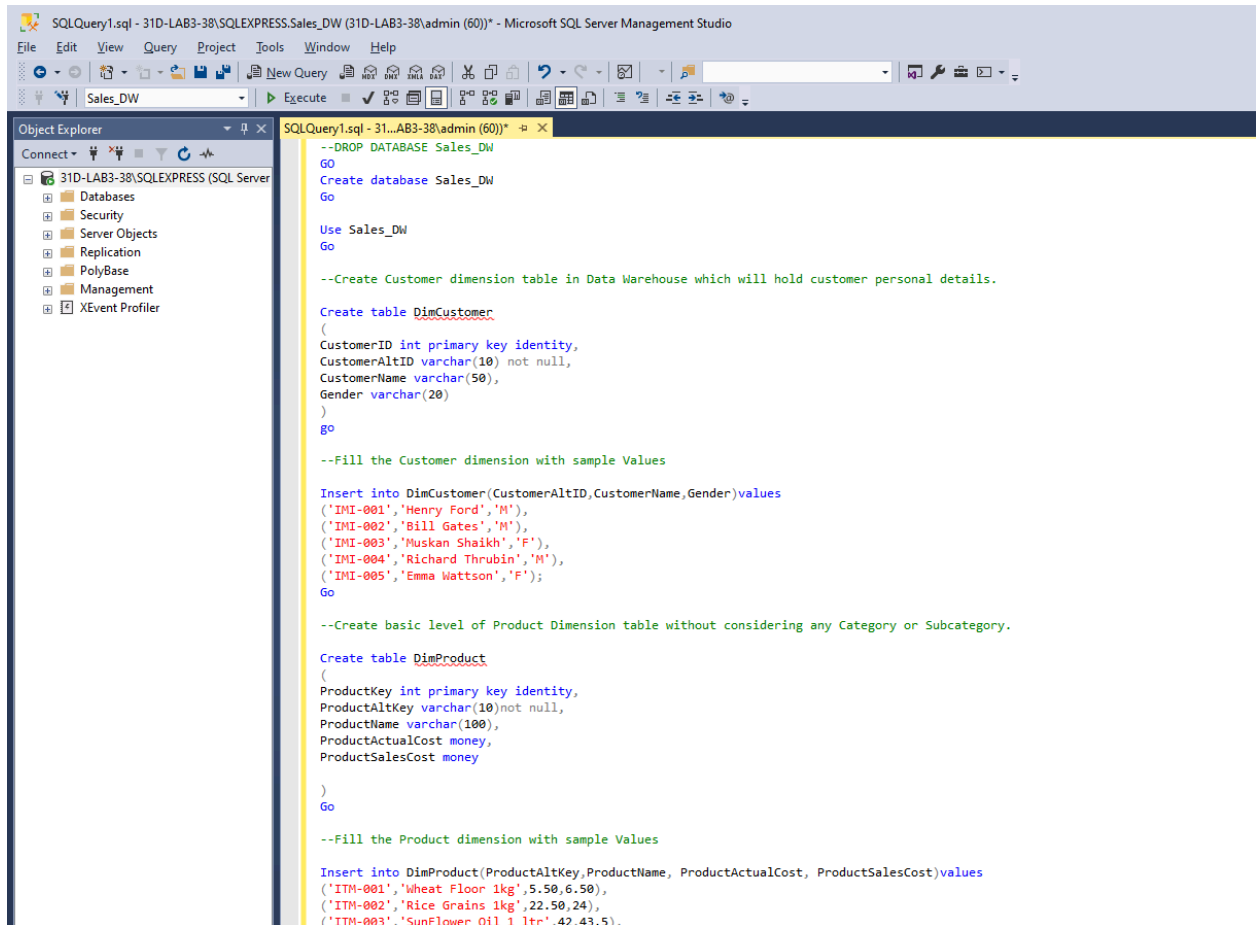
c. Open New Query editor.



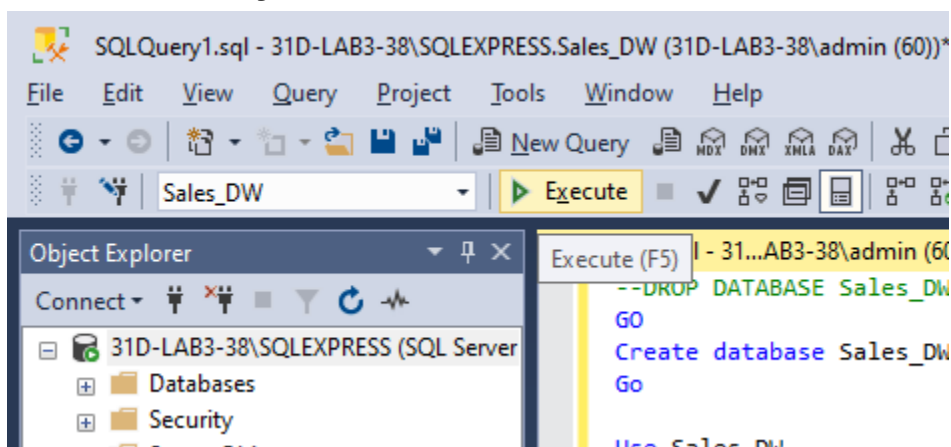


d. New SQL Query Editor will open copy and paste the Sales_DW Database.

(https://drive.google.com/file/d/1FkrnI7LAJXTWaVN29JEtWI7y9xBhZMU3/view?usp=classroom_web&authuser=0) : - this file(script inside this file). Paste this script into the new query editor



e. Click on execute or press F5 by selecting query one by one or directly click on Execute.



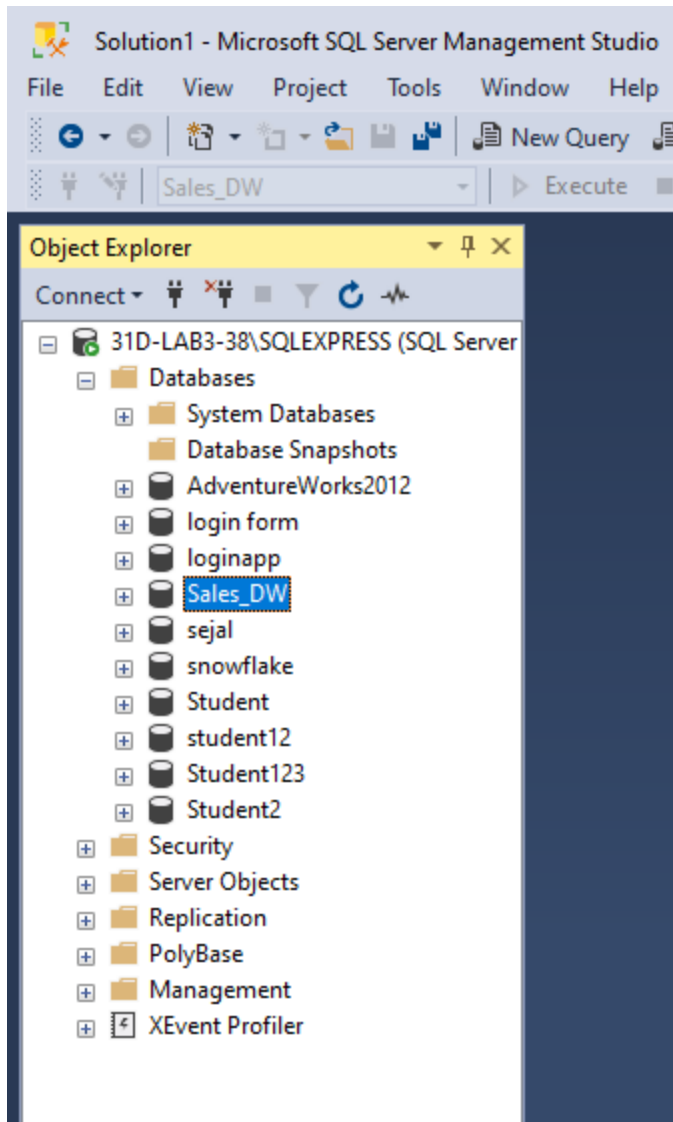
```
--Fill the Customer dimension with sample Values
Insert into DimCustomer (CustomerAltID, CustomerName, Gender) values
('IM1-001', 'Henry Ford', 'M'),
('IM1-002', 'Bill gates', 'M'),
('IM1-003', 'Huskan Shaikh', 'F');
```

DateKey	Date	FullDateUK	FullDateUSA	DayOfMonth	DaySuffix	DayName	DayOfWeekUSA	DayOfWeekUK	DayOfWeekInMonth	DayOfWeekInYear	DayOfQuarter	DayOfYear	WeekOfMonth	WeekOfQuarter	WeekOf
1	20130101	2013-01-01 00:00:00.000	01/01/2013	01/01/2013	1	1st	Tuesday	3	2	1	1	1	1	1	1
2	20130102	2013-01-02 00:00:00.000	02/01/2013	01/02/2013	2	2nd	Wednesday	4	3	1	1	2	1	1	1
3	20130103	2013-01-03 00:00:00.000	03/01/2013	01/03/2013	3	3rd	Thursday	5	4	1	1	3	1	1	1
4	20130104	2013-01-04 00:00:00.000	04/01/2013	01/04/2013	4	4th	Friday	6	5	1	1	4	1	1	1
5	20130105	2013-01-05 00:00:00.000	05/01/2013	01/05/2013	5	5th	Saturday	7	6	1	1	5	1	1	1
6	20130106	2013-01-06 00:00:00.000	06/01/2013	01/06/2013	6	6th	Sunday	1	7	1	1	6	2	1	2
7	20130107	2013-01-07 00:00:00.000	07/01/2013	01/07/2013	7	7th	Monday	2	1	1	1	7	2	1	2
8	20130108	2013-01-08 00:00:00.000	08/01/2013	01/08/2013	8	8th	Tuesday	3	2	2	2	8	2	2	2
9	20130109	2013-01-09 00:00:00.000	09/01/2013	01/09/2013	9	9th	Wednesday	4	3	2	2	9	2	2	2
10	20130110	2013-01-10 00:00:00.000	10/01/2013	01/10/2013	10	10th	Thursday	5	4	2	2	10	2	2	2
11	20130111	2013-01-11 00:00:00.000	11/01/2013	01/11/2013	11	11th	Friday	6	5	2	2	11	2	2	2
12	20130112	2013-01-12 00:00:00.000	12/01/2013	01/12/2013	12	12th	Saturday	7	6	2	2	12	2	2	2
13	20130113	2013-01-13 00:00:00.000	13/01/2013	01/13/2013	13	13th	Sunday	1	7	2	2	13	3	2	3
14	20130114	2013-01-14 00:00:00.000	14/01/2013	01/14/2013	14	14th	Monday	2	1	2	2	14	3	2	3
15	20130115	2013-01-15 00:00:00.000	15/01/2013	01/15/2013	15	15th	Tuesday	3	2	3	3	15	3	3	3
16	20130116	2013-01-16 00:00:00.000	16/01/2013	01/16/2013	16	16th	Wednesday	4	3	3	3	16	3	3	3
17	20130117	2013-01-17 00:00:00.000	17/01/2013	01/17/2013	17	17th	Thursday	5	4	3	3	17	3	3	3
18	20130118	2013-01-18 00:00:00.000	18/01/2013	01/18/2013	18	18th	Friday	6	5	3	3	18	3	3	3
19	20130119	2013-01-19 00:00:00.000	19/01/2013	01/19/2013	19	19th	Saturday	7	6	3	3	19	3	3	3

Query executed successfully.

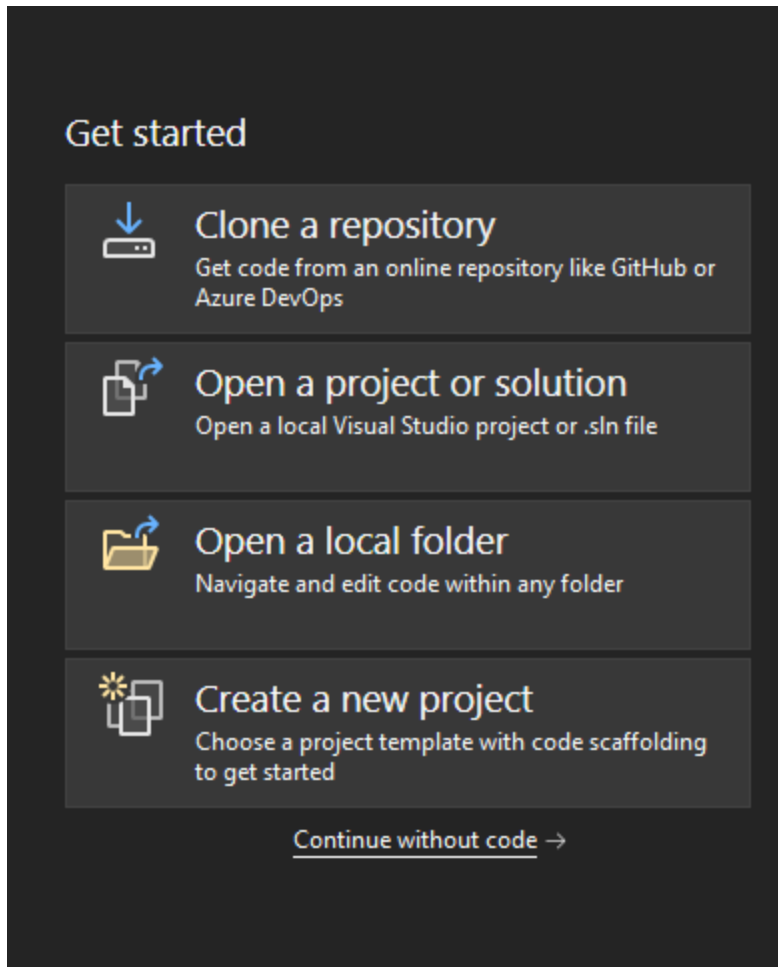
31D-LAB3-38\SQLEXPRESS (15... 31D-LAB3-38\admin (60) Sales_DW | 00:00:07 730 rows

f. After completing execution save and close SQL Server Management studio & Reopen to see Sales_DW in Databases Tab.

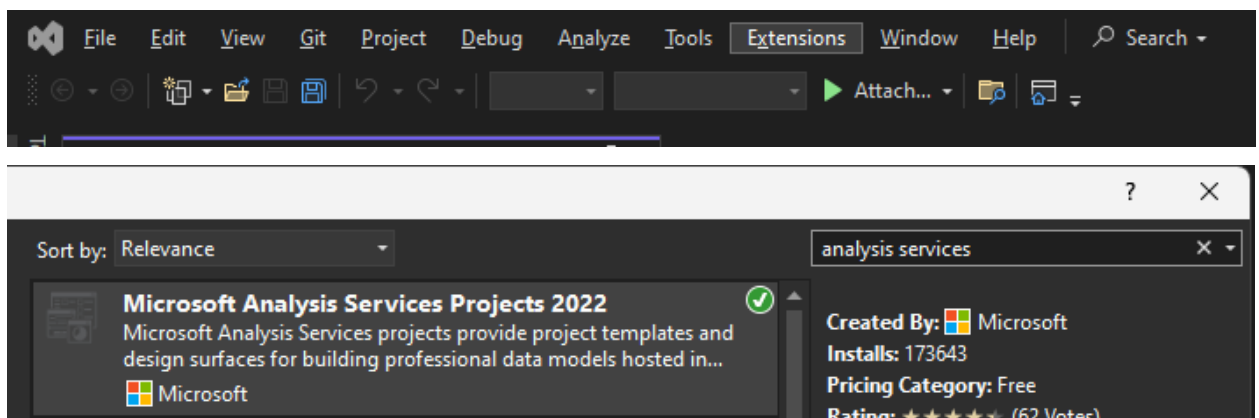


2. Adding Microsoft Analysis Services 2022 extension in visual studios

a. Open Visual studio 2022 and continue without code



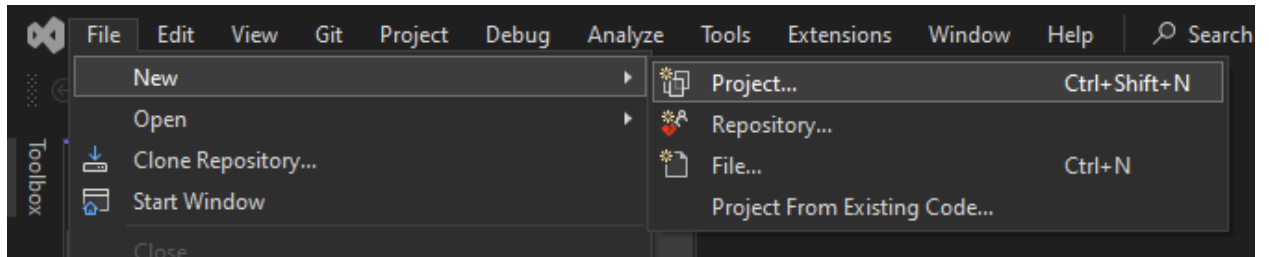
b. Go to Extension >Manage Extensions >Online >Search for Analysis Services download and install the extension



3. Creating NEW DATA SOURCE

a. Open Visual Studios.

Click on File ->New->Project-> Analysis Services
Multidimensional Project->Next->appropriate project name
->Create.



analysis services



Clear all

C#

Windows

Web



ASP.NET Core Web App (Model-View-Controller)

A project template for creating an ASP.NET Core application with example ASP.NET Core MVC Views and Controllers. This template can also be used for RESTful HTTP [services](#).

C#

Linux

macOS

Windows

Cloud

Service

Web



WCF Service

A Web site for creating WCF [services](#). This template does not produce a project file and has limited MSBuild support.

C#

Windows

Web

Service

gRPC

ASP.NET Core gRPC Service

A project template for creating a gRPC service using ASP.NET Core, with optional support for publishing as native AOT.

C#

Linux

macOS

Windows

Cloud

Service

Web



WCF Service Application

A project for creating WCF Service Application that is hosted in IIS/WAS

C#

Windows

Web

Service



WCF Service Library

A project for creating a host-independent WCF service class library (.dll)

C#

Windows

Library

Web

Service

Other results based on your search



[Analysis Services](#) Multidimensional Project

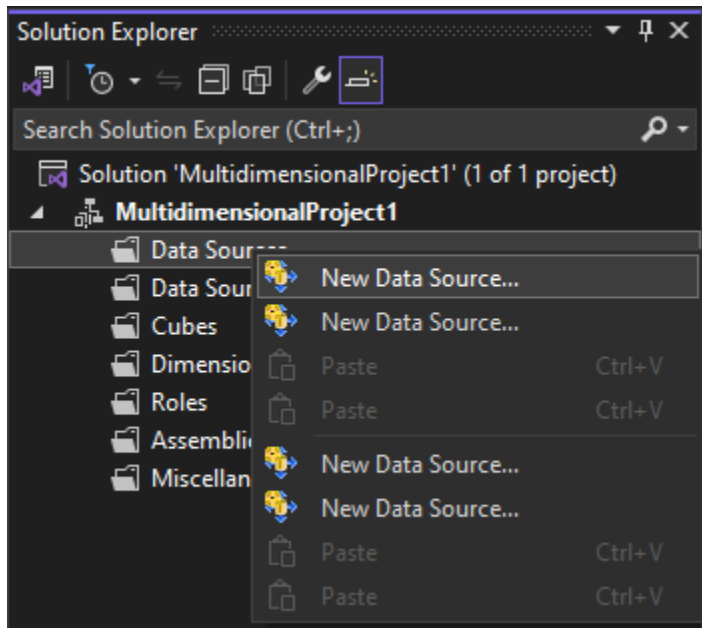
An [Analysis Services](#) project for creating multidimensional models.



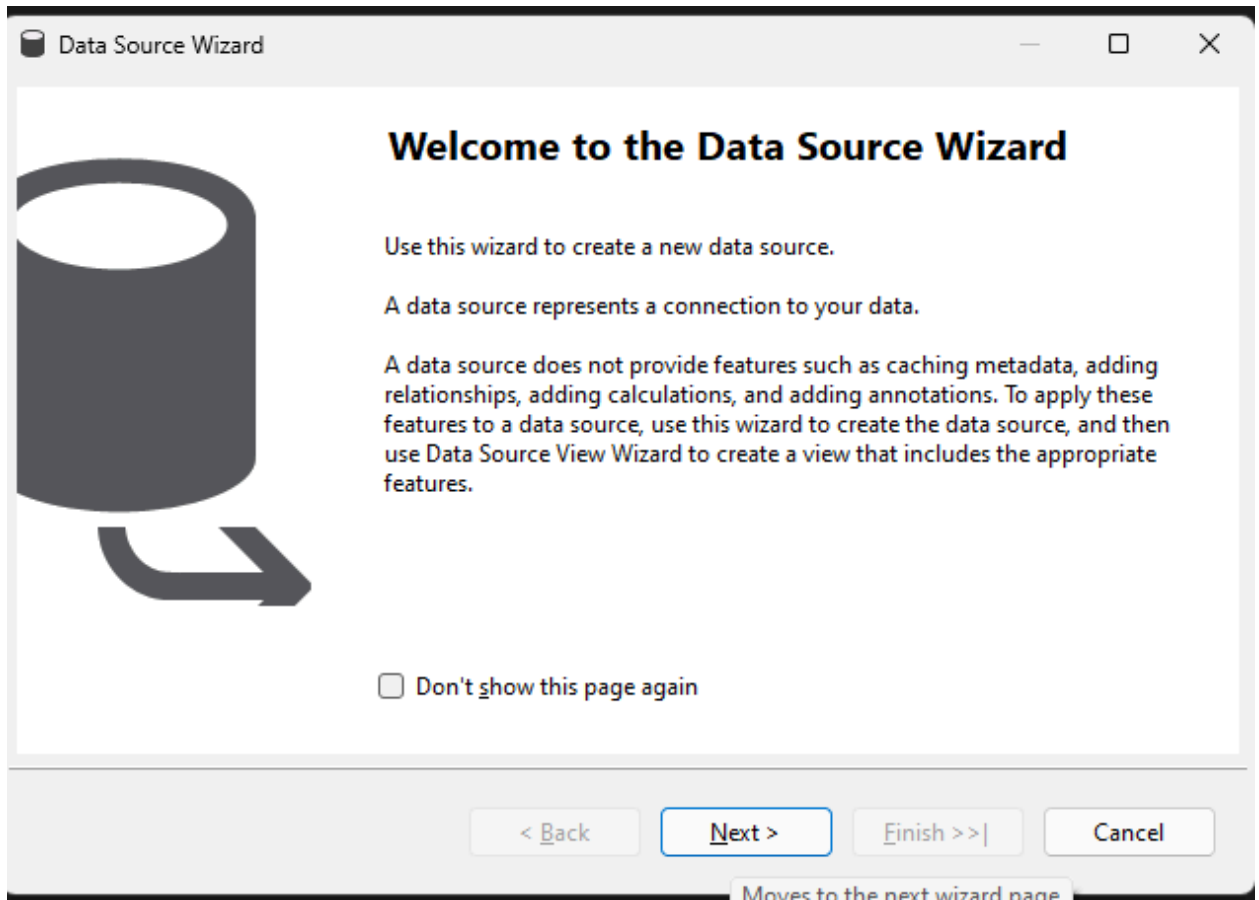
[Analysis Services](#) Multidimensional Project

An [Analysis Services](#) project for creating multidimensional models.

b. Right click on Data Sources in solution explorer ->New Data Source



c. Data Source Wizard appears ->Next



d. Select Server Name (Your PC Name)->select Use Windows Authentication ->Select or enter a database name (Sales_DW) -> Click Ok->Test Connection

Data Source Wizard

Select how to define the connection

You can select from a number of ways in which your data source will define its connection



☐ Create a data source based on another object

☒ Create a data source based on an existing or new connection

Data connections:

Data connection properties:

Property	Value	

New...

Delete

 A valid connection must be selected.

< Back

Next >

Finish >>|

Cancel

Connection Manager

Provider: Native OLE DB\SQL Server Native Client 11.0

Connection

Server name: 31D-LAB3-38\SQLEXPRESS Refresh

Log on to the server

Authentication: Windows Authentication

User name: Password: Save my password

Connect to a database

Select or enter a database name: Sales_DW

Attach a database file: Browse... Logical name:

Test Connection OK Cancel Help

Data Source Wizard

Select how to define the connection
You can select from a number of ways in which your data source will define its connection

☐ Create a data source based on another object

☒ Create a data source based on an existing or new connection

Data connections:

31D-LAB3-38\SQLEXPRESS.Sales_DW

Data connection properties:

Property	Value
Data Source	31D-LAB3-38\SQLEXP...
Initial Catalog	Sales_DW
Integrated Se...	SSPI
Provider	SQLNCLI11.1

New... Delete

< Back Next > Finish >>| Cancel

e. Click Next

f. Select Inherit ->Next

Data Source Wizard

Impersonation Information

You can define what Windows credentials Analysis Services will use to connect to the data

☐ Use a specific Windows user name and password

User name:

Password:

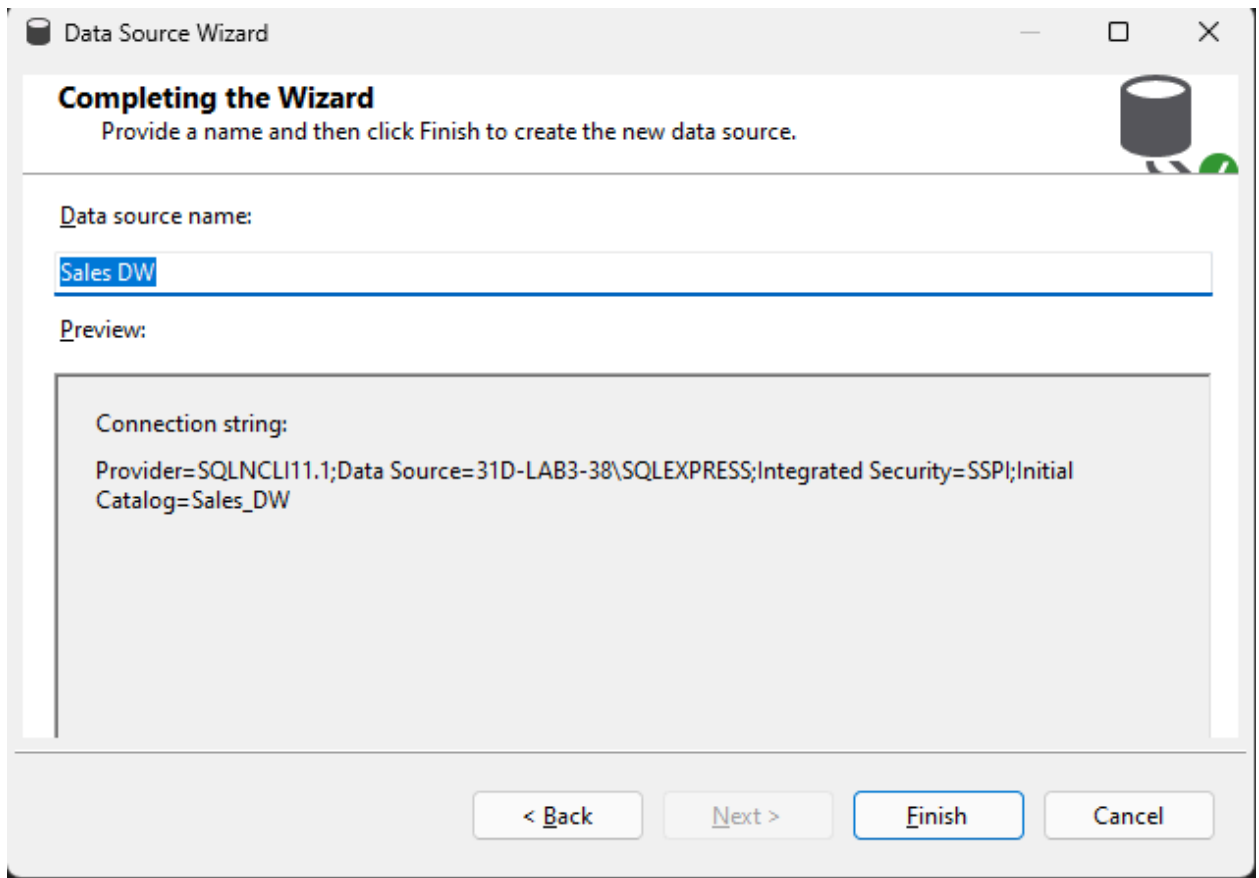
☐ Use the service account

☐ Use the credentials of the current user

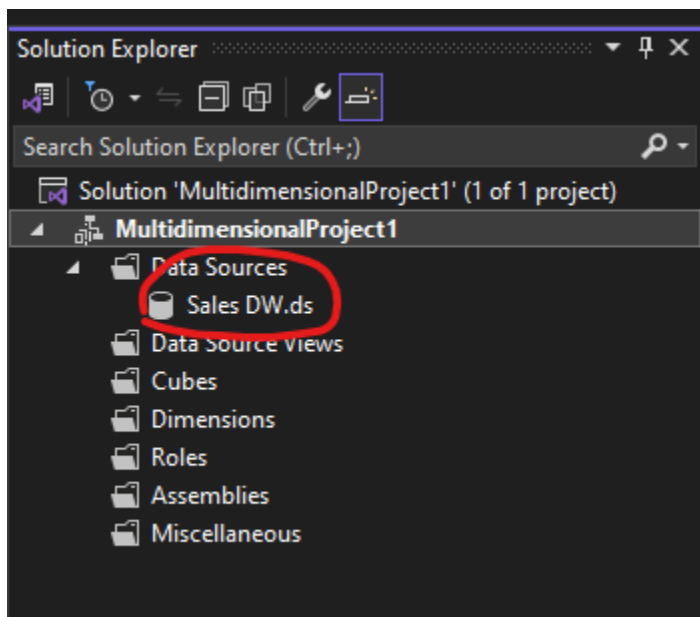
☒ Inherit

< Back Next > Finish >> | Cancel

g. Click Finish



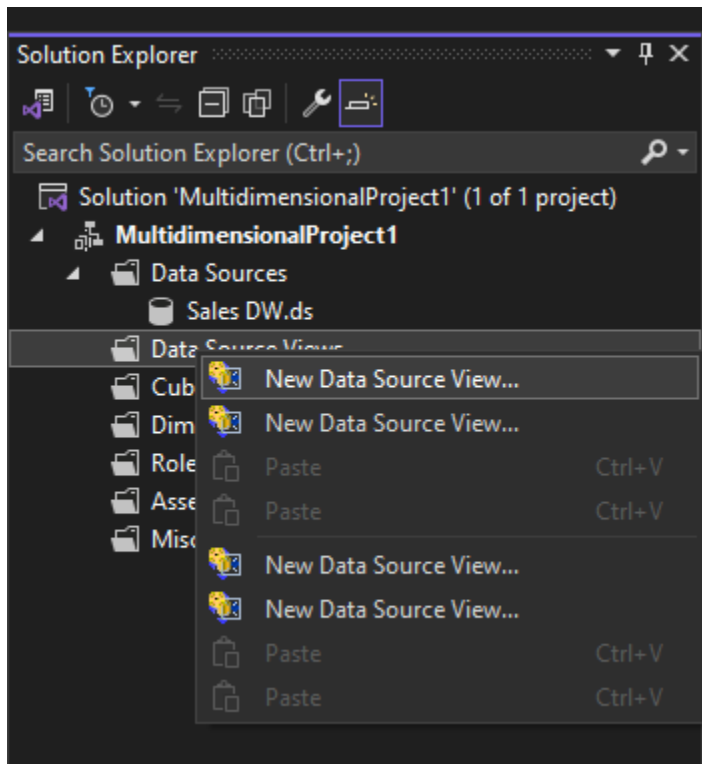
h.Sales DW.ds gets created under Data Sources in Solution Explorer



4. Creating New Data Source View

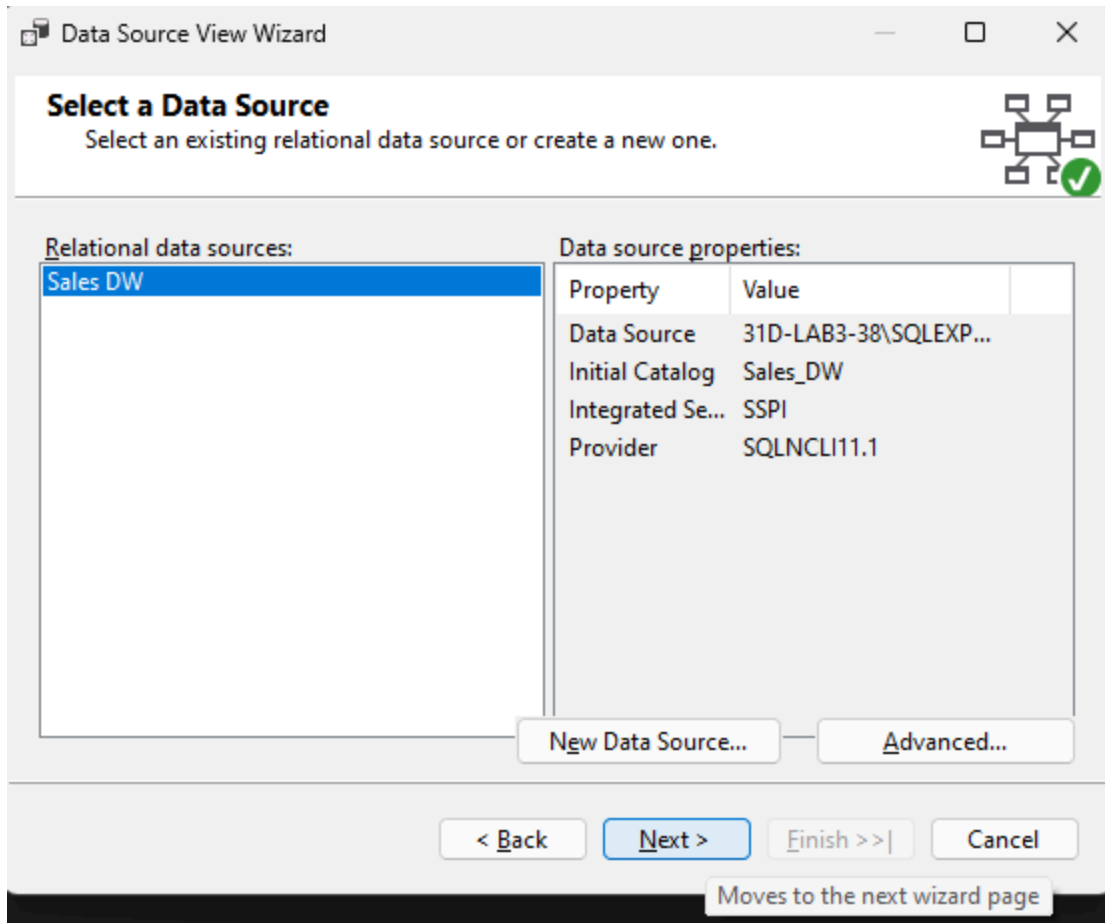
a. In Solution explorer right click on Data Source View

->Select New Data Source View



b. Click Next

c. in select data source Sales_DW is selected, click next



d. Select FactProductSales(dbo) from Available objects and put in Includes Objects by clicking on (>)

Data Source View Wizard

Select Tables and Views
Select objects from the relational database to be included in the data source view.

Available objects:

Name	Type
DimCustomer (dbo)	Table
DimDate (dbo)	Table
DimProduct (dbo)	Table
DimSalesPerson (dbo)	Table
DimStores (dbo)	Table
DimTime (dbo)	Table

Filter:

☐ Show system objects

Included objects:

Name	Type
FactProductSales (dbo)	Table

Add Related Tables

< Back Next > Finish >> | Cancel

e. Click on Add Related Tables

Data Source View Wizard

Select Tables and Views

Select objects from the relational database to be included in the data source view.



Available objects:

Name	Type
DimCustomer (dbo)	Table
DimDate (dbo)	Table
DimProduct (dbo)	Table
DimSalesPerson (dbo)	Table
DimStores (dbo)	Table
DimTime (dbo)	Table

Filter:

☐ Show system objects

Included objects:

Name	Type
FactProductSales (dbo)	Table

> < >> <<

Add Related Tables

< Back Next > Finish >>| Cancel

Data Source View Wizard

Select Tables and Views
Select objects from the relational database to be included in the data source view.

Available objects:

Name	Type
------	------

Filter:

☐ Show system objects

Included objects:

Name	Type
FactProductSales (dbo)	Table
DimTime (dbo)	Table
DimProduct (dbo)	Table
DimStores (dbo)	Table
DimCustomer (dbo)	Table
DimSalesPerson (dbo)	Table
DimDate (dbo)	Table

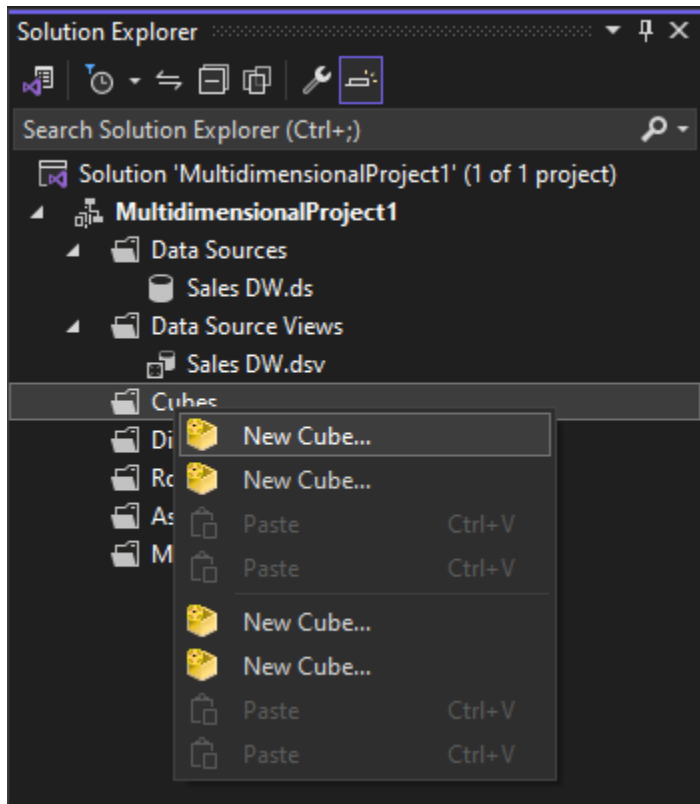
Add Related Tables

< Back **Next >** Finish >>| Cancel

- f. Click next and finish
- g. Sales DW.dsv appears in Data Source Views in Solution Explorer.

5. Creating New Cube

a. Right click on Cubes -> New Cube



b. Select Use existing tables in Select Creation Method

->Next

Cube Wizard

Select Creation Method

Cubes can be created by using existing tables, creating an empty cube, or generating tables in the data source.

How would you like to create the cube?

☒ Use existing tables

☐ Create an empty cube

☐ Generate tables in the data source

Template:

(None)

Description:

Create a cube based on one or more tables in a data source.

< Back Next > Finish >> | Cancel

c. In Select Measure Group Tables → Select FactProductSales → Click Next

Cube Wizard

Select Measure Group Tables

Select a data source view or diagram and then select the tables that will be used for measure groups.

Data source view:
Sales DW

Measure group tables:

☒ FactProductSales

☐ DimTime

☐ DimProduct

☐ DimStores

☐ DimCustomer

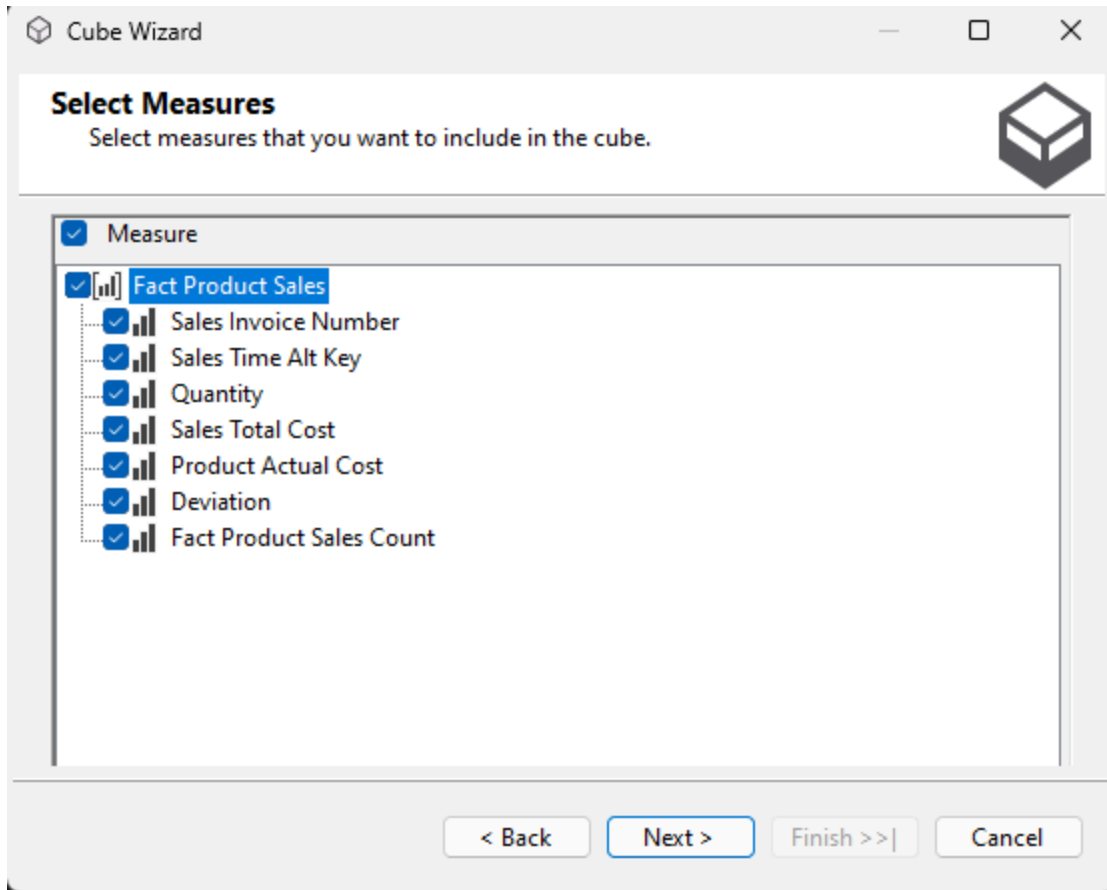
☐ DimSalesPerson

☐ DimDate

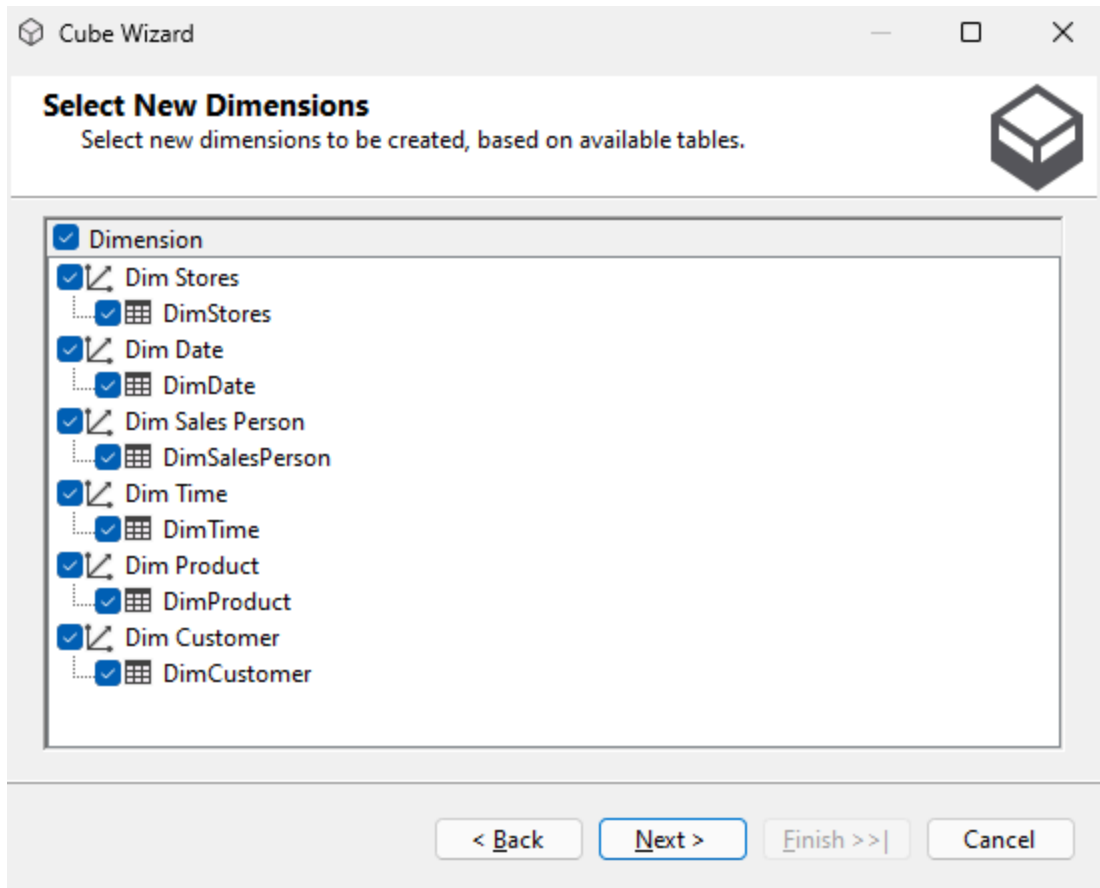
Suggest

< Back Next > Finish >> | Cancel

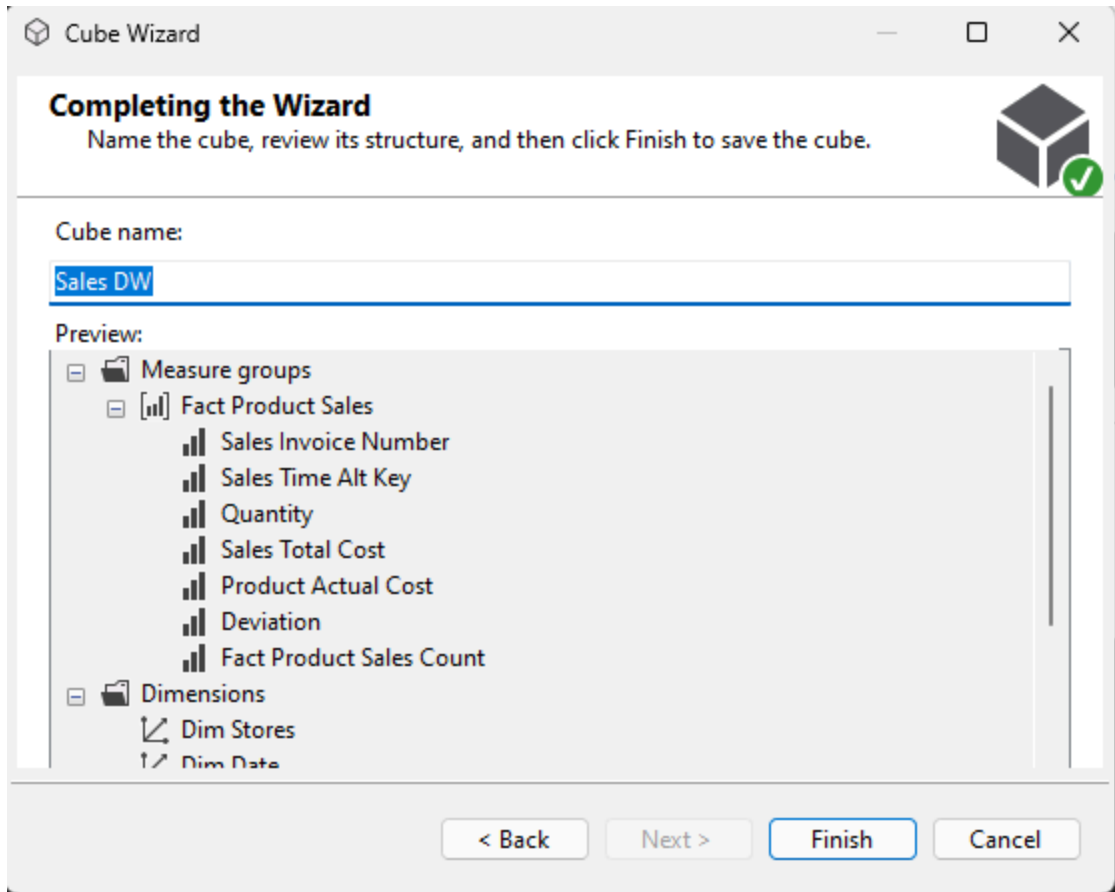
d. In Select Measures → check all measures → Next



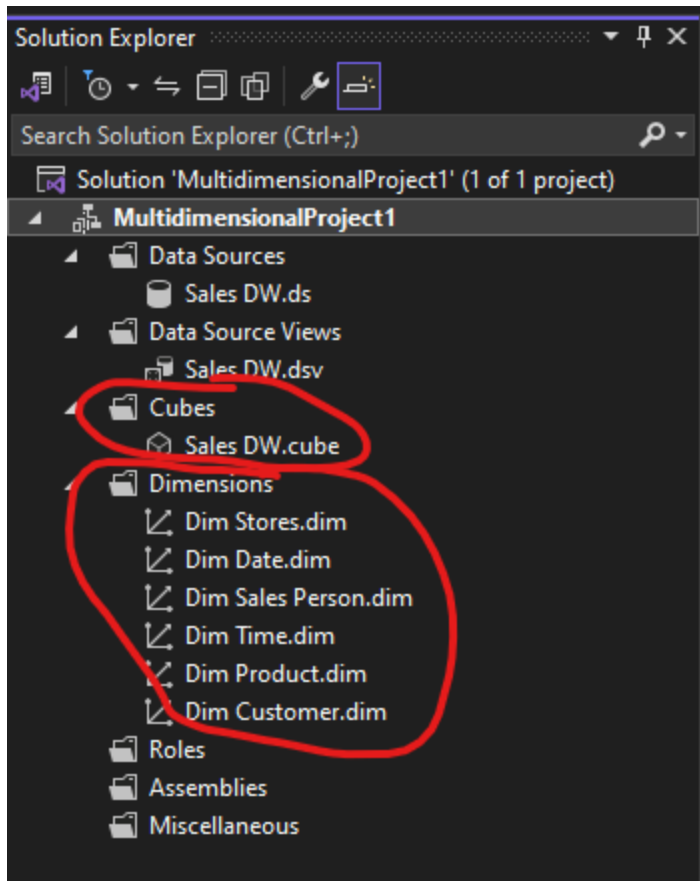
**e. In Select New Dimensions → Check all Dimensions →
Next**



f. Click on Finish

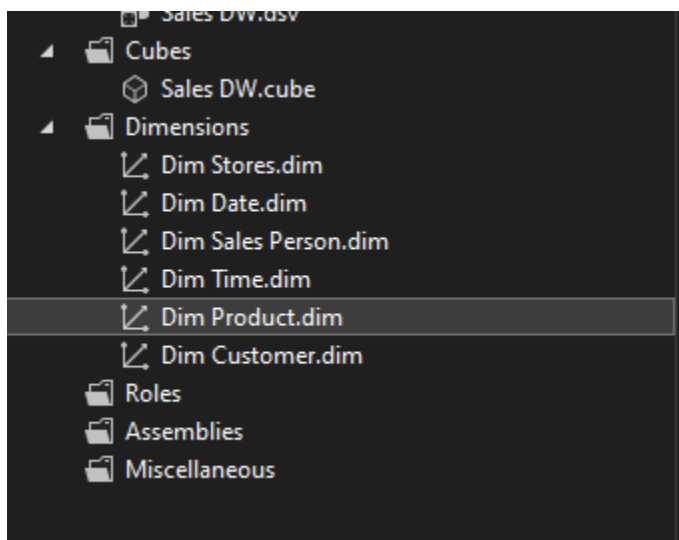


g. Sales_DW.cube is created.

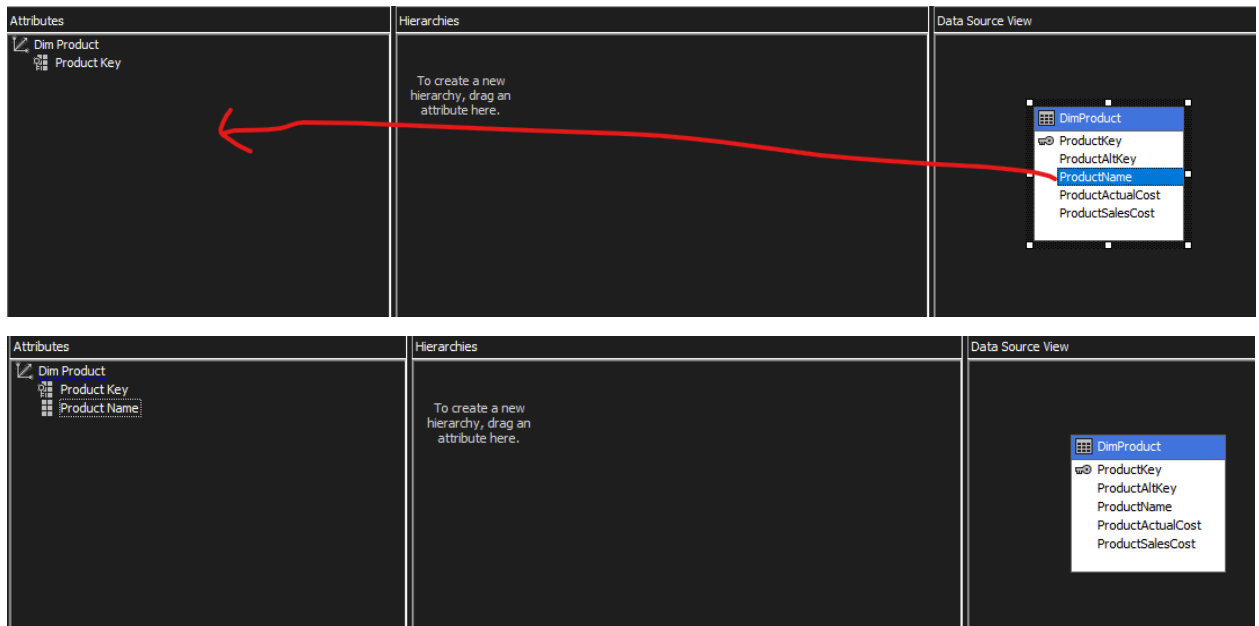


6. Dimension Modification

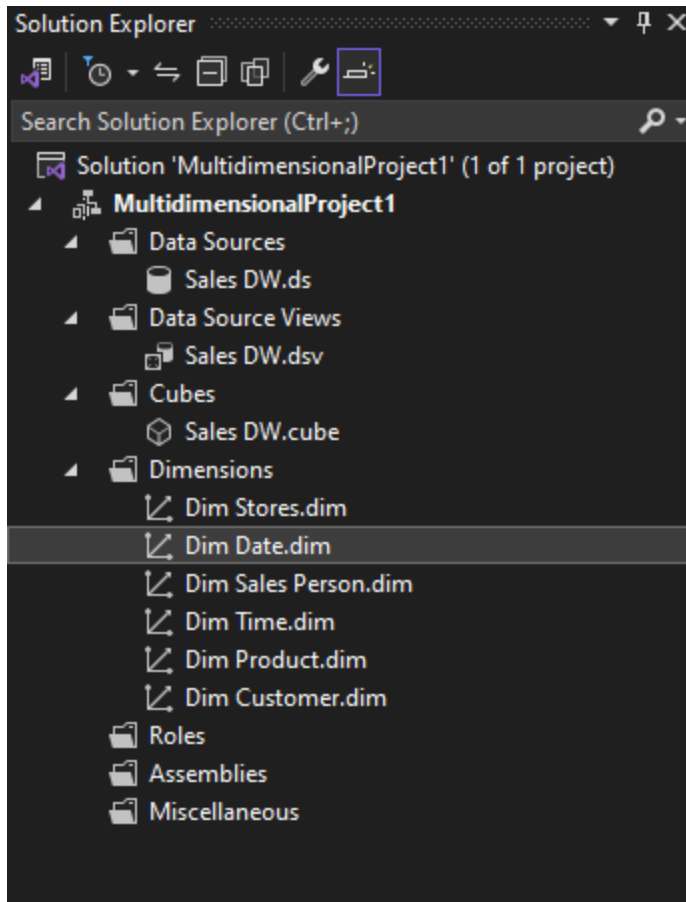
a. In dimension tab → Double Click Dim Product.dim



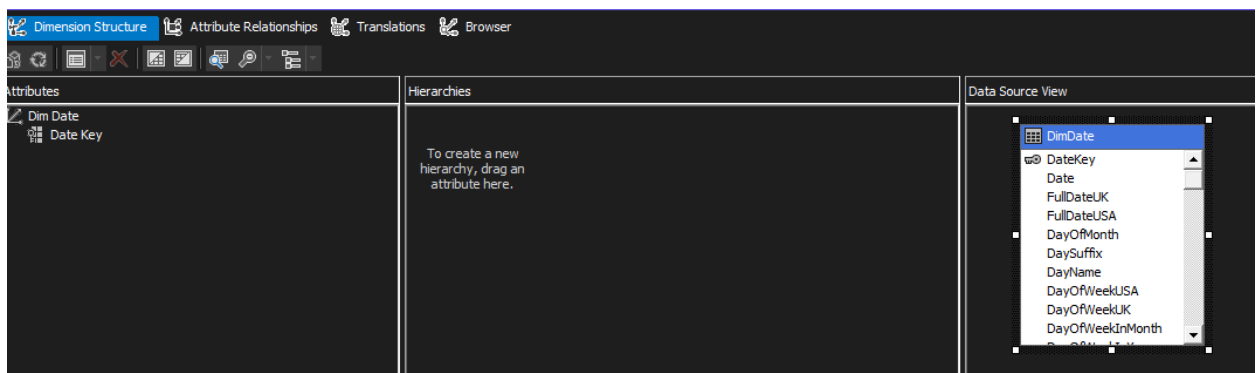
b. Drag and Drop Product Name from Table in Data Source View and Add in Attribute Pane at left side



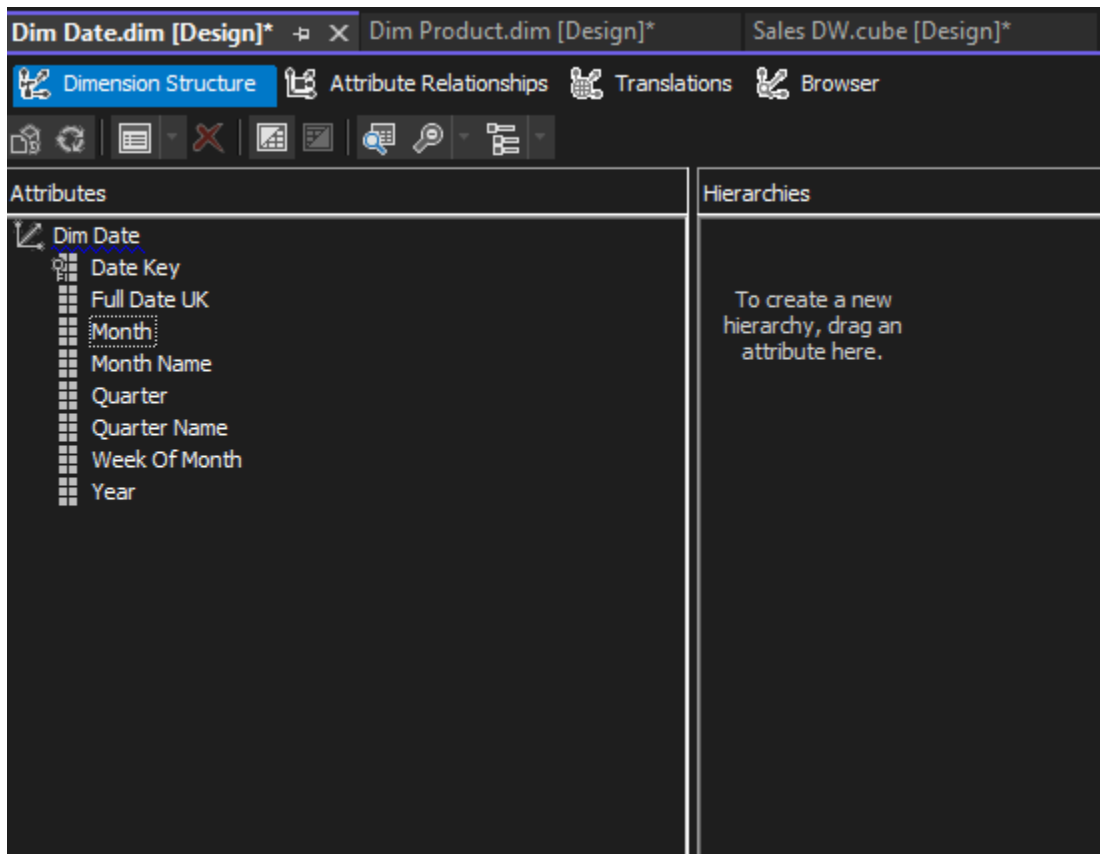
**c. Creating Attribute Hierarchy in Date Dimension.
Double click On Dim Datedimension ->**



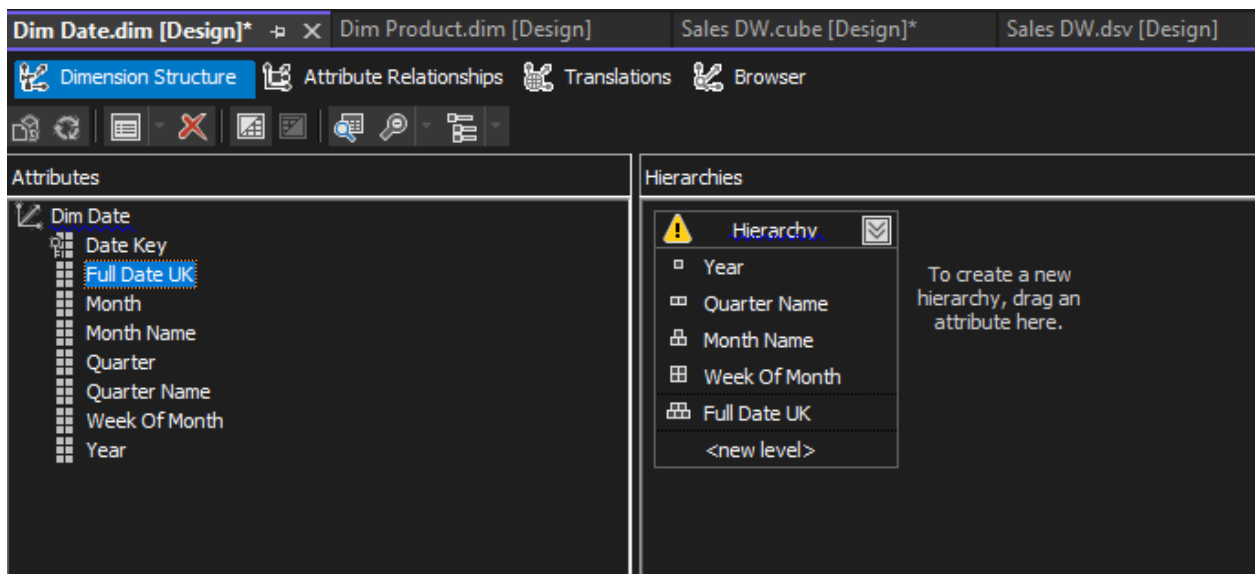
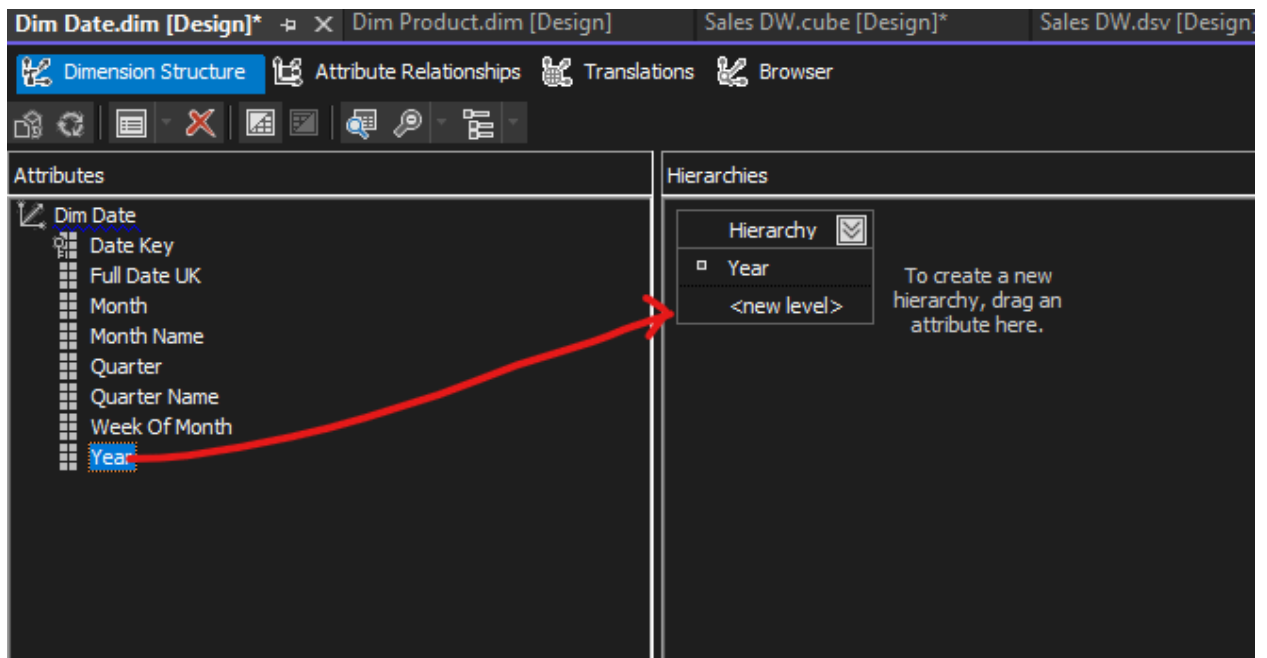
Drag and Drop Fields from Table shown in Data Source View to Attributes-> Drag and Drop attributes from leftmost pane of attributes to middle pane of Hierarchy.



Drag fields in sequence from Attributes to Hierarchy window (Year, Quarter Name, Month Name, Week of the Month, Full Date UK) (some issue in hierarchy i directly dropped in the attributes section)

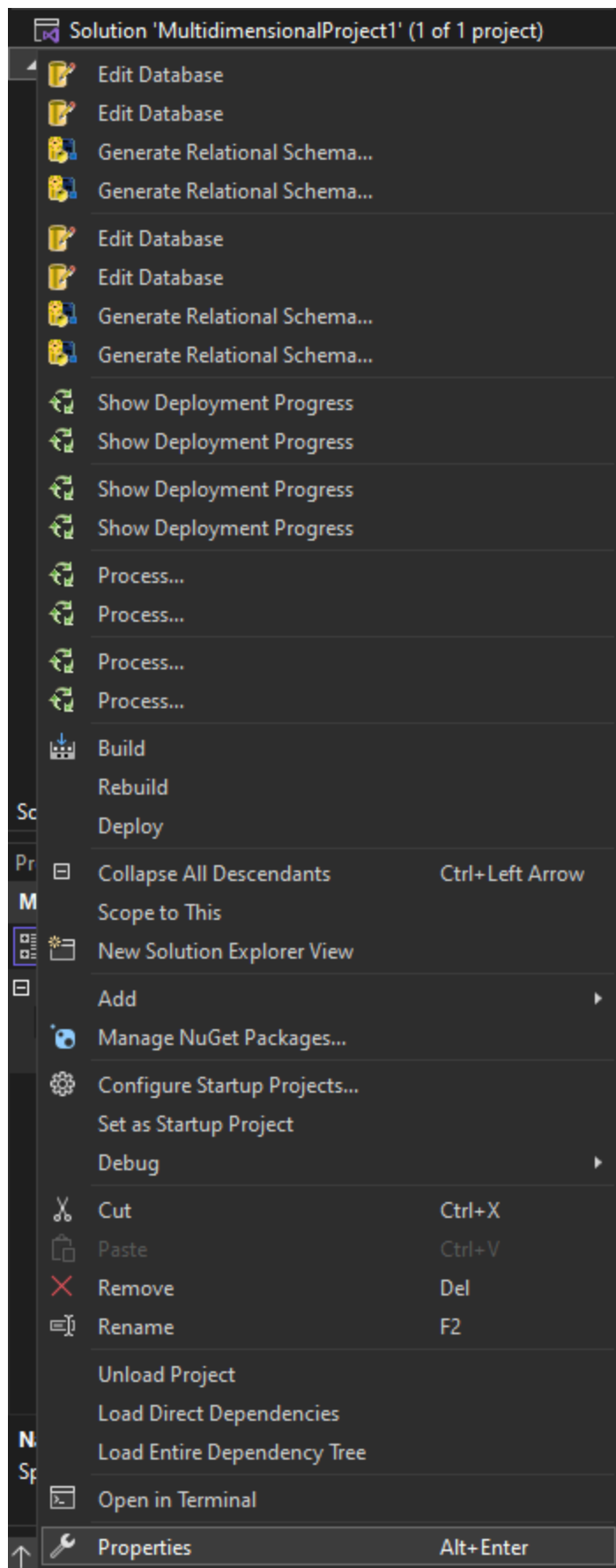


First drag it into attributes and then into hierarchies



7. Deploy Cube

a. Right click on Project name ->Properties



b. Deployment[Server Mode(Deploy All)]->Apply->Ok

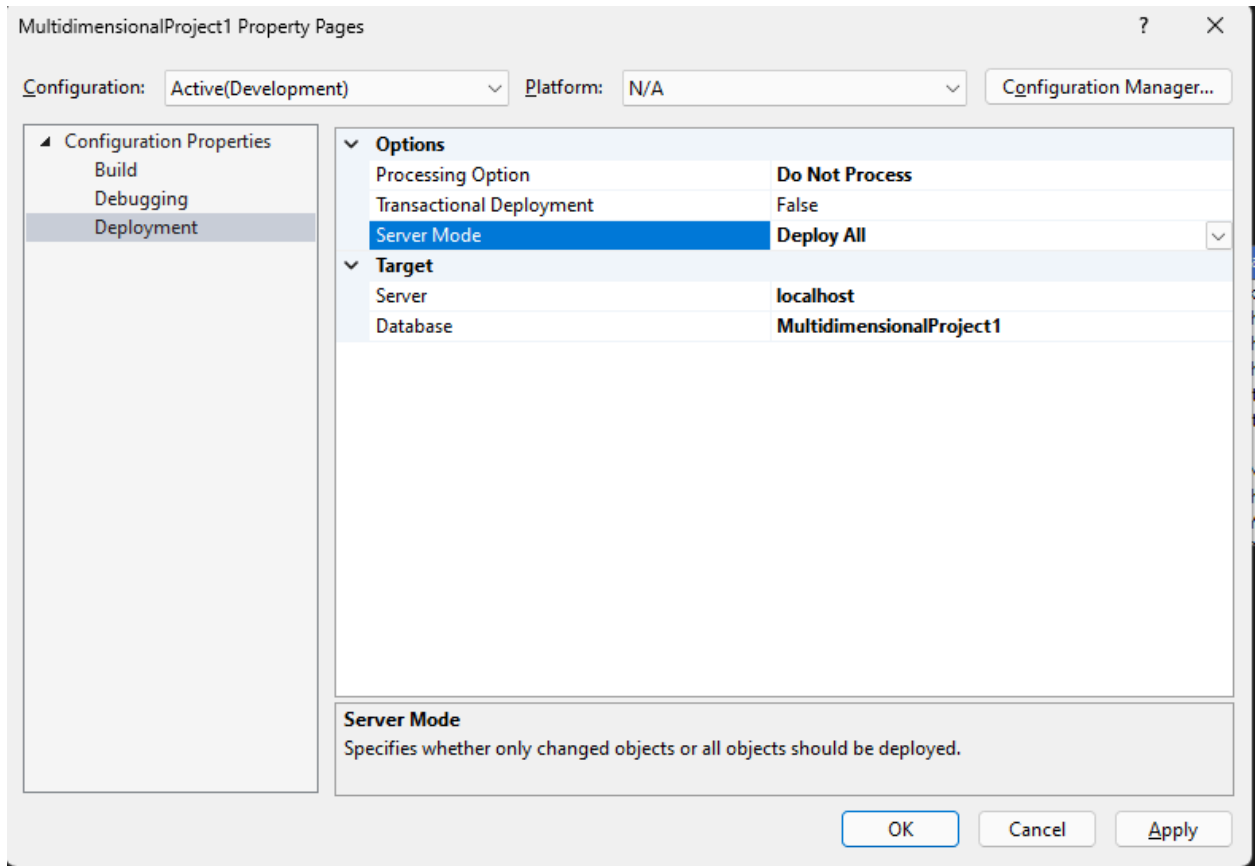
MultidimensionalProject1 Property Pages

Configuration: Active(Development) Platform: N/A Configuration Manager...

Configuration Properties	Deployment Server
Build	Deployment Server Edition Developer
Debugging	Deployment Server Version 15.0
Deployment	Outputs
	Output Path bin\
	Remove Passwords True

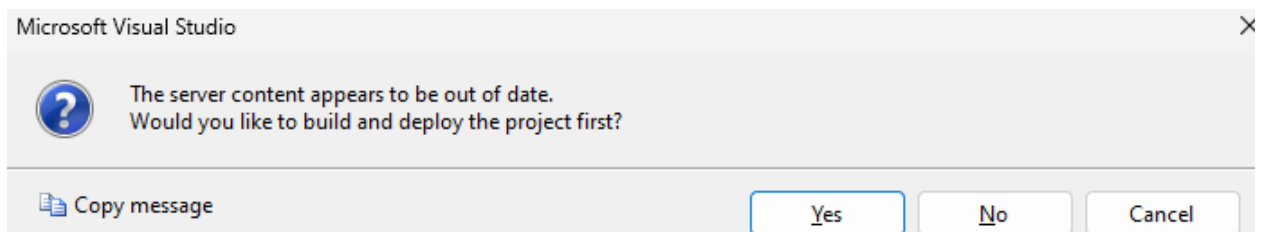
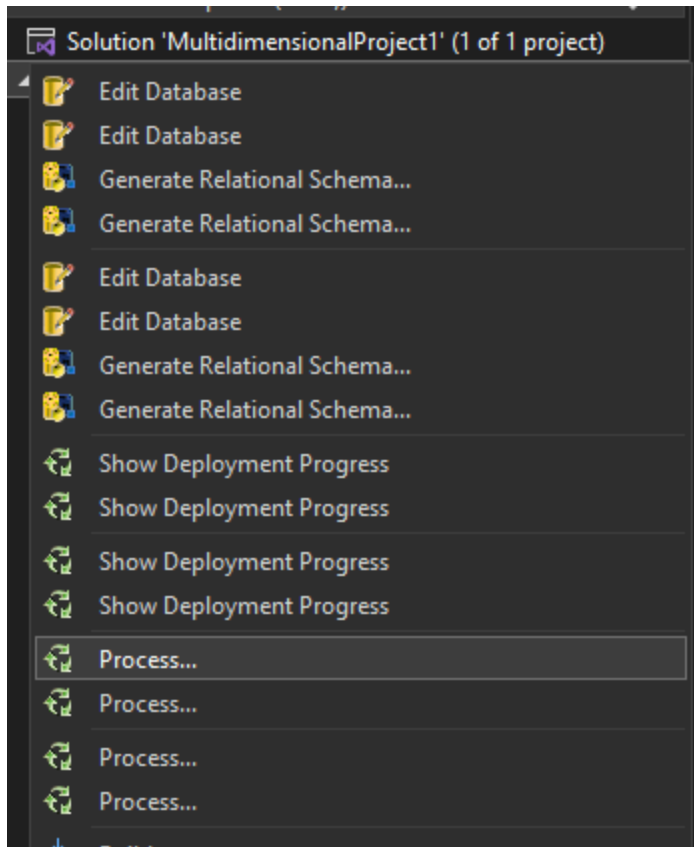
Deployment Server Edition
The edition of the Analysis Services instance to which the project will be deployed.

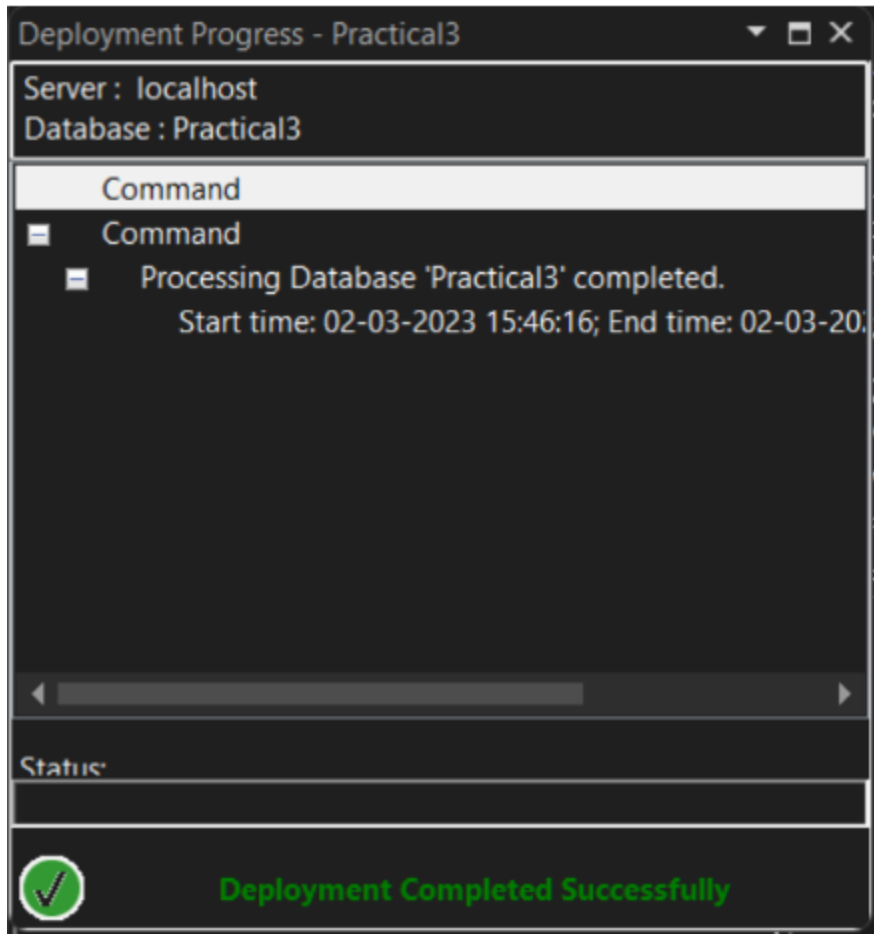
OK Cancel Apply



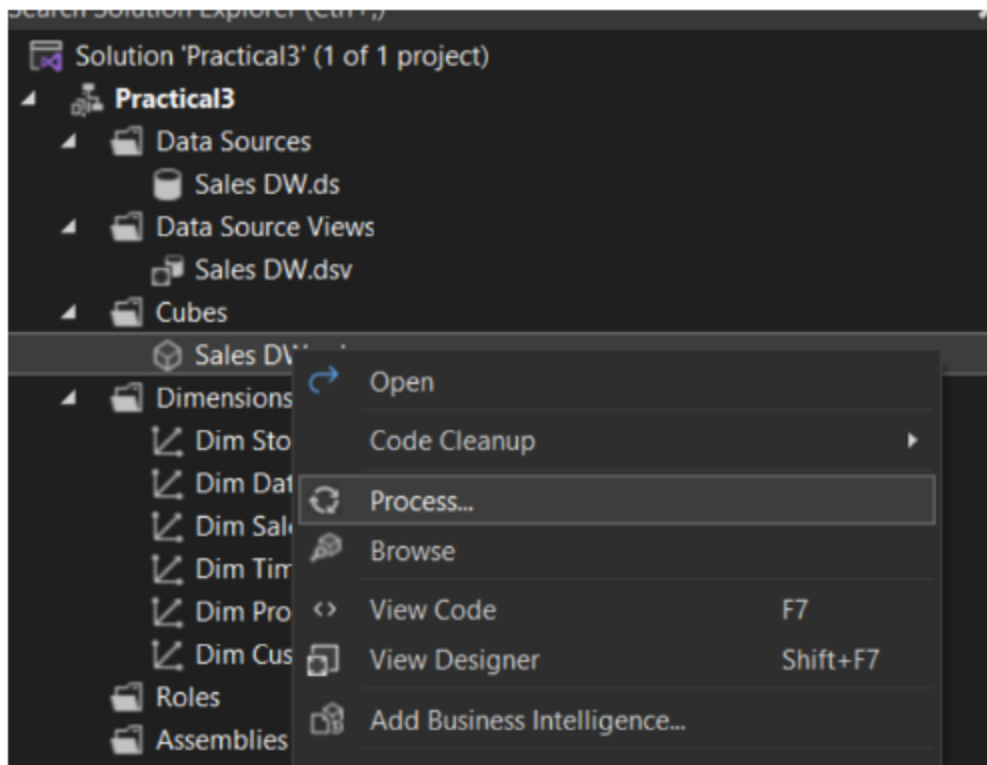
(apply,ok)

c. Right click on project name ->Process





d. To process cube right click on Sales_DW.cube ->Process



e. Click run

Process Cube - Sales DW

Object list:

Object Name	Type	Process Options	Settings
 Sales DW	Cube	Process Full	

Remove

Impact Analysis...

Batch Settings Summary

Processing order:

Parallel

Transaction mode:

(Default)

Dimension errors:

(Default)

Dimension key error log path :

(Default)

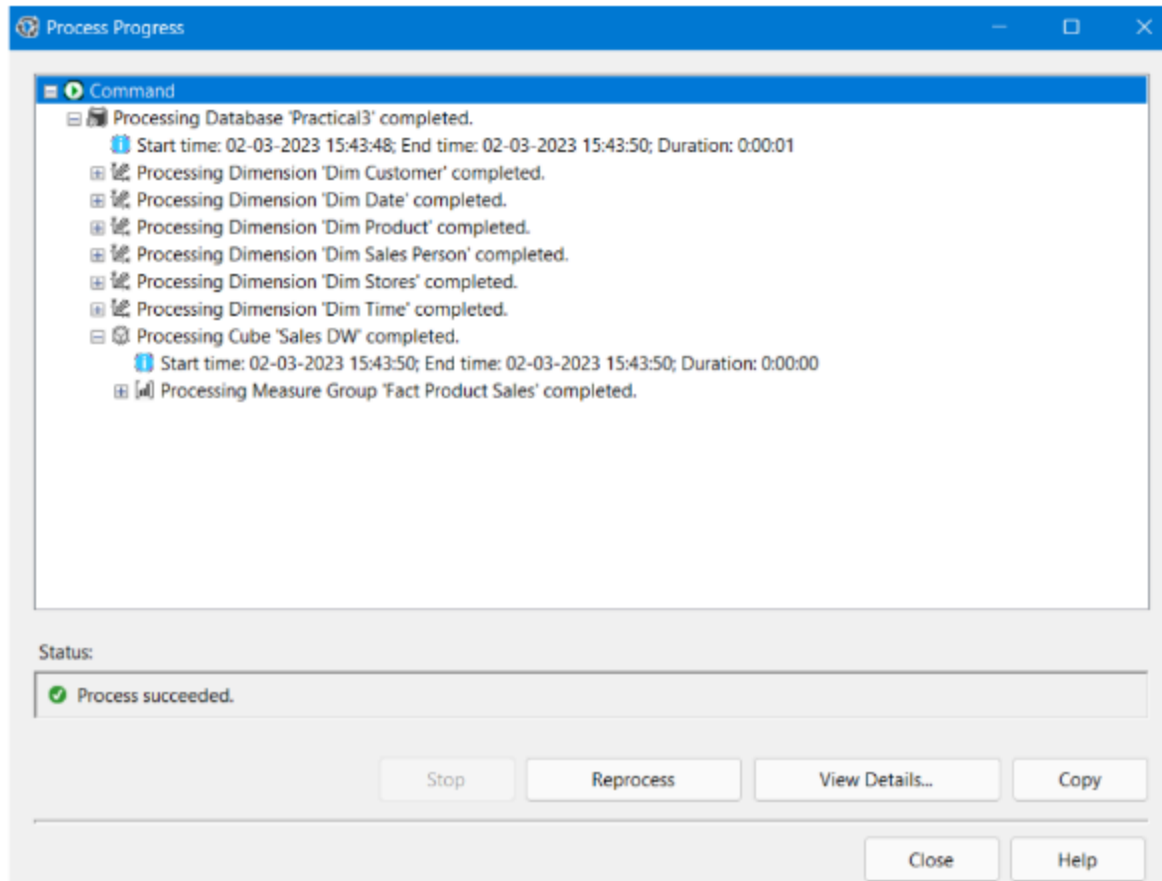
Process affected objects:

Do not process

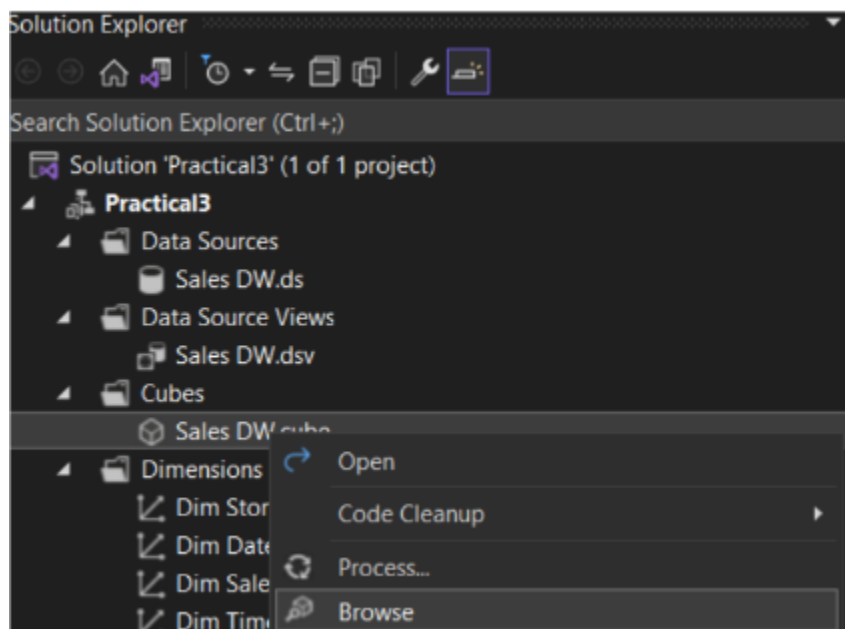
Change Settings...

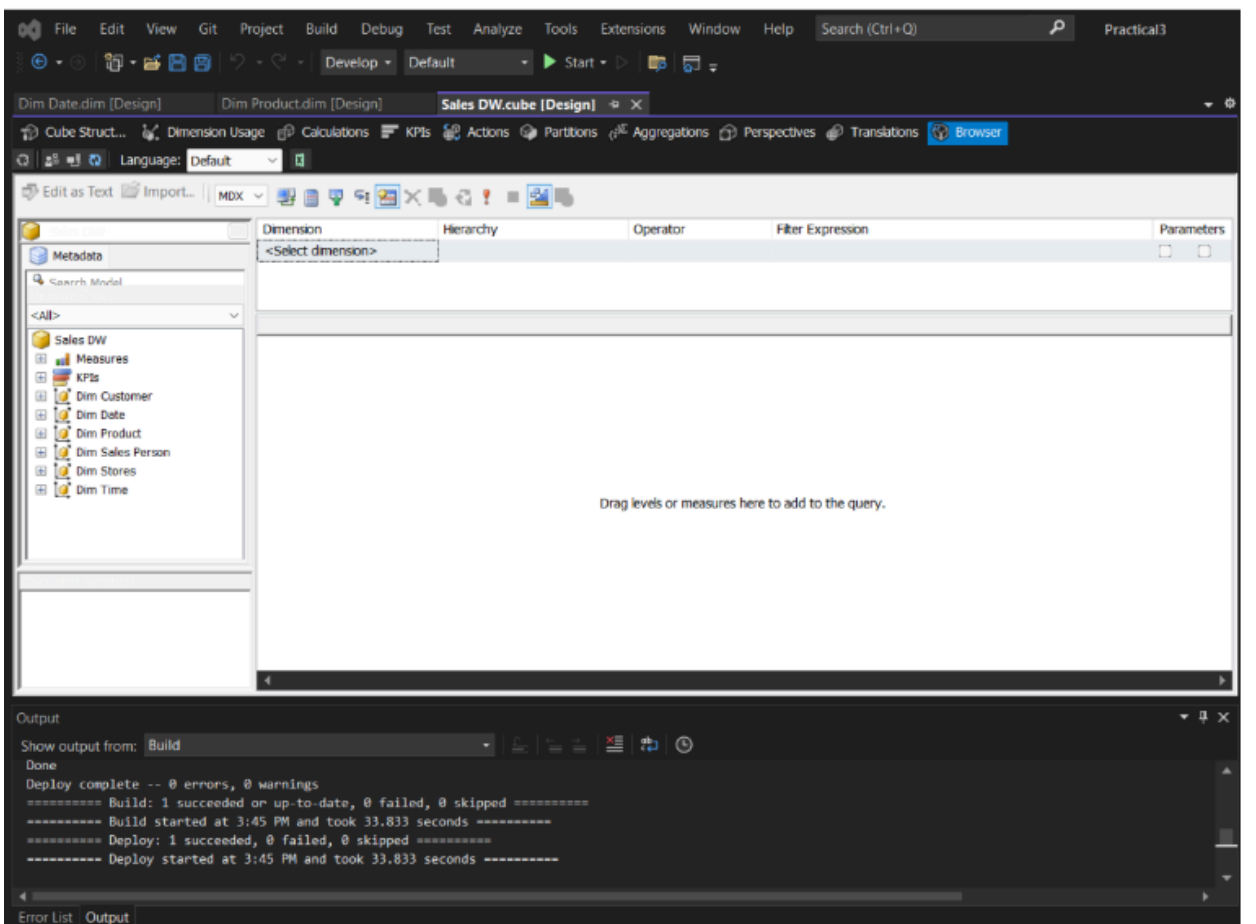
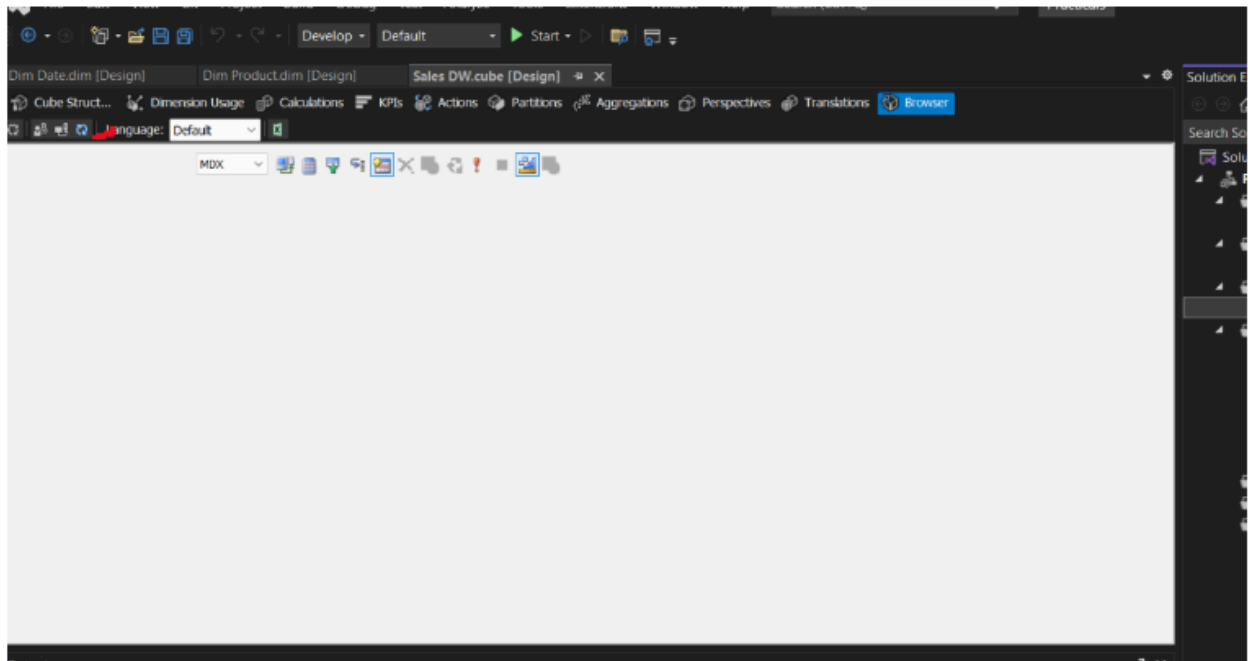
Run...

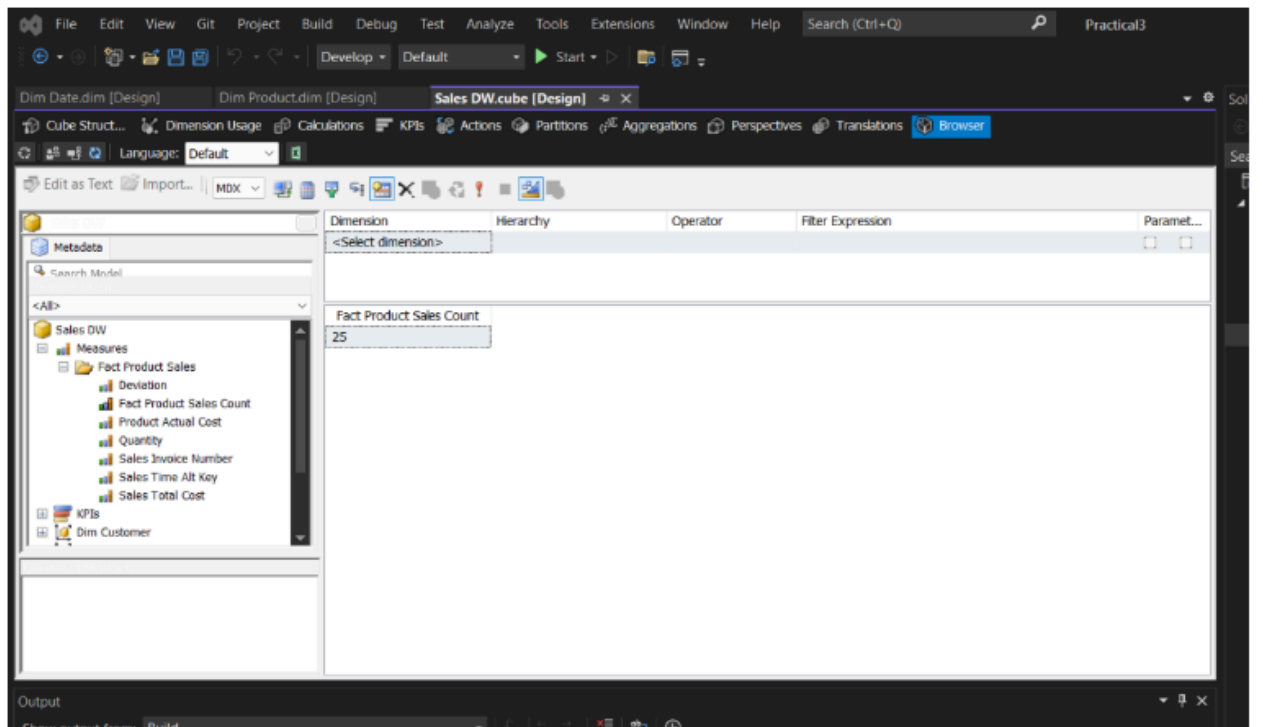
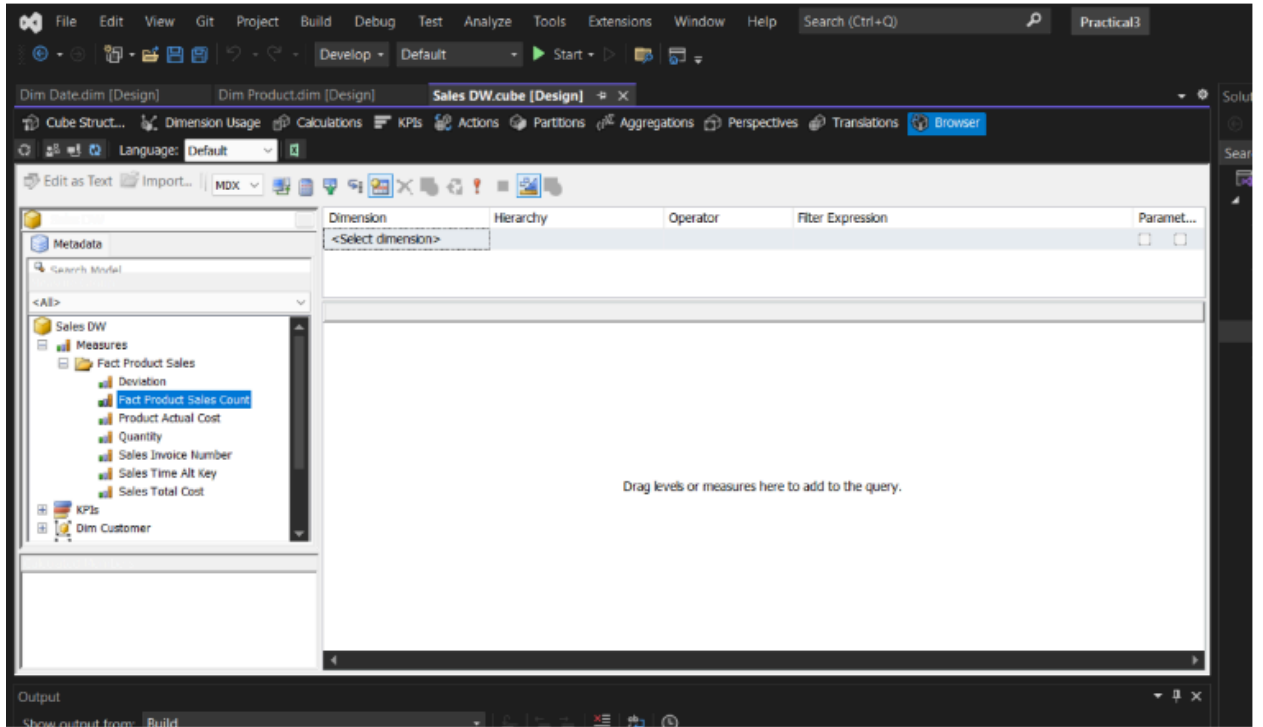
Close



f. Browse the cube for analysis in solution explorer



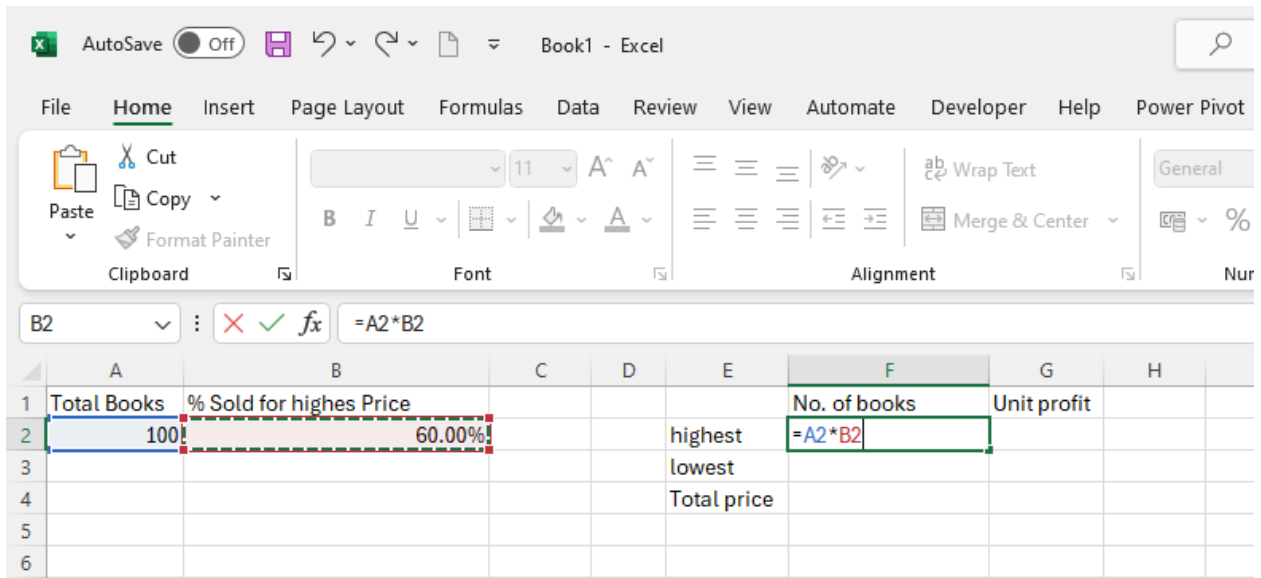




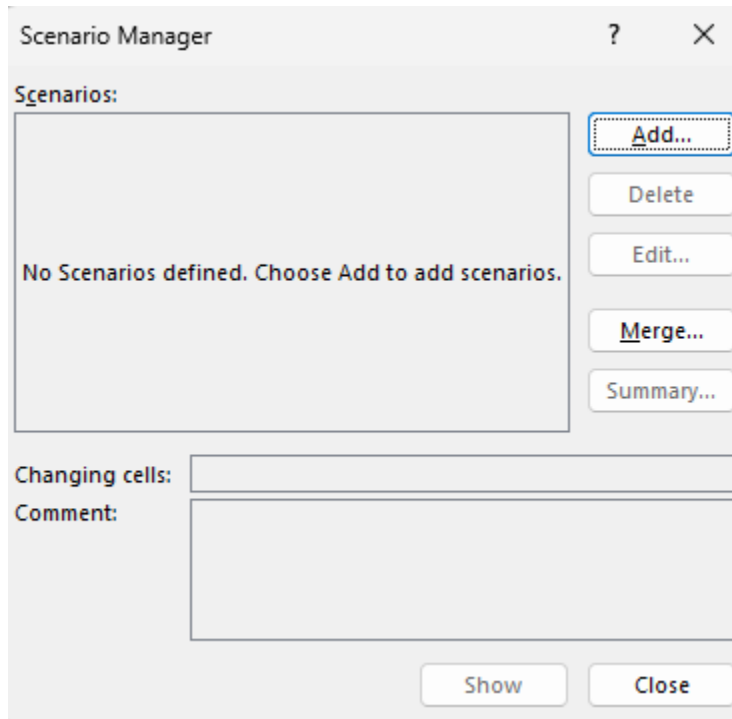
PRACTICAL 4

Aim: Apply the what-if Analysis for data visualization.

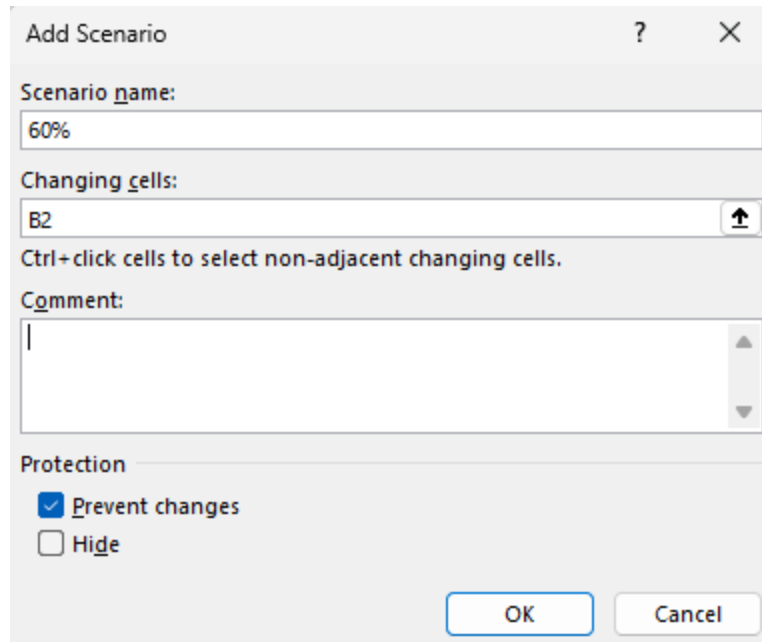
1: Open Excel Sheet ->Enter data {Total Books(100) %Sold for Highest Price(60.00%) Highest[Unit Profit(50)] Lowest[Unit Profit(20)]}



3: Add a scenario by clicking on Add.



Step 4: Type a name (60percent), select cell \$F\$4 (% sold for the highest price) for the Changing cells and click on OK.



Add Scenario

Scenario name:
60%

Changing cells:
B2

Ctrl+click cells to select non-adjacent changing cells.

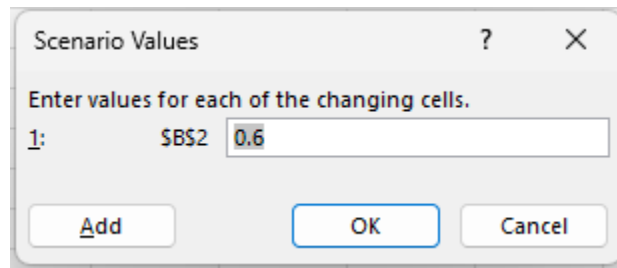
Comment:
|

Protection

☒ Prevent changes
☐ Hide

OK Cancel

Step 5: Enter the corresponding value 0.6 and click on OK again.



Scenario Values

Enter values for each of the changing cells.

1: \$B\$2 0.6

Add OK Cancel

**Step 6: Next, add 4 other scenarios (70%, 80%, 90% and 100%)
Finally, your Scenario Manager should be consistent with the picture below.**

Add Scenario ? X

Scenario name:
70%

Changing cells:
B2 

Ctrl+click cells to select non-adjacent changing cells.

Comment:
|

Protection

☒ Prevent changes
☐ Hide

OK Cancel

Scenario Values ? X

Enter values for each of the changing cells.

1: \$B\$2 0.7

Add OK Cancel

Same for 80-90-100 (0.8,0.9,1)

Step 7: Click On Summary

Scenario Manager ? X

Scenarios:

60%
70%
80%
90%
100%

Add...
Delete
Edit...
Merge...
Summary...

Changing cells: \$B\$2

Comment:

Show Close

Scenario Summary ? X

Report type

☒ Scenario summary
☐ Scenario PivotTable report

Result cells:

= \$G\$4

OK Cancel

Scenario Summary

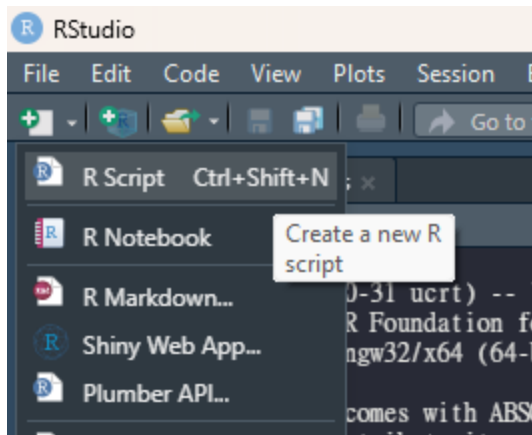
Current Values:	60%	70%	80%	90%	100%
Changing Cells:					
\$B\$2	60.00%	60.00%	70.00%	80.00%	90.00%
Result Cells:					
\$G\$4	3800	3800	4100	4400	5000

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

PRACTICAL 5

Aim: Perform the data classification using classification algorithm

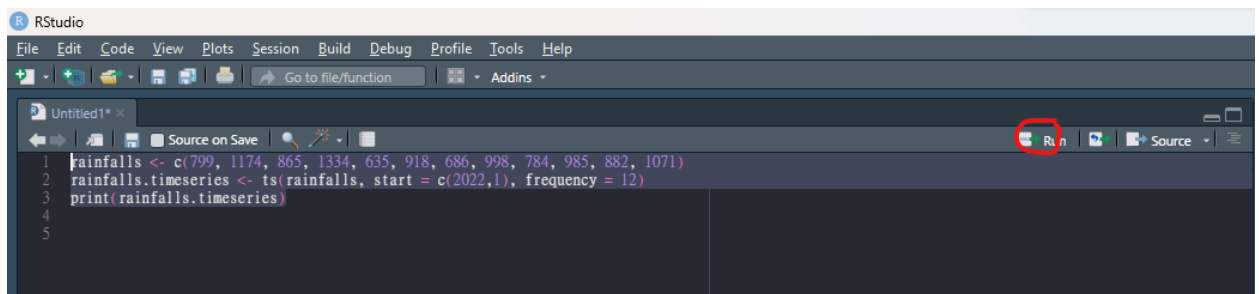
1. Open RStudio and click new script



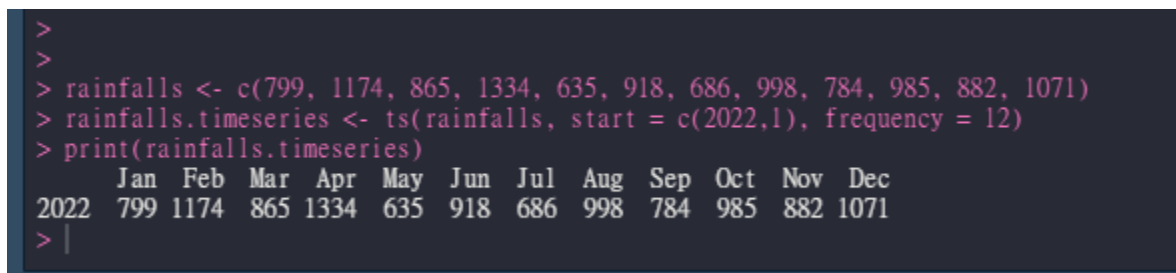
2. Type code and run

```
rainfalls <- c(799, 1174, 865, 1334, 635, 918, 686, 998, 784,
985, 882, 1071)
rainfalls.timeseries <- ts(rainfalls, start = c(2022,1),
frequency = 12)
print(rainfalls.timeseries)
```

Select three lines and run:-



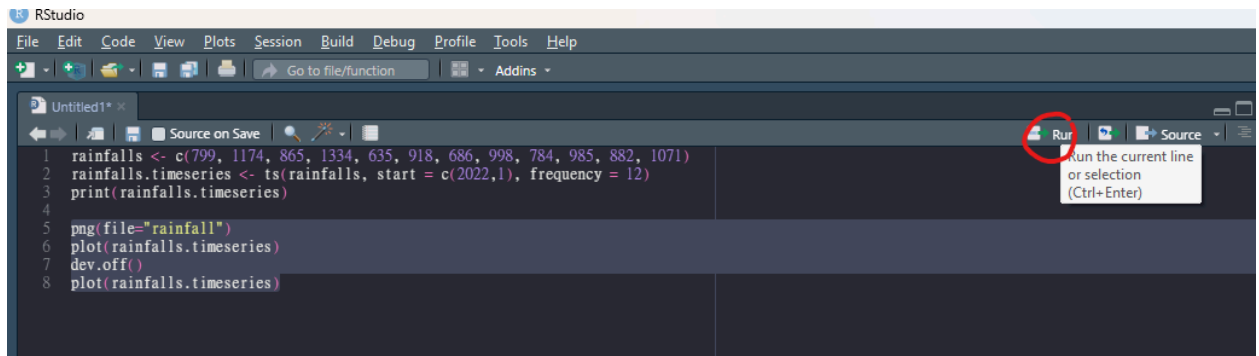
Output:-



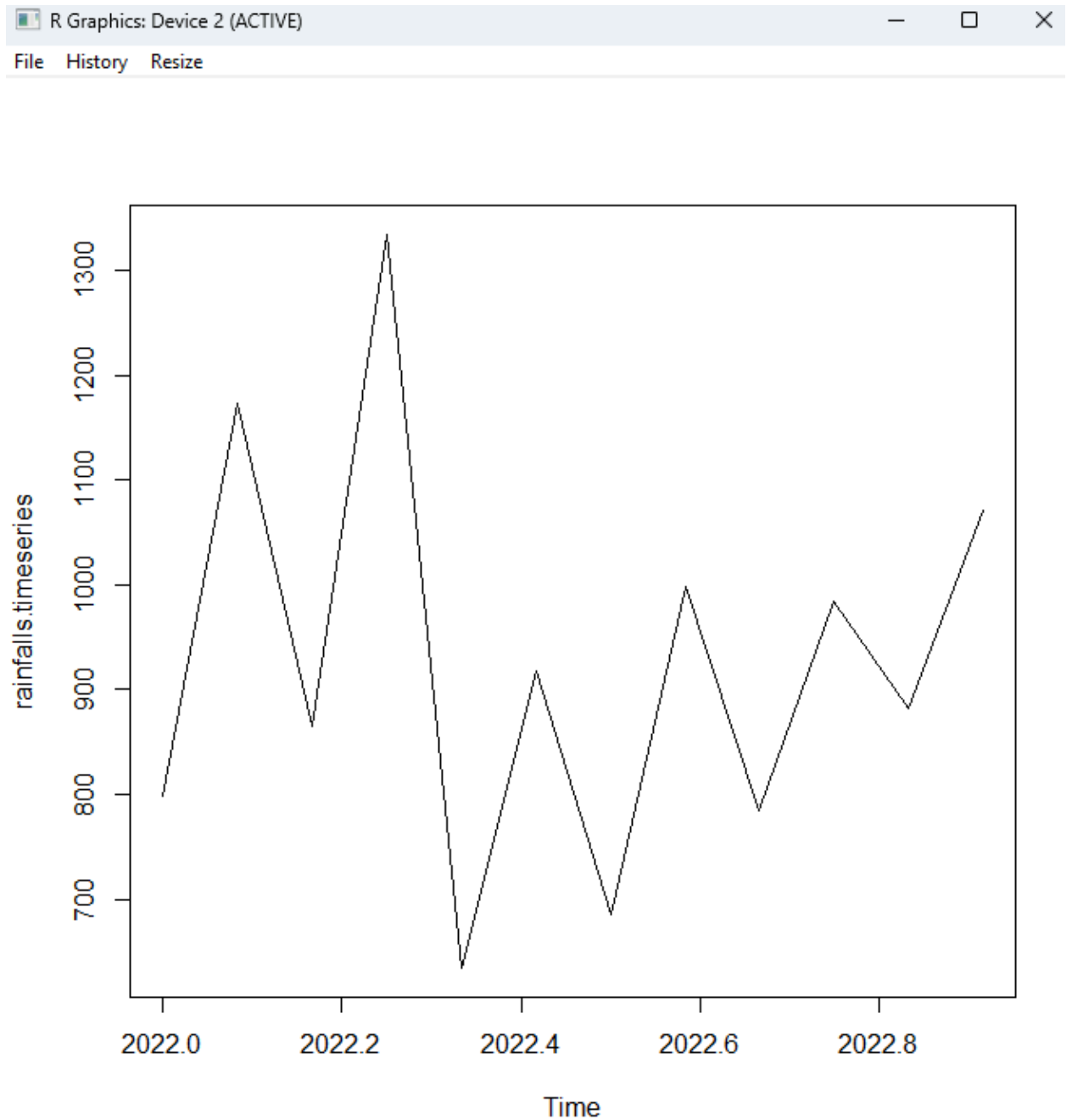
3. Second code;-

```
png(file="rainfall")
```

```
plot(rainfalls.timeseries)
dev.off()
plot(rainfalls.timeseries)
```



Output:-



PRACTICAL 6

Aim: K-Means clustering using R

Code:-

```
newiris <- iris  
newiris$Species <- NULL  
kc <- kmeans(newiris,3)  
kc
```

Output:-

[illegible]

Code:-

```
table(iris$Species, kc$cluster)
```

Output:-

```
> table(iris$Species, kc$cluster)
```

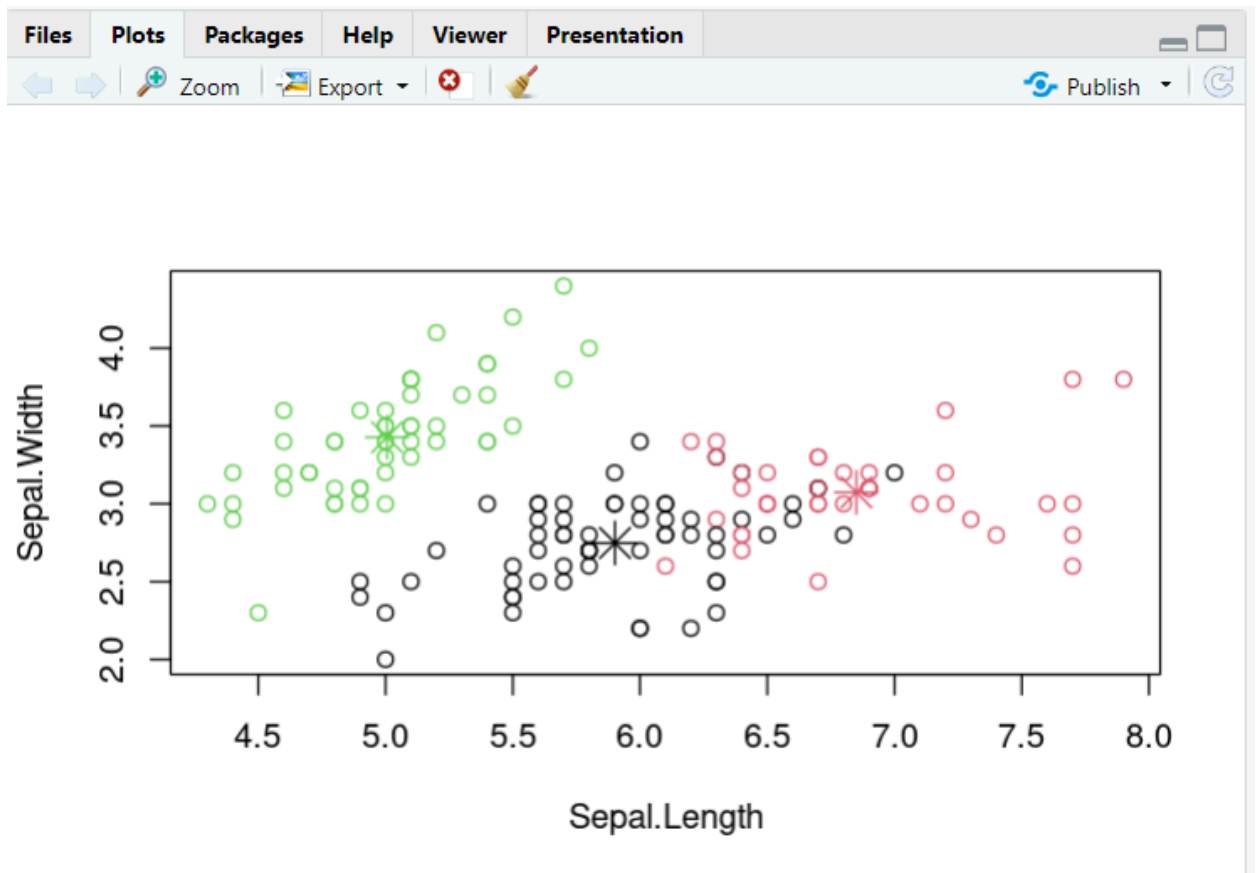
	1	2	3
setosa	50	0	0
versicolor	0	48	2
virginica	0	14	36

```
> |
```

Code:-

```
plot(newiris[c("Sepal.Length", "Sepal.Width"),
col=kc$cluster)
points(kc$centers[, c("Sepal.Length", "Sepal.Width")],
col=1:3, pch = 8, cex = 2)
dev.off()
```

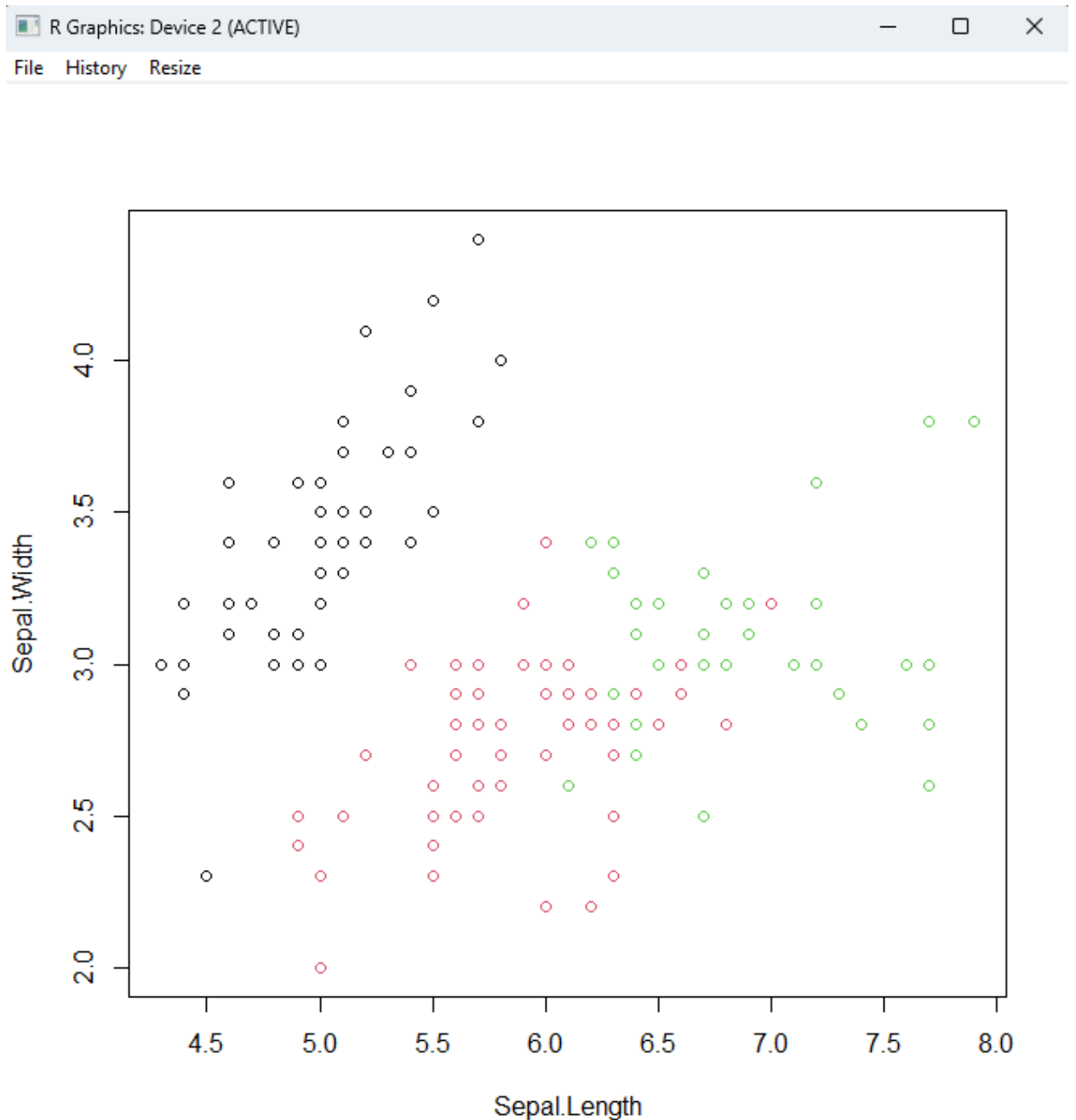
Output;-



Code;-

```
plot(newiris[c("Sepal.Length", "Sepal.Width"]),  
col=kc$cluster)
```

Output;-



PRACTICAL 7

Aim: Prediction Using Linear Regression

Code:-

```
x<-c(151,174,138,186,128,136,179,163,152,131)
y<-c(63,81,56,91,47,57,76,72,62,48)
relation<-lm(y~x)
print(relation)
```

Output:-

```
> plot(newiris[c("Sepal.Length", "Sepal.Width")], col=kc$cluster)
> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> print(relation)
```

Call:

lm(formula = y ~ x)

Coefficients:

(Intercept)	x
-38.4551	0.6746

Code:-

```
x<-c(151,174,138,186,128,136,179,163,152,131)
y<-c(63,81,56,91,47,57,76,72,62,48)
relation<-lm(y~x)
print(summary(relation))
```

Output:-

```
> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> print(summary(relation))
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.3002	-1.6629	0.0412	1.8944	3.9775

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-38.45509	8.04901	-4.778	0.00139	**
x	0.67461	0.05191	12.997	1.16e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.253 on 8 degrees of freedom

Multiple R-squared: 0.9548, Adjusted R-squared: 0.9491

F-statistic: 168.9 on 1 and 8 DF, p-value: 1.164e-06

Code:-

```
x<-c(151,174,138,186,128,136,179,163,152,131)
```

```
y<-c(63,81,56,91,47,57,76,72,62,48)
```

```
relation<-lm(y~x)
```

```
a <-data.frame(x=170)
```

```
result<-predict(relation,a)
```

```
print(result)
```

Output:-

```

> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> a <-data.frame(x=170)
> result<-predict(relation,a)
> print(result)
      1
76.22869

```

Code:-

```

x<-c(151,174,138,186,128,136,179,163,152,131)
y<-c(63,81,56,91,47,57,76,72,62,48)
relation<-lm(y~x)
png(file="linearregression.png")
plot(y,x,col="blue",main="Height & Weight
Regression",abline(lm(x~y)),cex=1.3,pch=16,xlab="Weight in
Kg",ylab="Height in
cm")
dev.off()

```

Output:-

```

> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> png(file="linearregression.png")
> plot(y,x,col="blue",main="Height & Weight
+ Regression",abline(lm(x~y)),cex=1.3,pch=16,xlab="Weight in Kg",ylab="Height in
+ cm")
> dev.off()
RStudioGD
      2

```

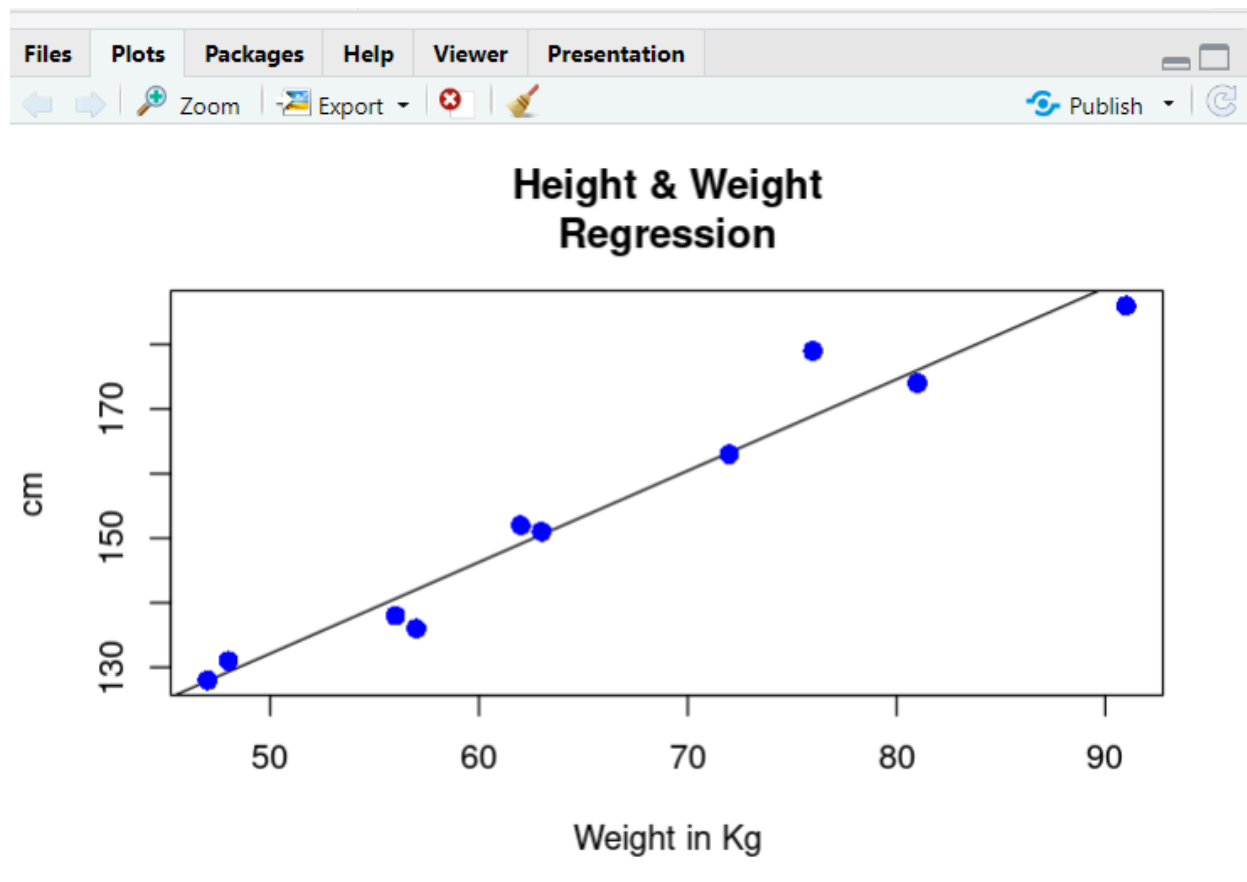
Code:-

```

plot(y,x,col="blue",main="Height & Weight
Regression",abline(lm(x~y)),cex=1.3,pch=16,xlab="Weight in
Kg",ylab="Height in
cm")

```

Output:-



PRACTICAL 8

Aim: Perform the logistic regression on the given data warehouse data

Code:-

```
quality <- read.csv("C/quality.csv")
str(quality)
table(quality$PoorCare)
98/131
install.packages("caTools")
library(caTools)
set.seed(88)
```

Output:-

```
> quality <- read.csv("quality.csv")
> str(quality)
'data.frame': 131 obs. of 14 variables:
 $ MemberID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ InpatientDays  : int  0 1 0 0 8 2 16 2 2 4 ...
 $ ERVisits       : int  0 1 0 1 2 0 1 0 1 2 ...
 $ OfficeVisits   : int  18 6 5 19 19 9 8 8 4 0 ...
 $ Narcotics      : int  1 1 3 0 3 2 1 0 3 2 ...
 $ DaysSinceLastERVisit: num  731 411 731 158 449 ...
 $ Pain          : int  10 0 10 34 10 6 4 5 5 2 ...
 $ TotalVisits    : int  18 8 5 20 29 11 25 10 7 6 ...
 $ ProviderCount  : int  21 27 16 14 24 40 19 11 28 21 ...
 $ MedicalClaims  : int  93 19 27 59 51 53 40 28 20 17 ...
 $ ClaimLines     : int  222 115 148 242 204 156 261 87 98 66 ...
 $ StartedOnCombination: logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
 $ AcuteDrugGapSmall : int  0 1 5 0 0 4 0 0 0 0 ...
 $ PoorCare       : int  0 0 0 0 0 1 0 0 1 0 ...
> table(quality$PoorCare)

 0  1
98 33
> 98/131
[1] 0.7480916
> install.packages("caTools")
Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.3'
(as 'lib' is unspecified)
also installing the dependency 'bitops'

trying URL 'http://rspm/default/__linux__/focal/latest/src/contrib/bitops_1.0-7.tar.gz'
Content type 'application/x-gzip' length 24790 bytes (24 KB)
=====
downloaded 24 KB

trying URL 'http://rspm/default/__linux__/focal/latest/src/contrib/caTools_1.18.2.tar.g
z'
Content type 'application/x-gzip' length 199922 bytes (195 KB)
=====
downloaded 195 KB

* installing *binary* package 'bitops' ...
* DONE (bitops)
* installing *binary* package 'caTools' ...
* DONE (caTools)

The downloaded source packages are in
      '/tmp/RtmpVXhH64/downloaded_packages'
> library(caTools)
> set.seed(88)
```

split=sample.split(quality\$PoorCare,SplitRatio=0.75)
split

```

> split=sample.split(quality$PoorCare,SplitRatio=0.75)
> split
  [1]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE
 [14]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE
 [27]  TRUE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
 [40]  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE  TRUE  TRUE FALSE
 [53] FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
 [66] FALSE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
 [79]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
 [92]  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE
[105]  TRUE  TRUE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE
[118] FALSE  TRUE  TRUE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE
[131] FALSE
> |

```

qualityTrain=subset(quality,split==TRUE)

qualityTest=subset(quality,split==FALSE)

nrow(qualityTrain)

```

> qualityTrain=subset(quality,split==TRUE)
> qualityTest=subset(quality,split==FALSE)
> nrow(qualityTrain)
[1] 99

```

**QualityLog = glm(PoorCare ~ OfficeVisits +
Narcotics,data=qualityTrain,family=binomial)
summary(QualityLog)**

```
> QualityLog = glm(PoorCare ~ OfficeVisits + Narcotics,data=qualityTrain,family=binomial)
> summary(QualityLog)
```

```
Call:
glm(formula = PoorCare ~ OfficeVisits + Narcotics, family = binomial,
    data = qualityTrain)
```

```
Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.64613	0.52357	-5.054	4.33e-07	***
OfficeVisits	0.08212	0.03055	2.688	0.00718	**
Narcotics	0.07630	0.03205	2.381	0.01728	*

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 111.888  on 98  degrees of freedom
Residual deviance: 89.127  on 96  degrees of freedom
AIC: 95.127
```

```
Number of Fisher Scoring iterations: 4
```

```
predictTrain=predict(QualityLog,type="response")
summary(predictTrain)
```

```
> predictTrain=predict(QualityLog,type="response")
> summary(predictTrain)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.06623 0.11912 0.15967 0.25253 0.26765 0.98456
```

```
tapply(predictTrain,qualityTrain$PoorCare,mean)
```

```
table(qualityTrain$PoorCare,predictTrain>0.5)
```

```
table(qualityTrain$PoorCare,predictTrain>0.7)
```

```
8/25
```

```
73/74
```

```
table(qualityTrain$PoorCare,predictTrain>0.2)
```

```
16/25
```

```
54/74
```

```

> tapply(predictTrain,qualityTrain$PoorCare,mean)
      0      1
0.1894512 0.4392246
> table(qualityTrain$PoorCare,predictTrain>0.5)

      FALSE TRUE
0      70    4
1      15   10
> table(qualityTrain$PoorCare,predictTrain>0.7)

      FALSE TRUE
0      73    1
1      17    8
> 8/25
[1] 0.32
> 73/74
[1] 0.9864865
> table(qualityTrain$PoorCare,predictTrain>0.2)

      FALSE TRUE
0      54   20
1       9   16
> 16/25
[1] 0.64
> 54/74
[1] 0.7297297
> |

```

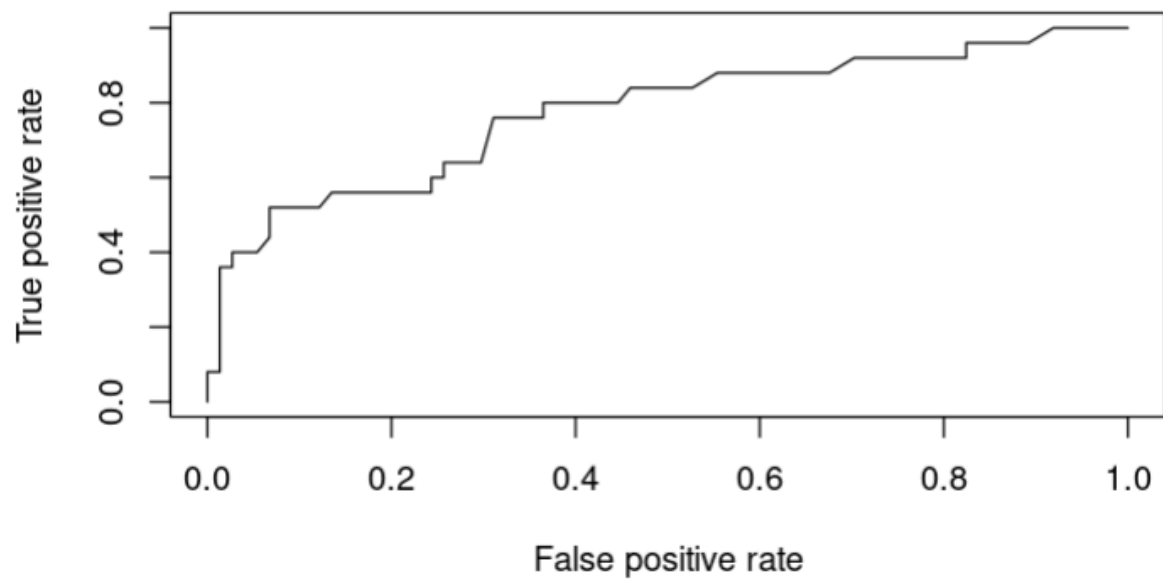
install.packages("ROCR")

library(ROCR)

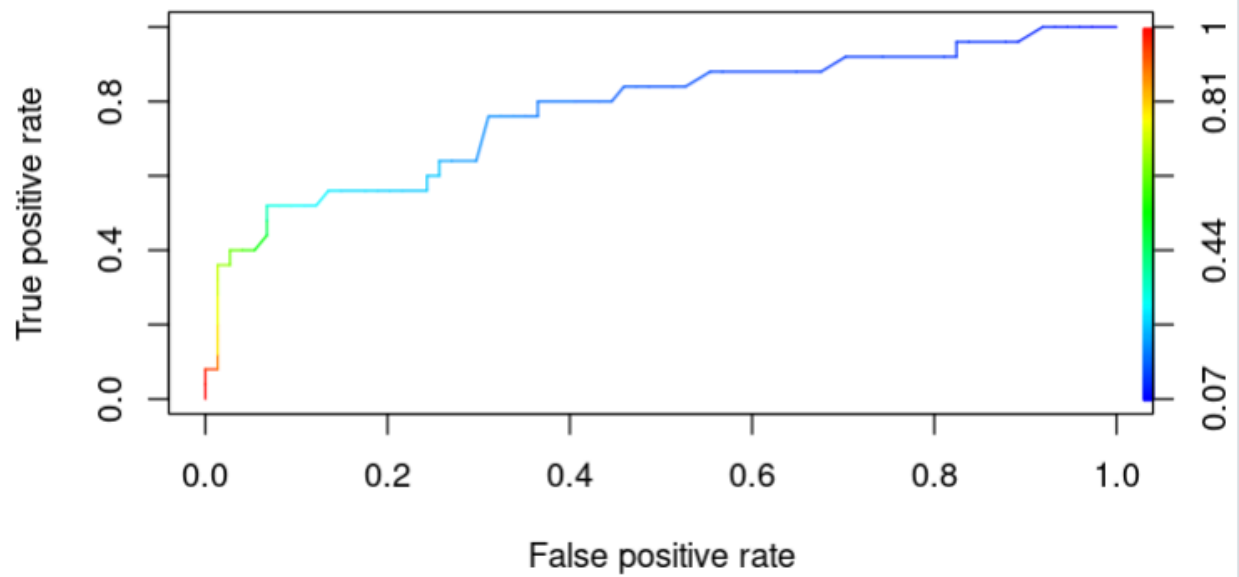
ROCRpred = prediction(predictTrain,qualityTrain\$PoorCare)

ROCRperf = performance(ROCRpred,"tpr","fpr")

plot(ROCRperf)



`plot(ROCRperf,colorize=TRUE)`



```
plot(ROCRperf,colorize=TRUE,print.cutoffs.at=seq(0,1,by=0.1),tex  
t.adj=c(-0.2,1.7))
```

